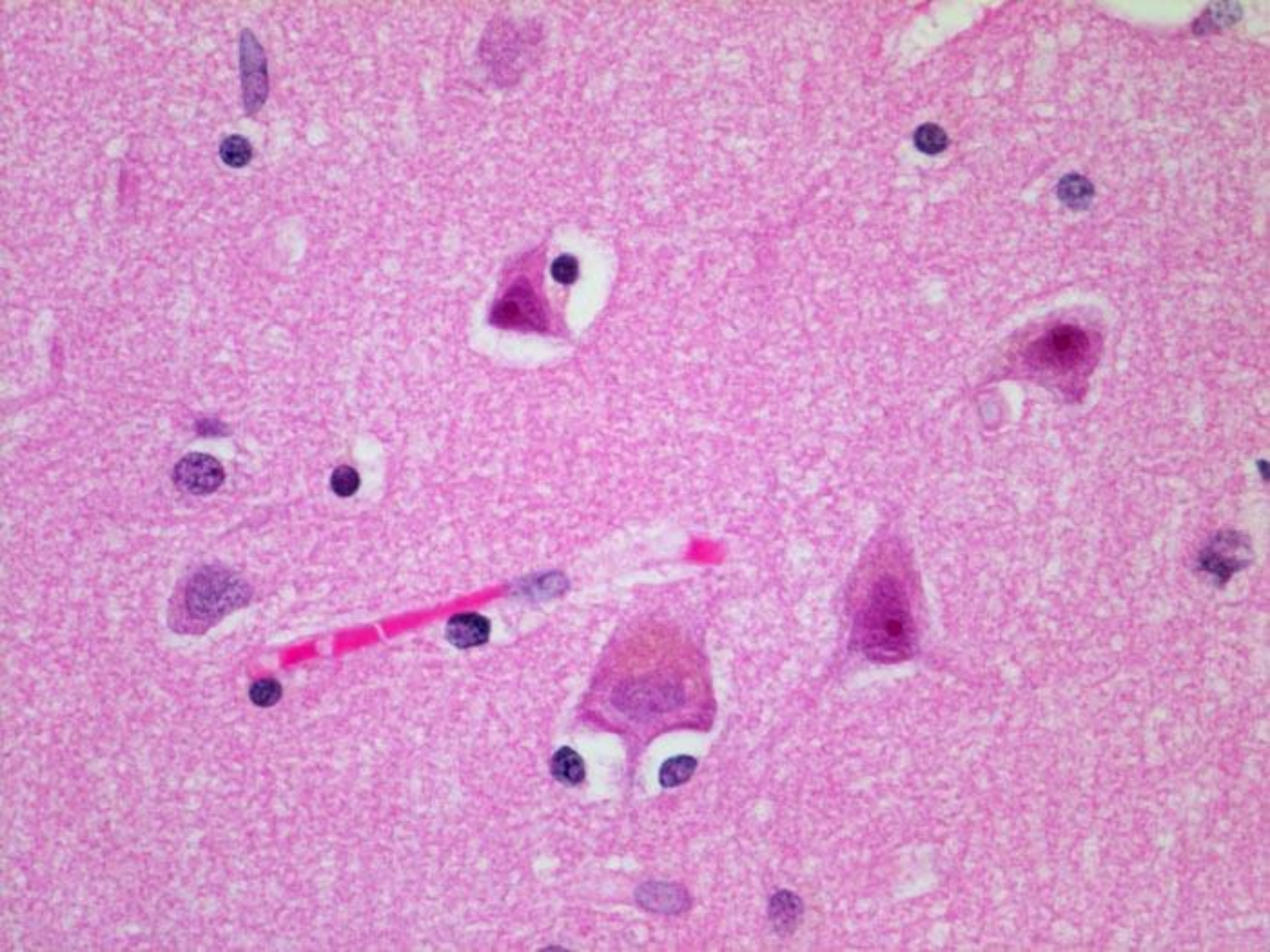
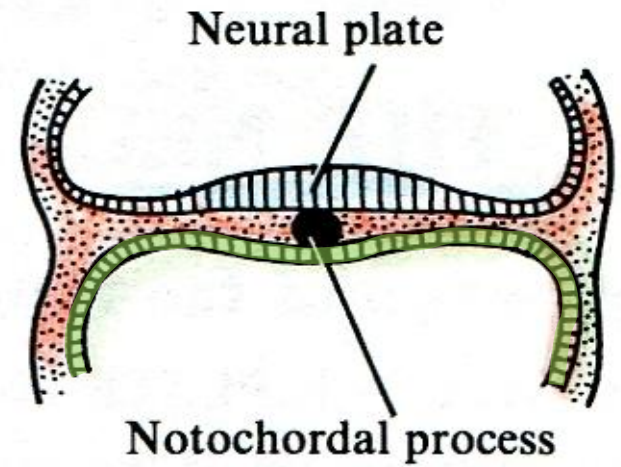
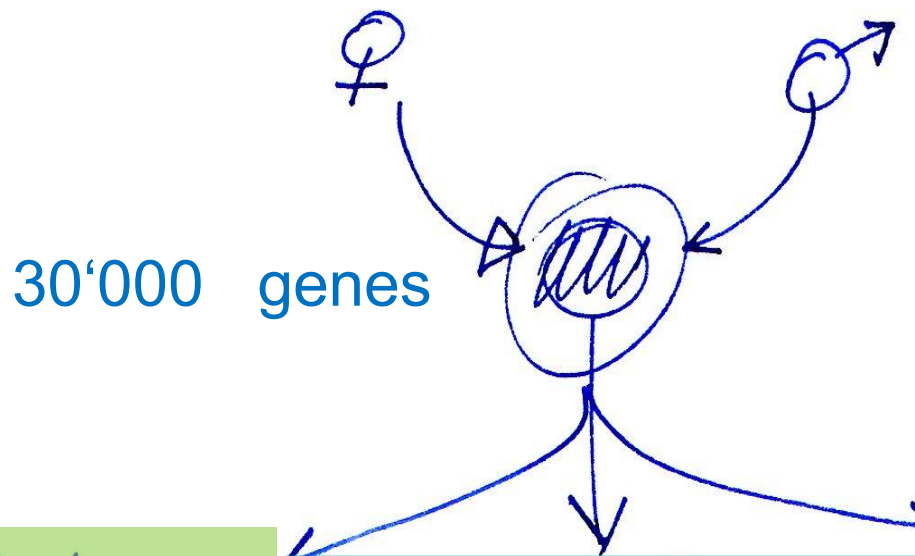


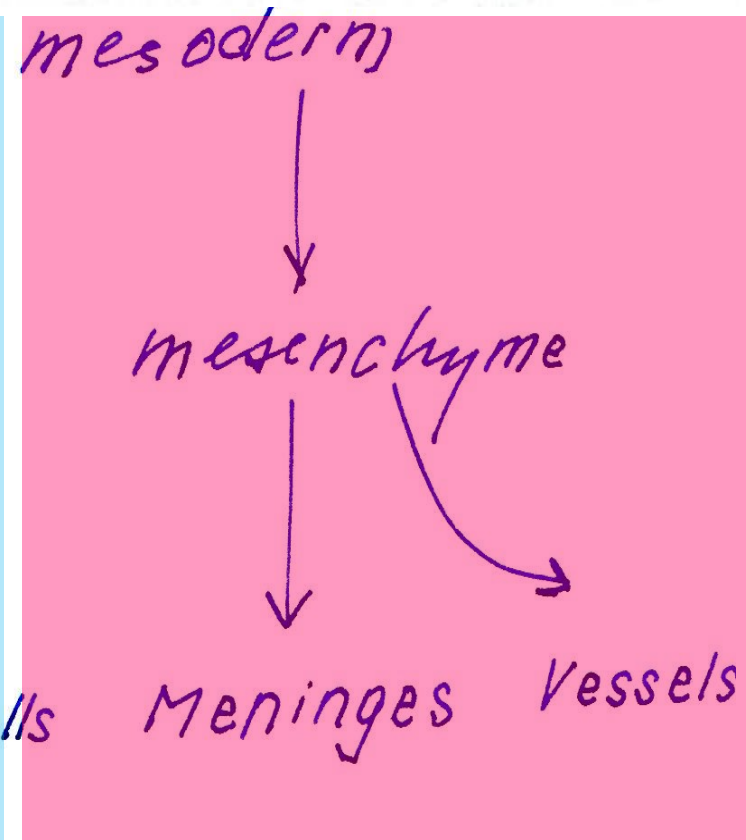
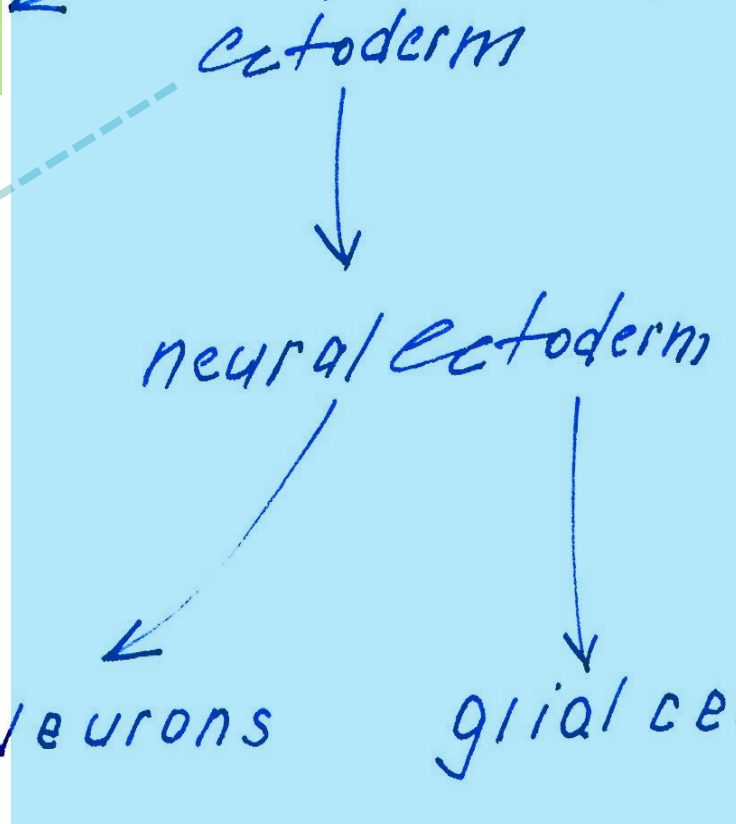
CNS III

Brain tumors

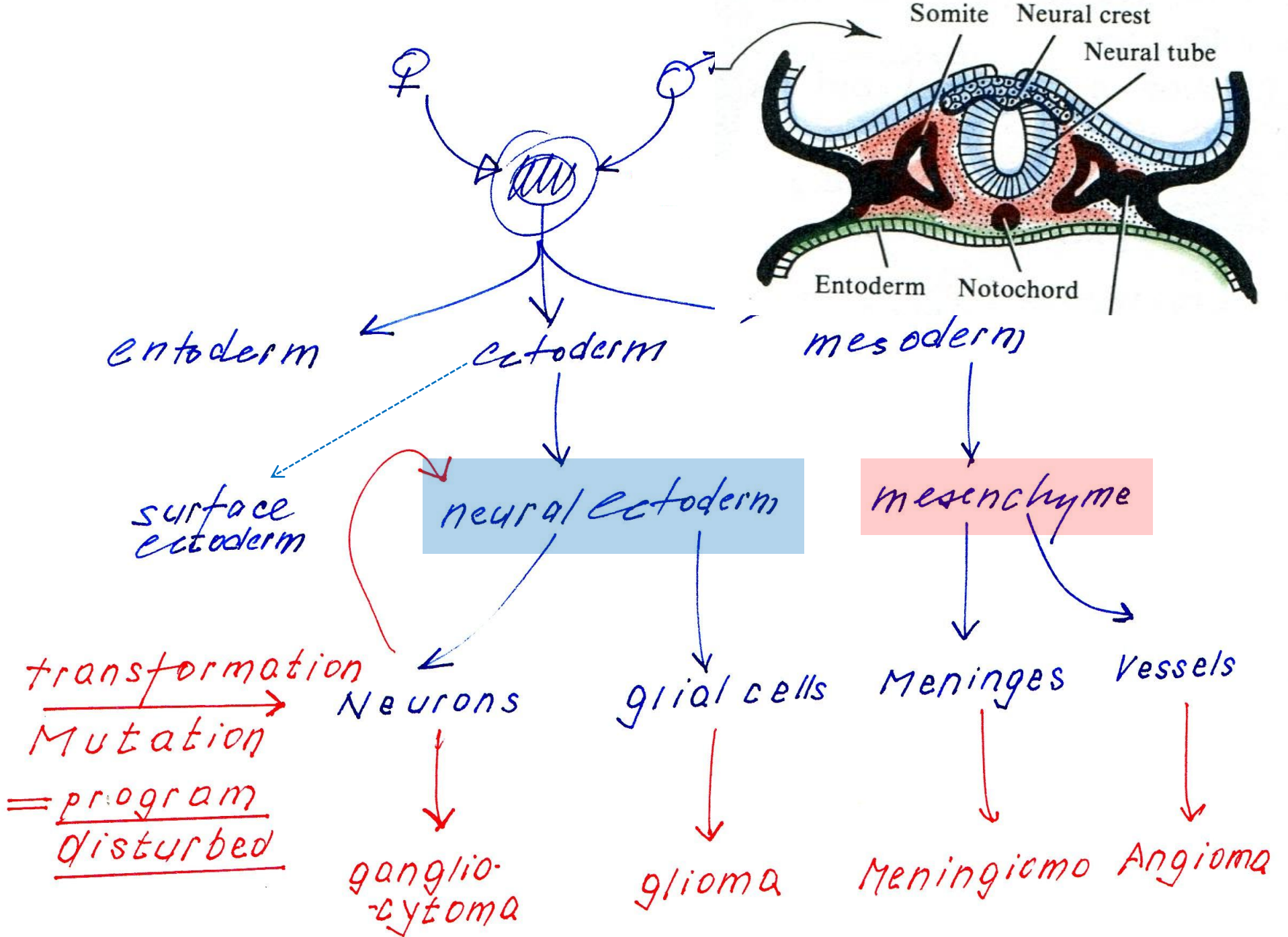




entoderm



surface ectoderm



Astrocyte

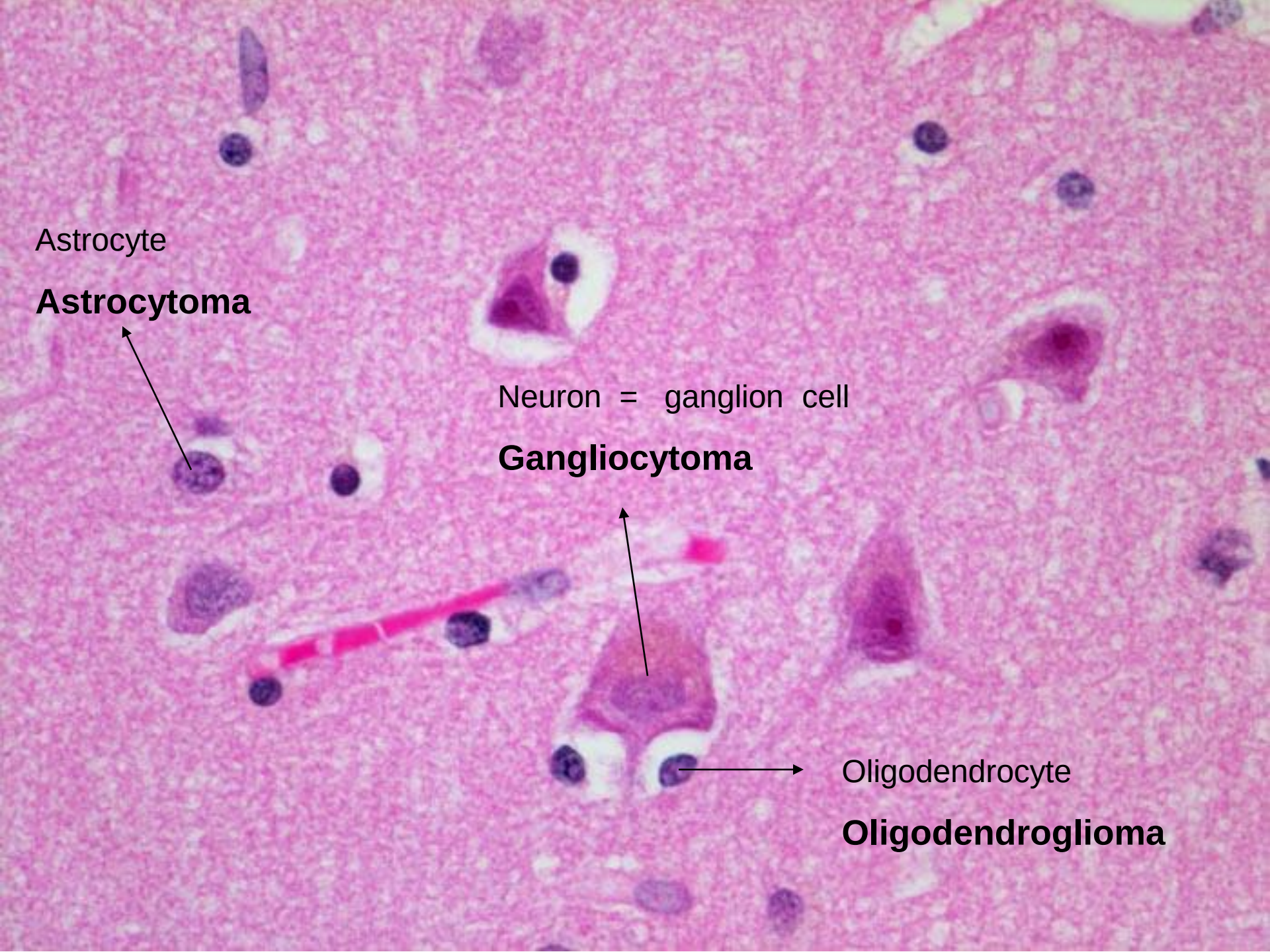
Astrocytoma

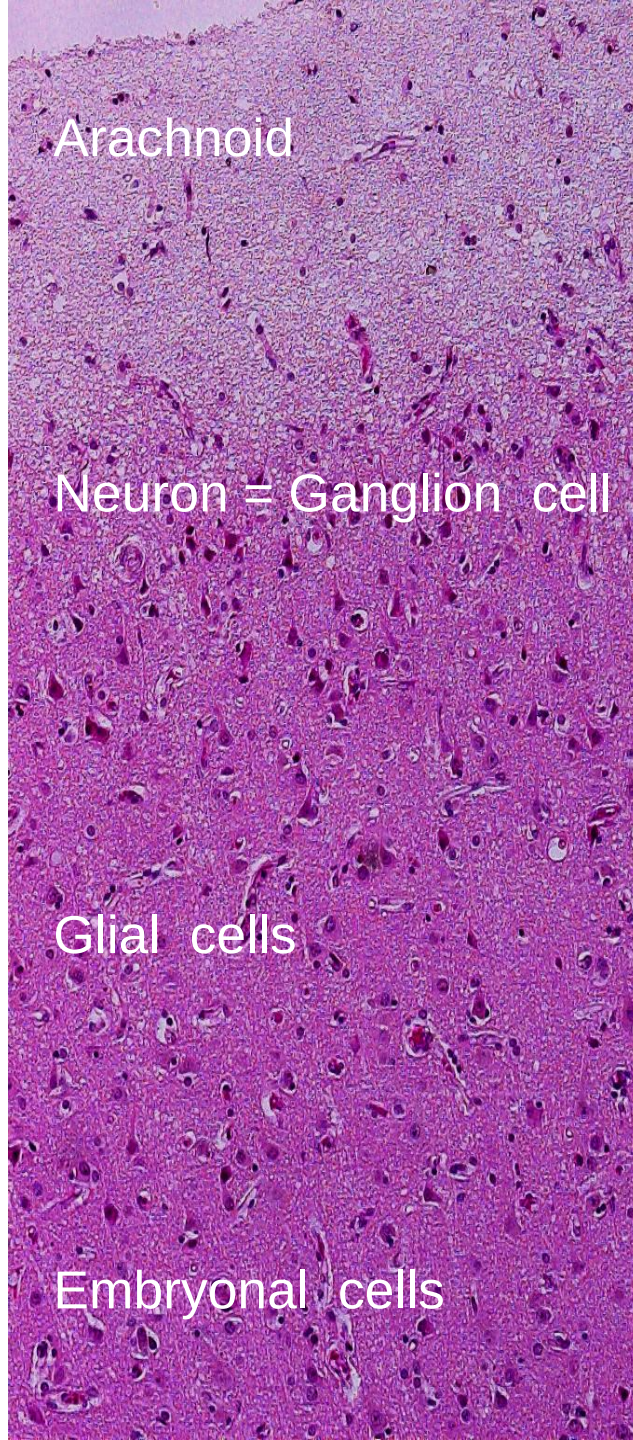
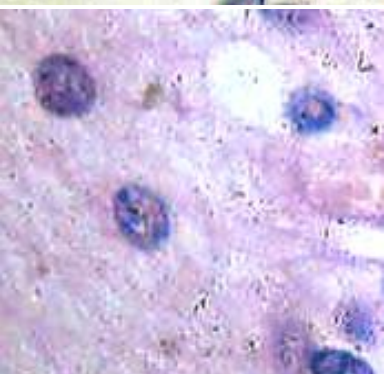
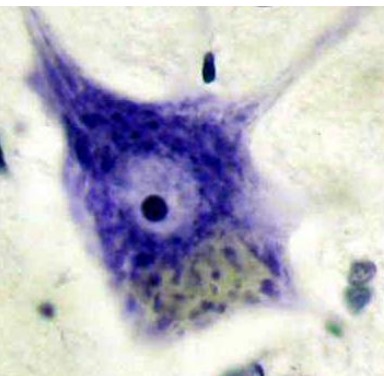
Neuron = ganglion cell

Gangliocytoma

Oligodendrocyte

Oligodendroglioma





Arachnoid

Neuron = Ganglion cell

Glial cells

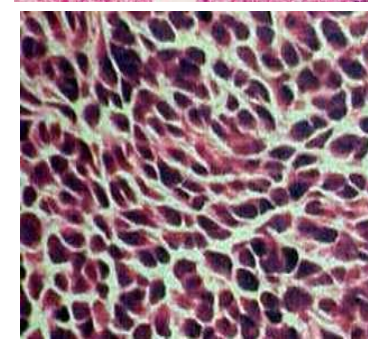
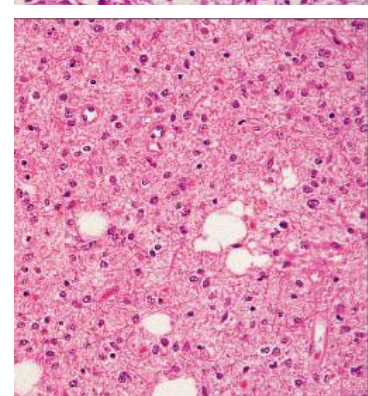
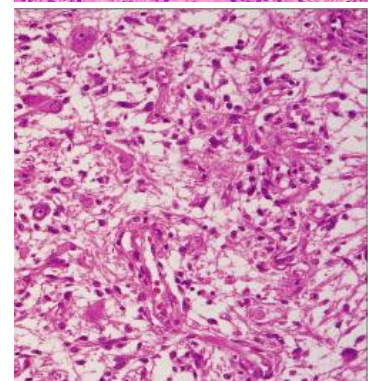
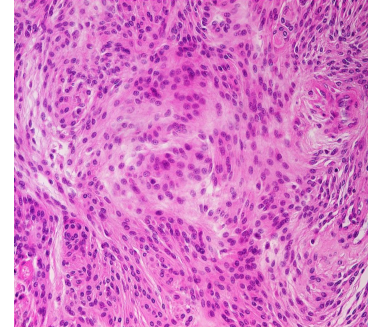
Embryonal cells

Meningioma

Gangliocytoma

Glioma

Blastoma



Arachnoid

Neuron = ganglion cell

Glial cells

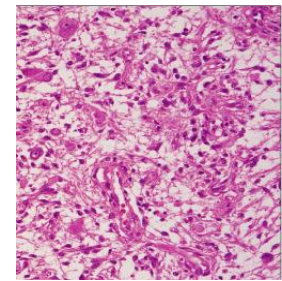
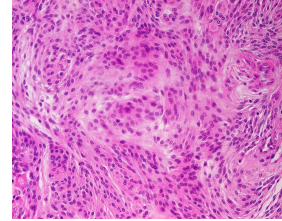
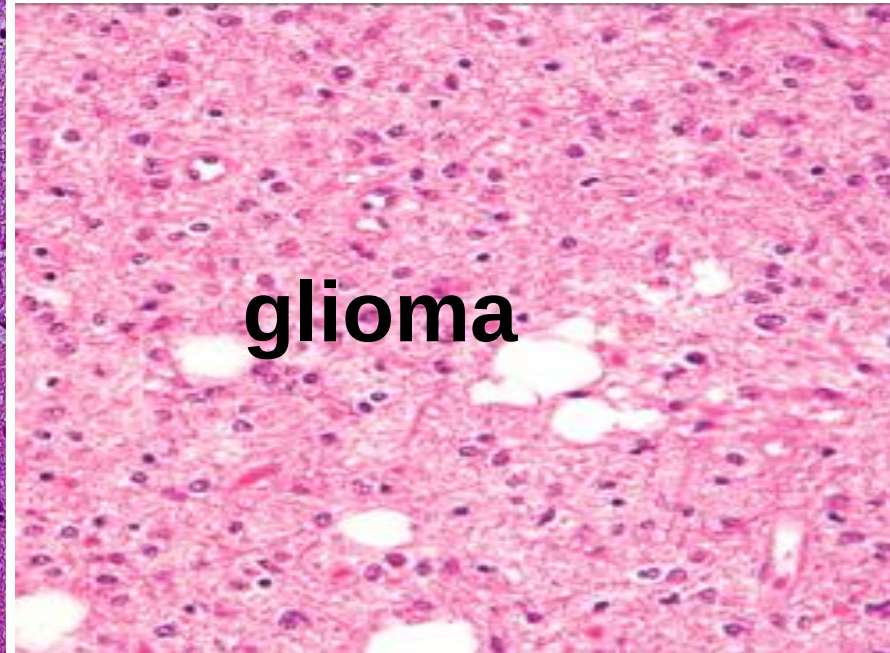
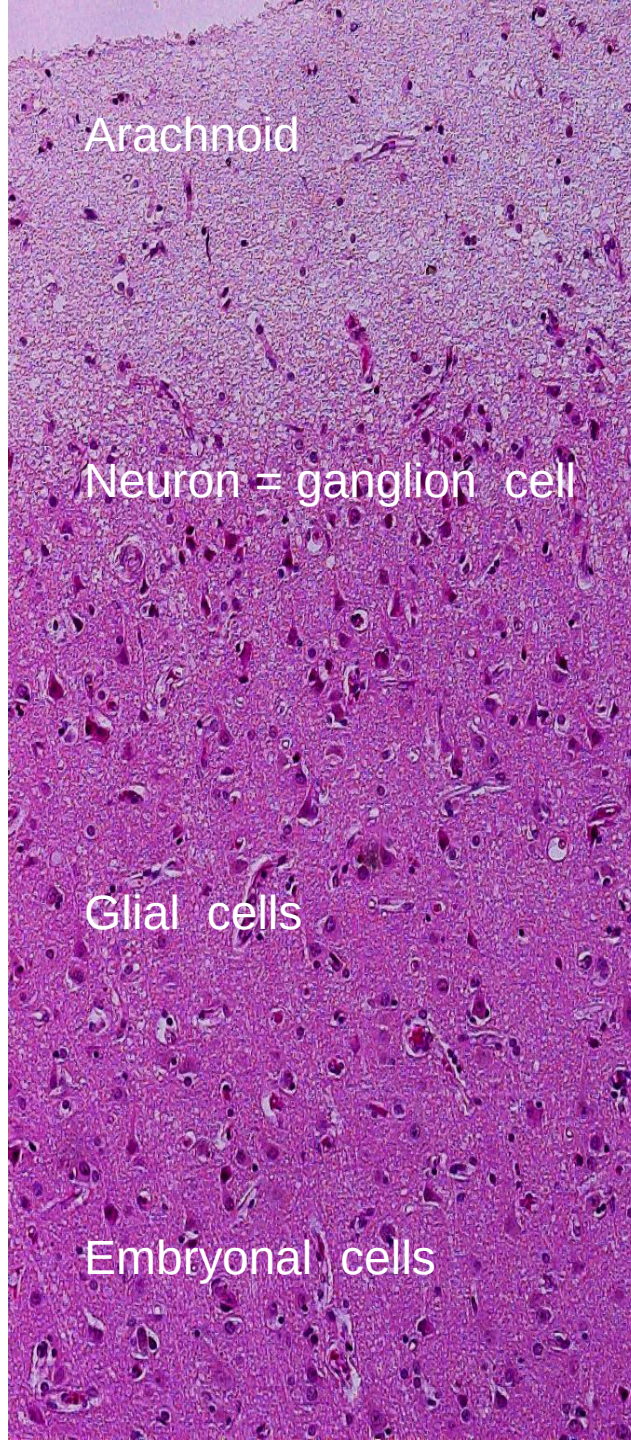
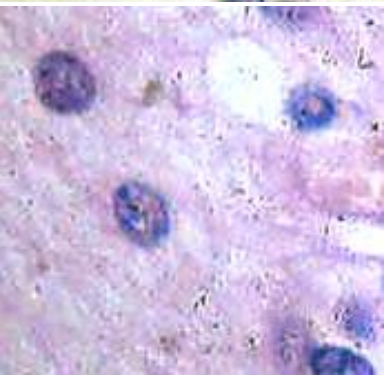
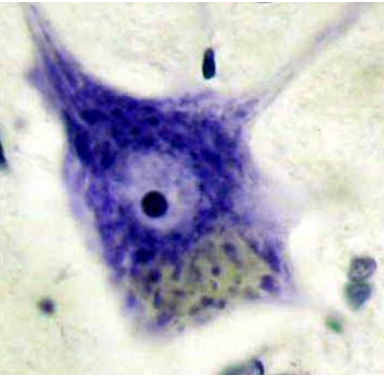
Embryonal cells

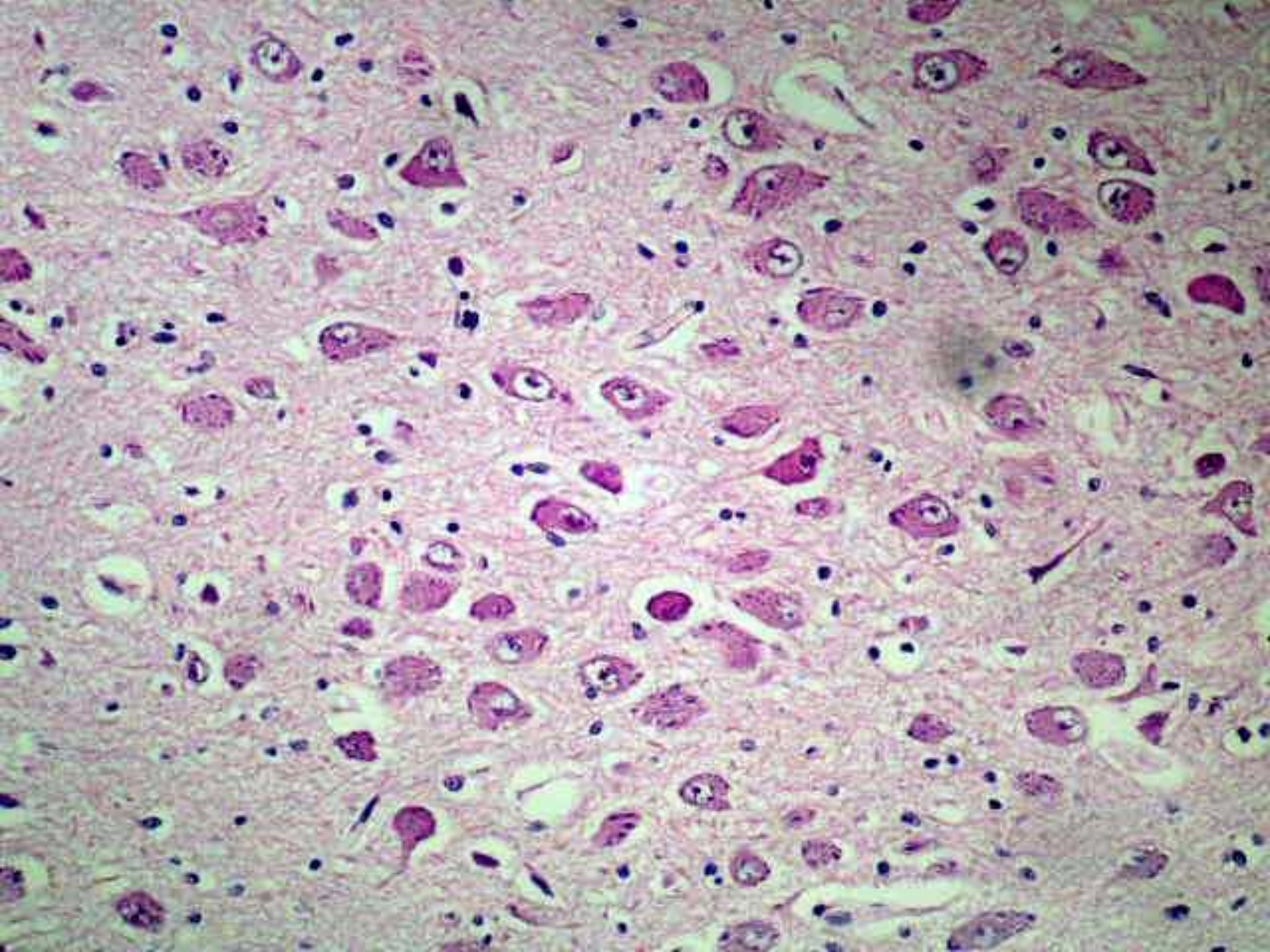
Meningioma

Gangliocytoma

glioma

Blastoma

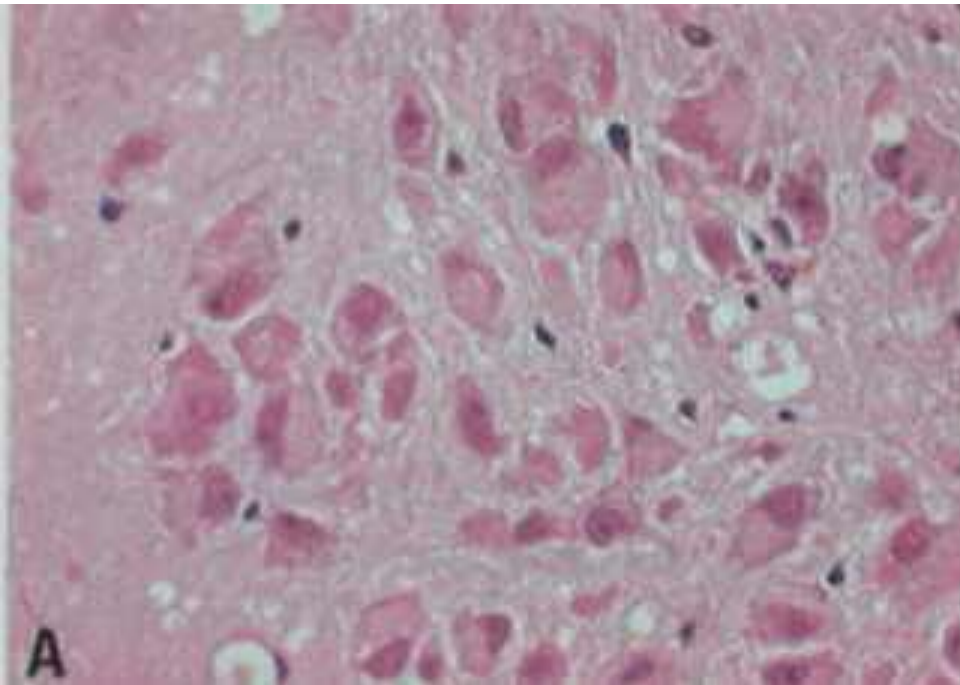




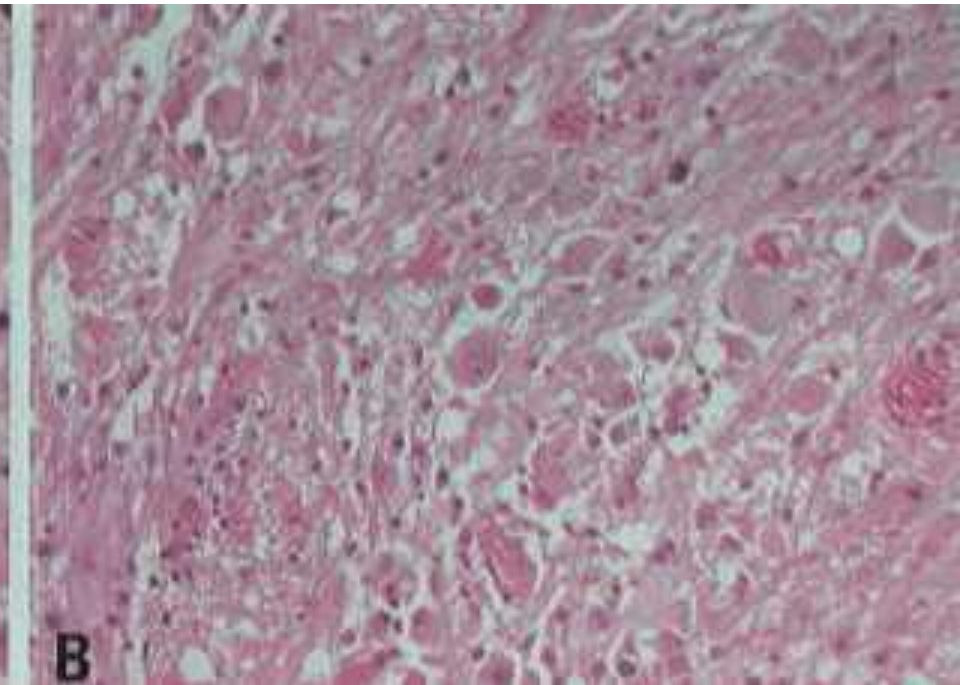
Neurons = Ganglion cells



Gangliocytoma



Gangliocytoma



Ganglioglioma

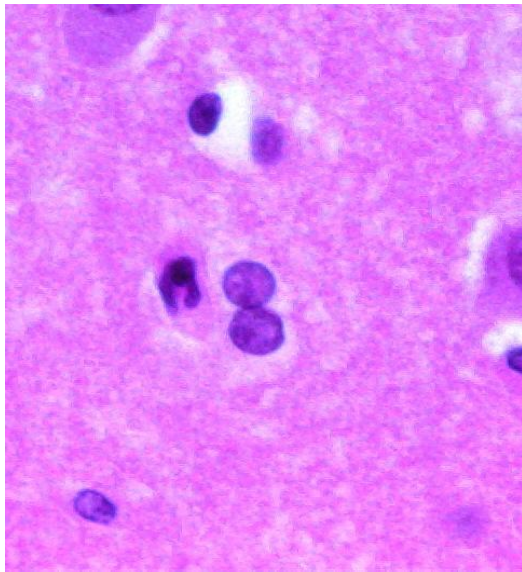
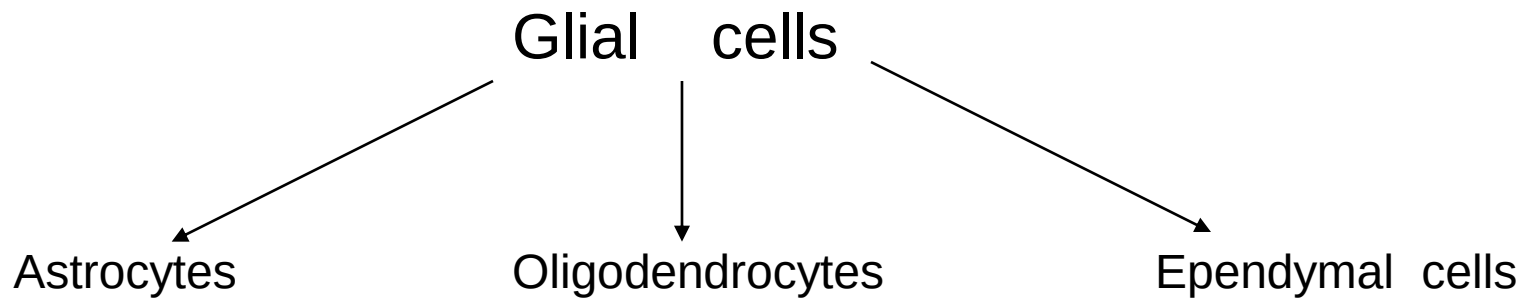


Gangliocytoma

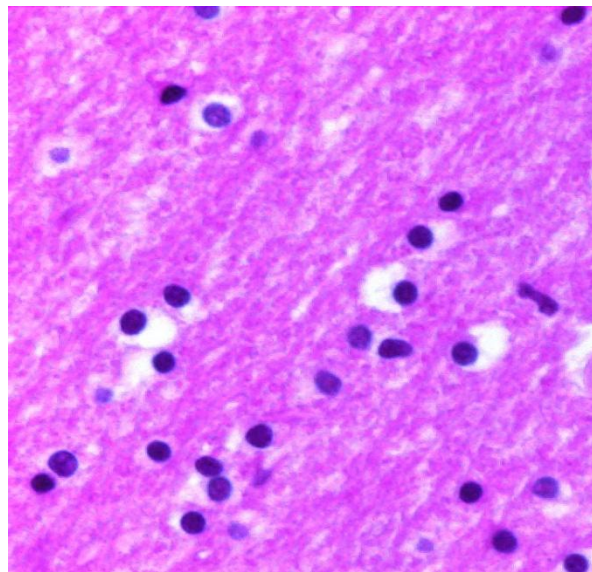
Ganglioglioma

rare: 1.3% of all brain tumors

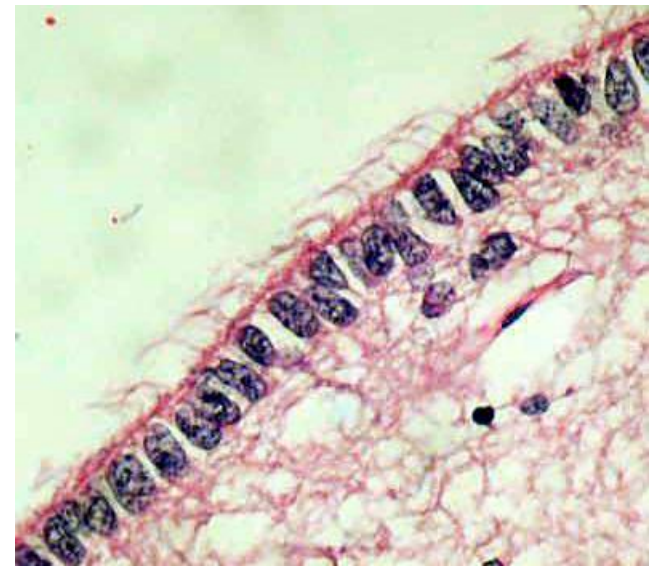
benign



Astrocytoma



Oligodendroglioma



Ependymoma

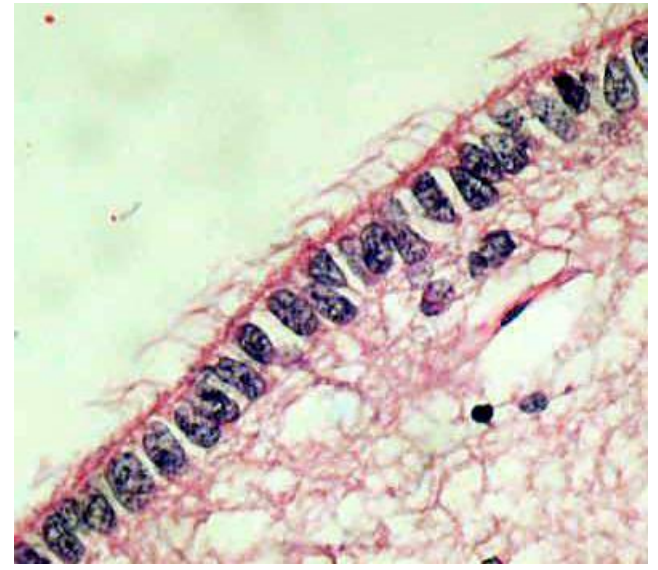
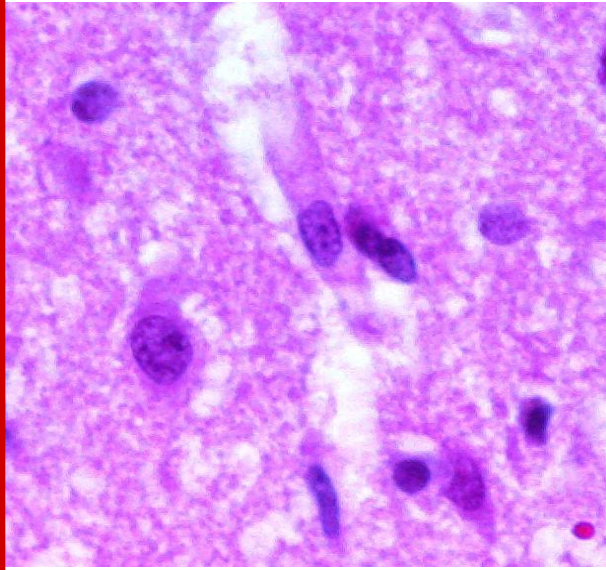
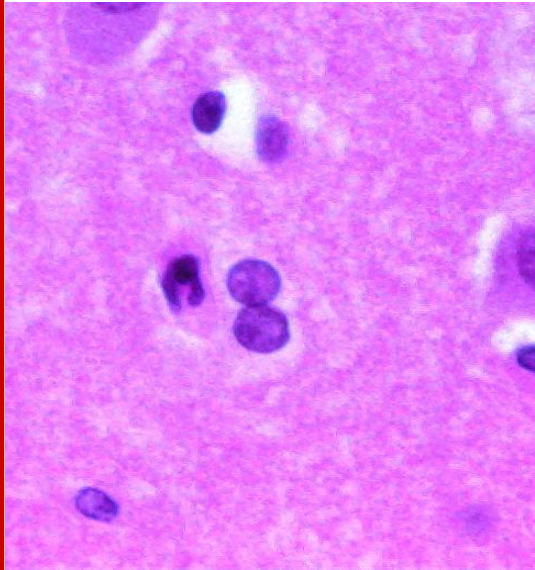
Glioma

Glial cells

Oligodendrocytes

Ependymal cells

Astrocytes

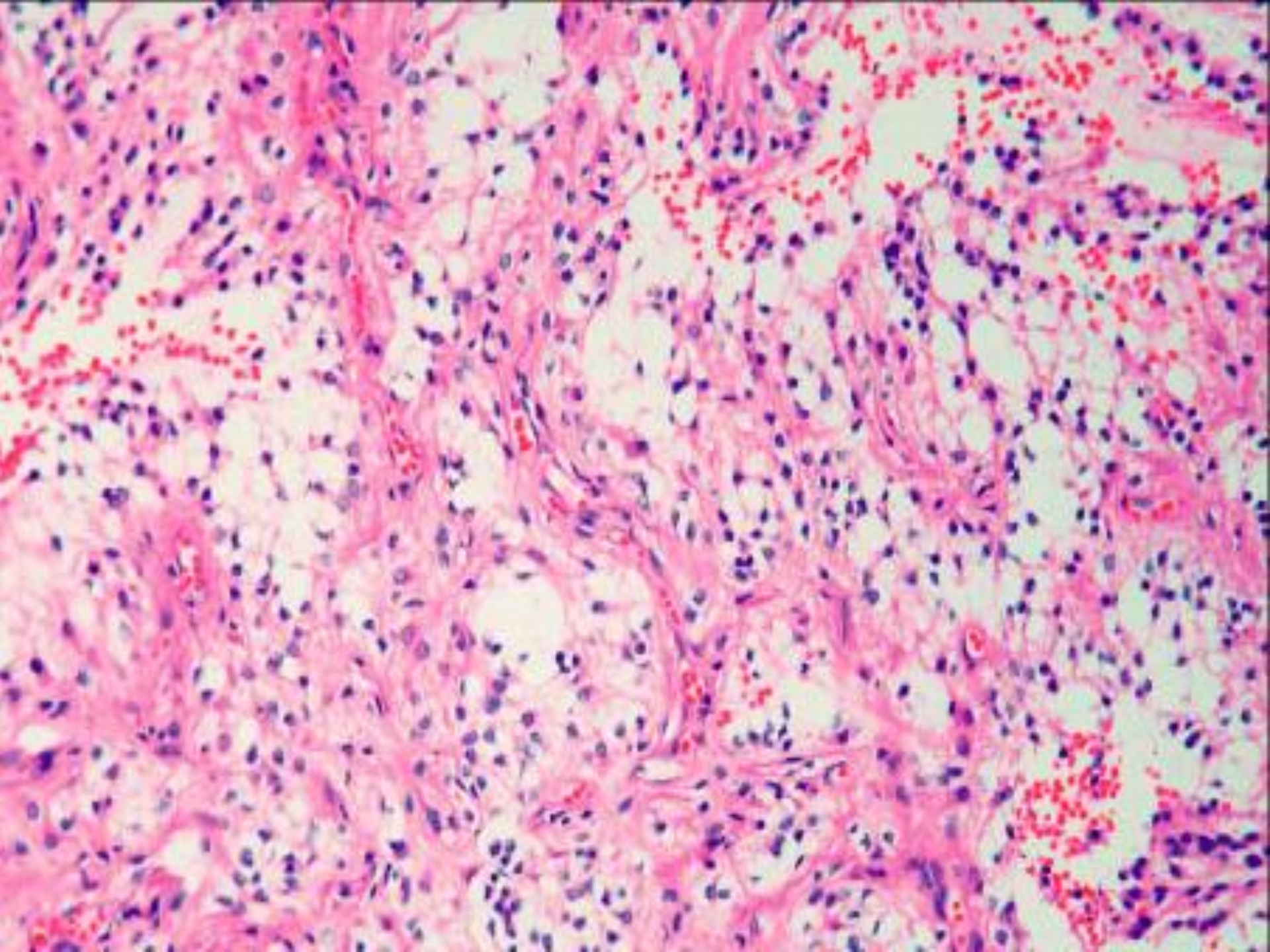


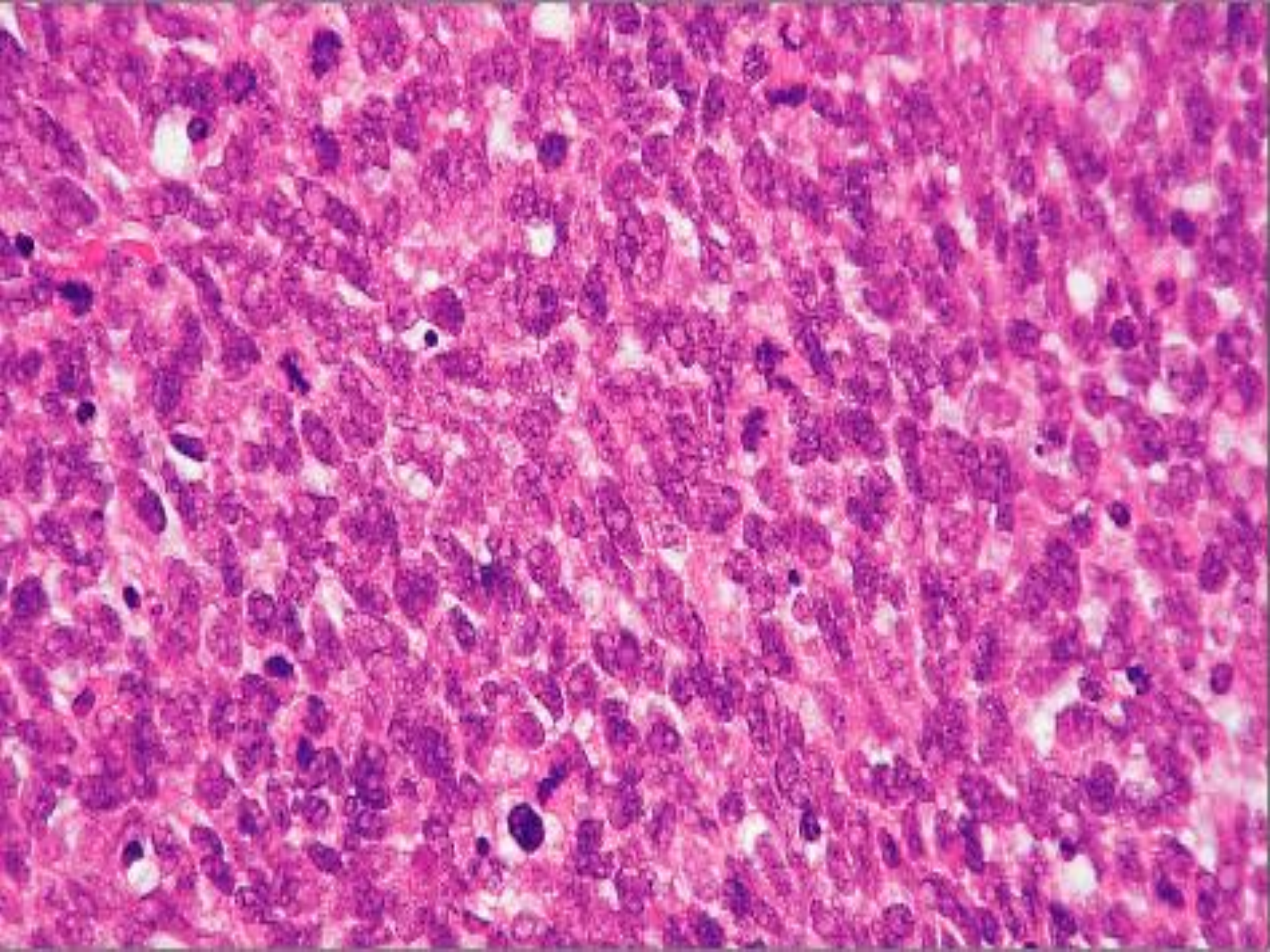
Astrocytoma

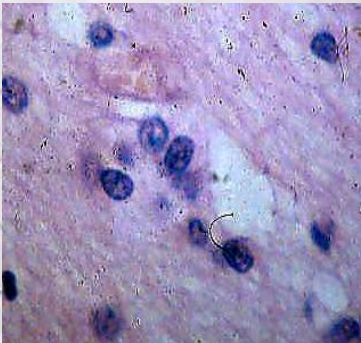
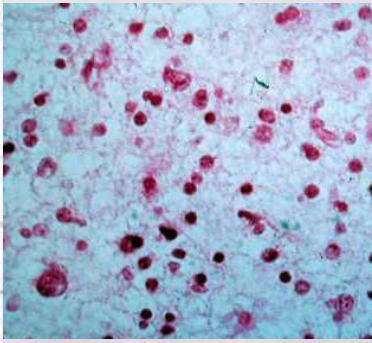
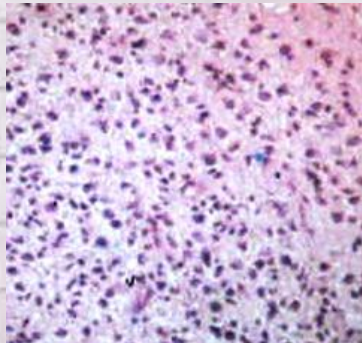
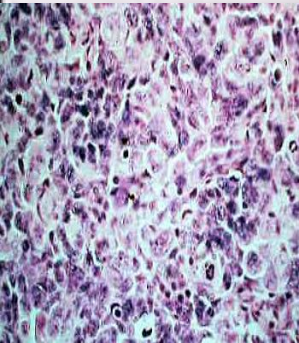
most common, 60%

Oligodendroglioma

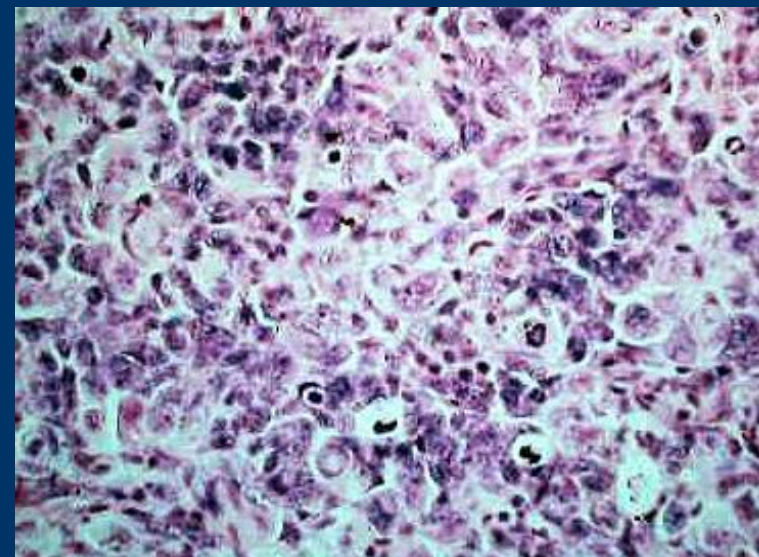
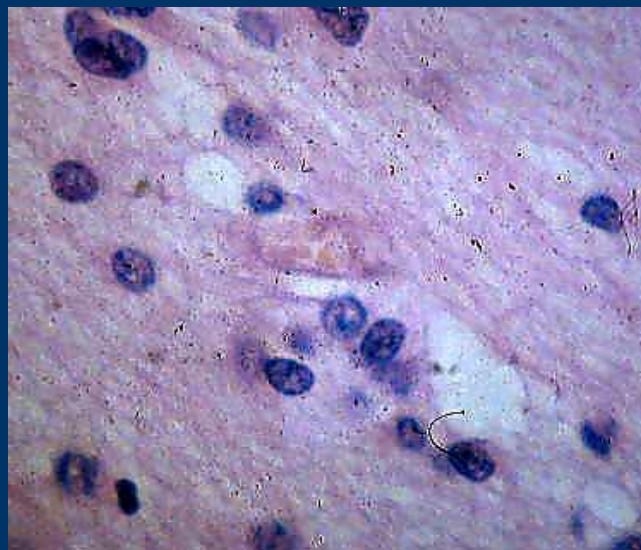
Ependymoma



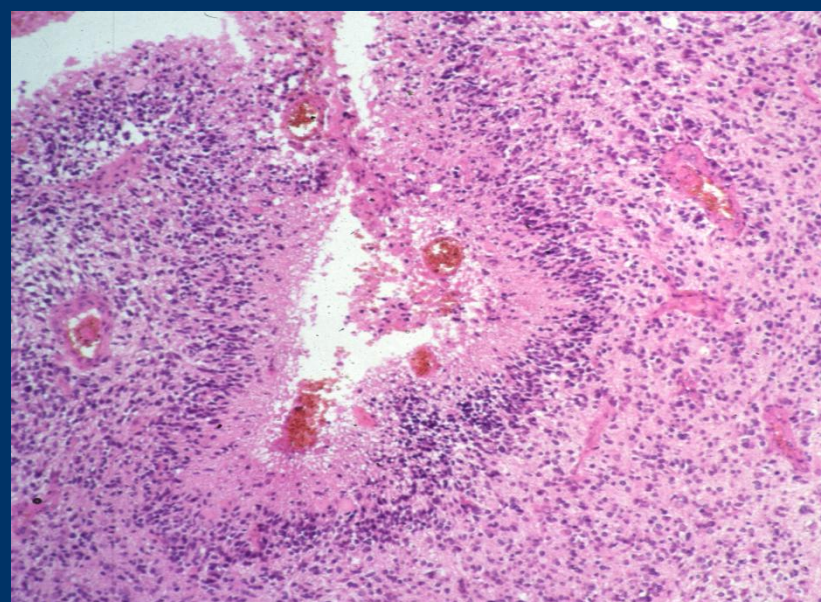
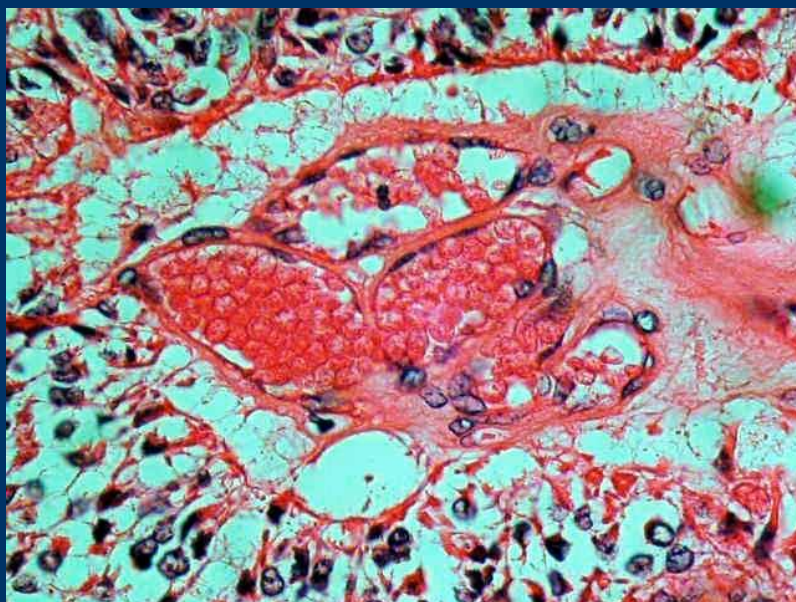


Grade	I	II	III	IV
cellularity				

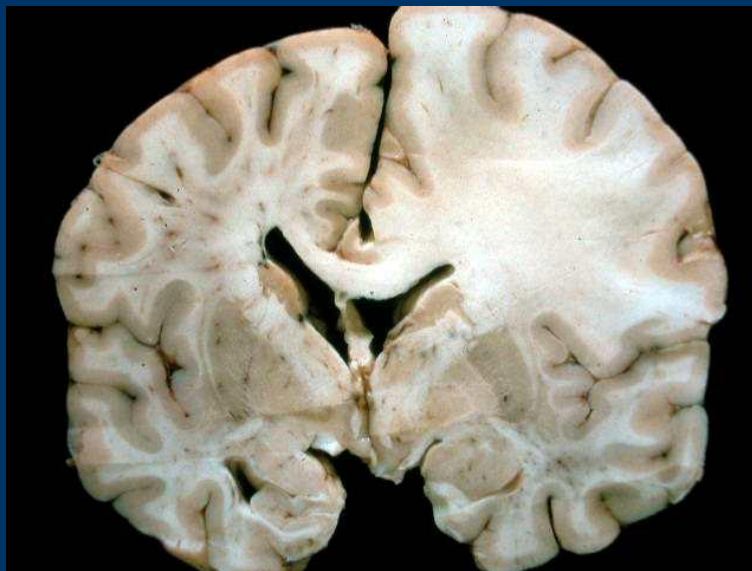
Grade	I	II	III	IV
			+ atypia + mitosis	
cellularity				

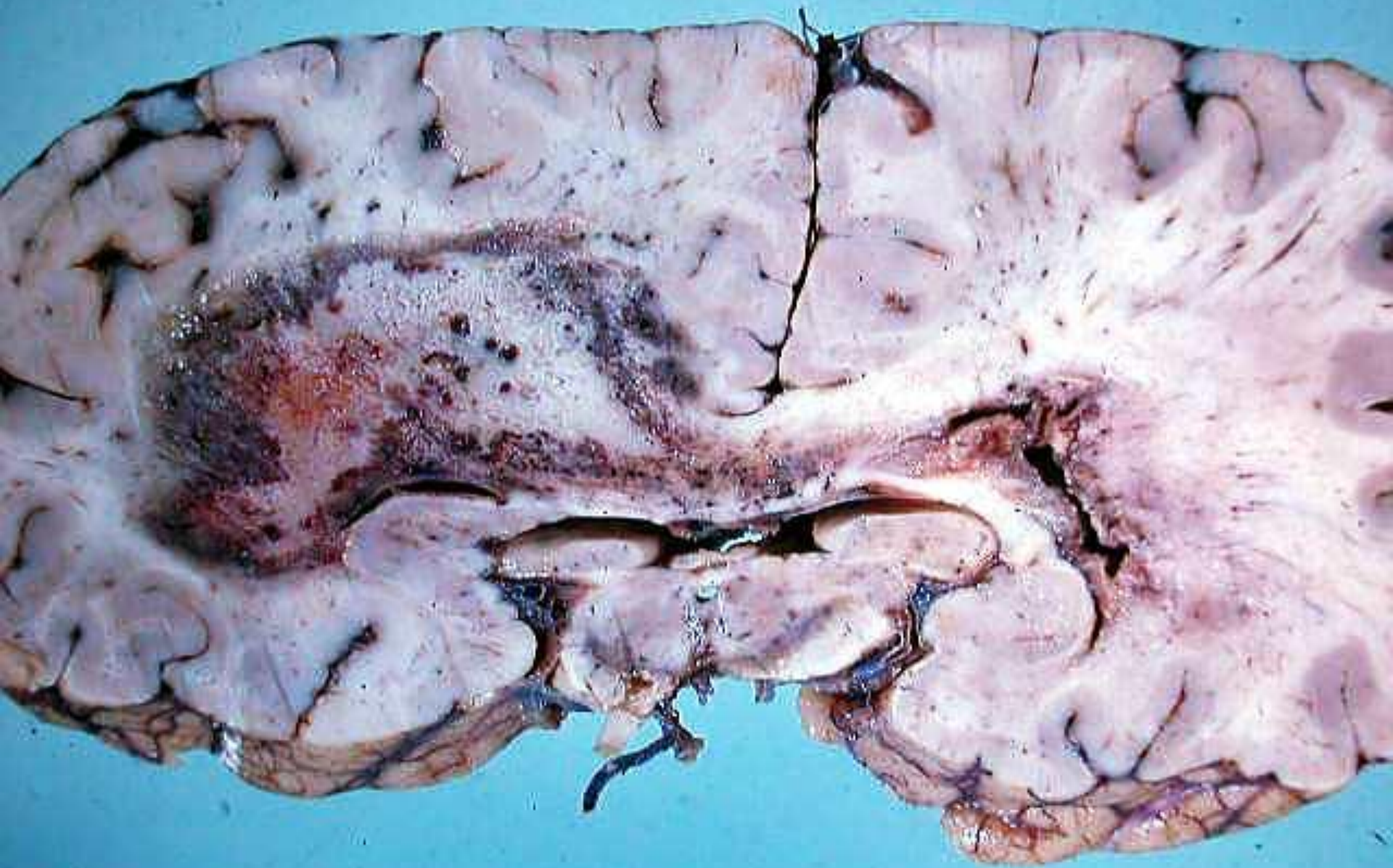


Grade	I	II	III	IV
			+ atypia + mitosis	+ vessels + necrosis
cellularity				



Grade	I	II	III	IV
	Diffuse astrocytoma		Anaplastic astrocytoma	Glioblastoma multiforme
			+ atypia + mitosis	+ vessels + necrosis
cellularity				





Grade	I	II	III	IV
	Diffuse astrocytoma		Anaplastic astrocytoma	Glioblastoma multiforme
			+ atypia + mitosis	+ vessels + necrosis
cellularity				

benign

malignant

Grade	I	II	III	IV
	Diffuse astrocytoma		Anaplastic astrocytoma	Glioblastoma multiforme
survival [years]		5.6	2.7	0.7 + vessels + necrosis
			+ atypia + mitosis	
cellularity				

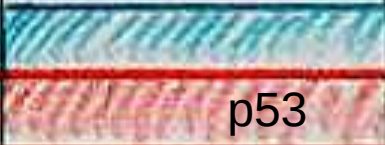
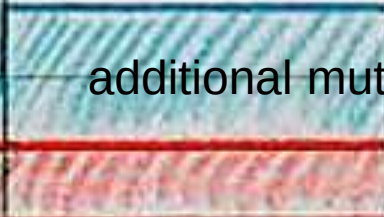
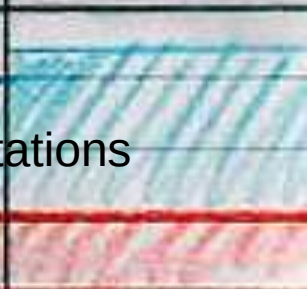
benign

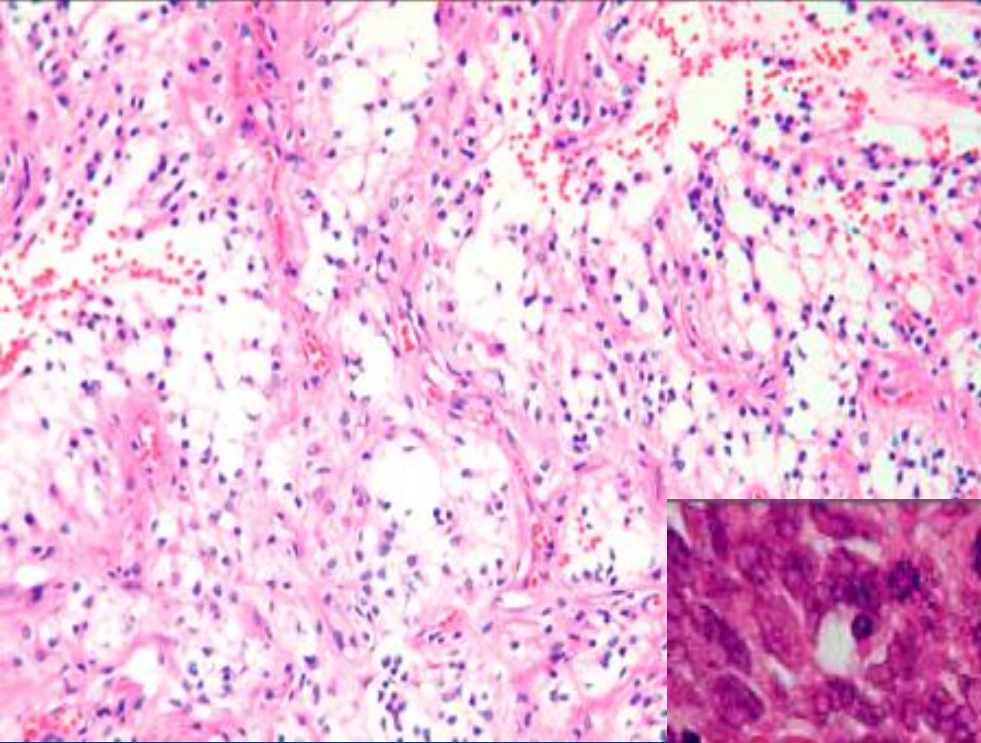
malignant

Grade	I	II	III	IV
	Diffuse	astrocytoma	Anaplastic astrocytoma	Glioblastoma
survival [years]		5.6	2.7	0.7
			+ atypia + mitosis	+ vessels + necrosis
cellularity				

benign

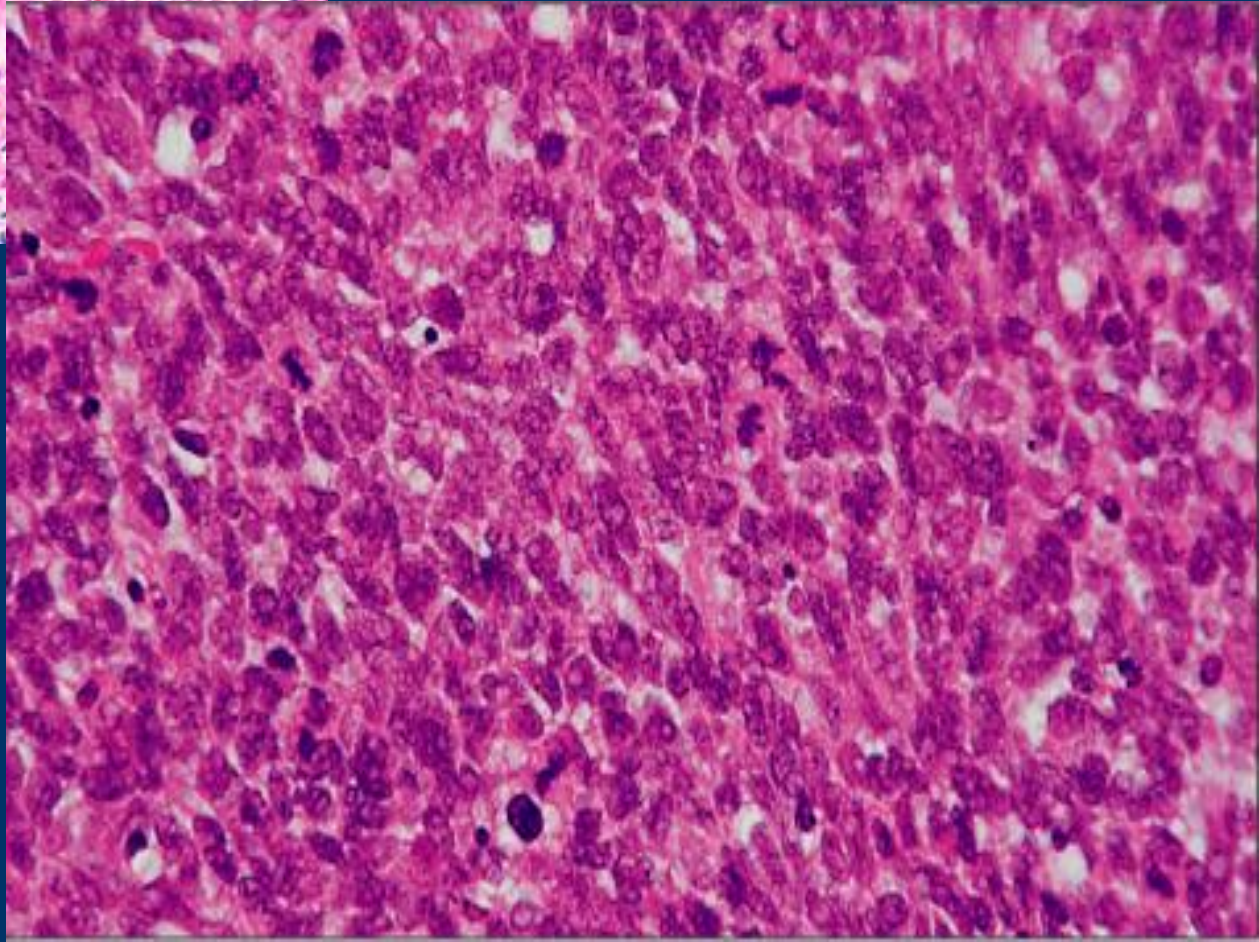
malignant

Grade	I	II	III	IV
			Anaplastic astrocytoma	Glioblastoma
	Diffuse astrocytoma		+ atypia + mitosis	+ vessels + necrosis
cellularity				
molecular		 p53	 additional mutations	



additional mutations

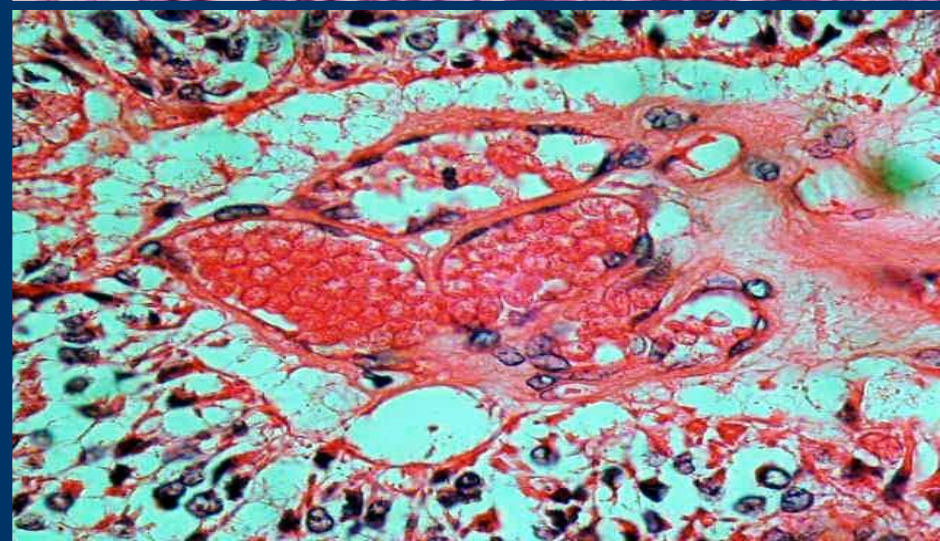
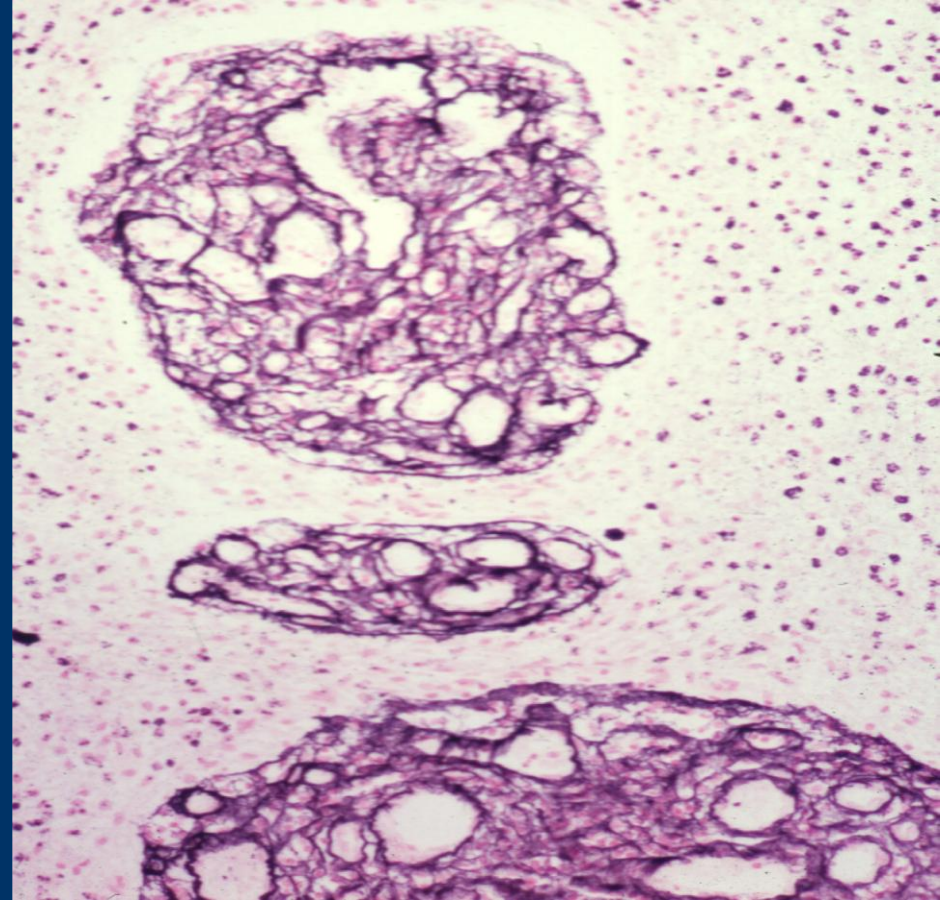
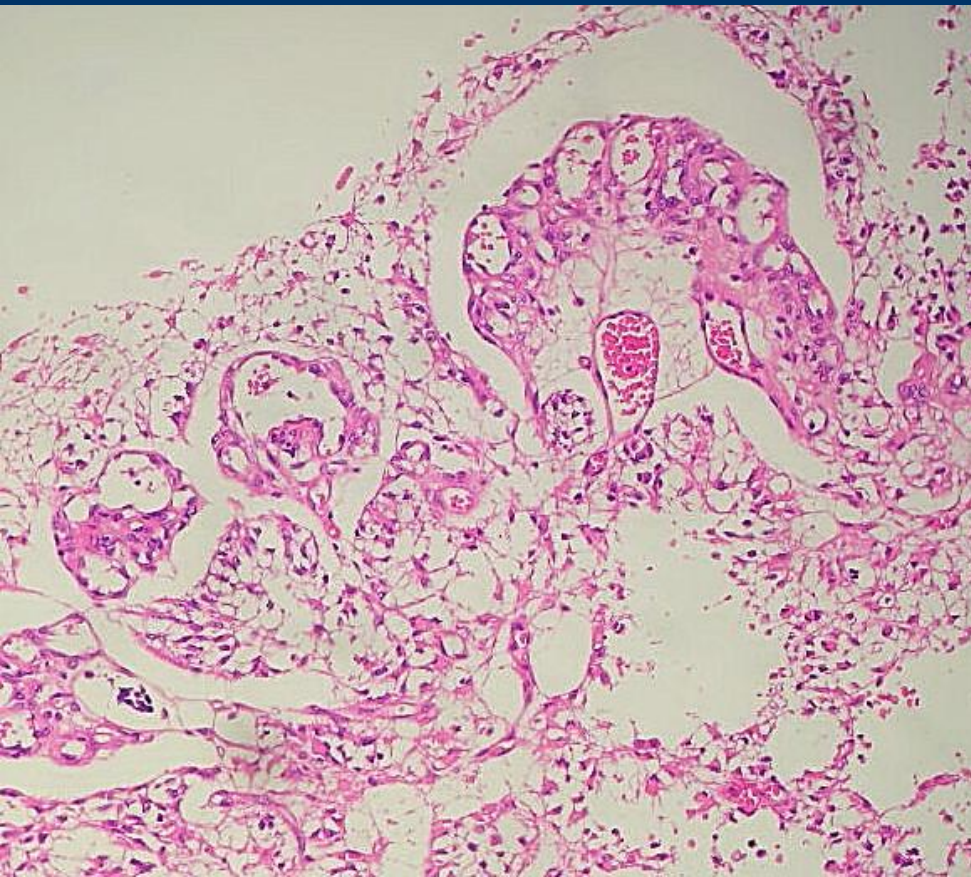
overexpression
of
PDGFR
= platelet-derived
growth factor



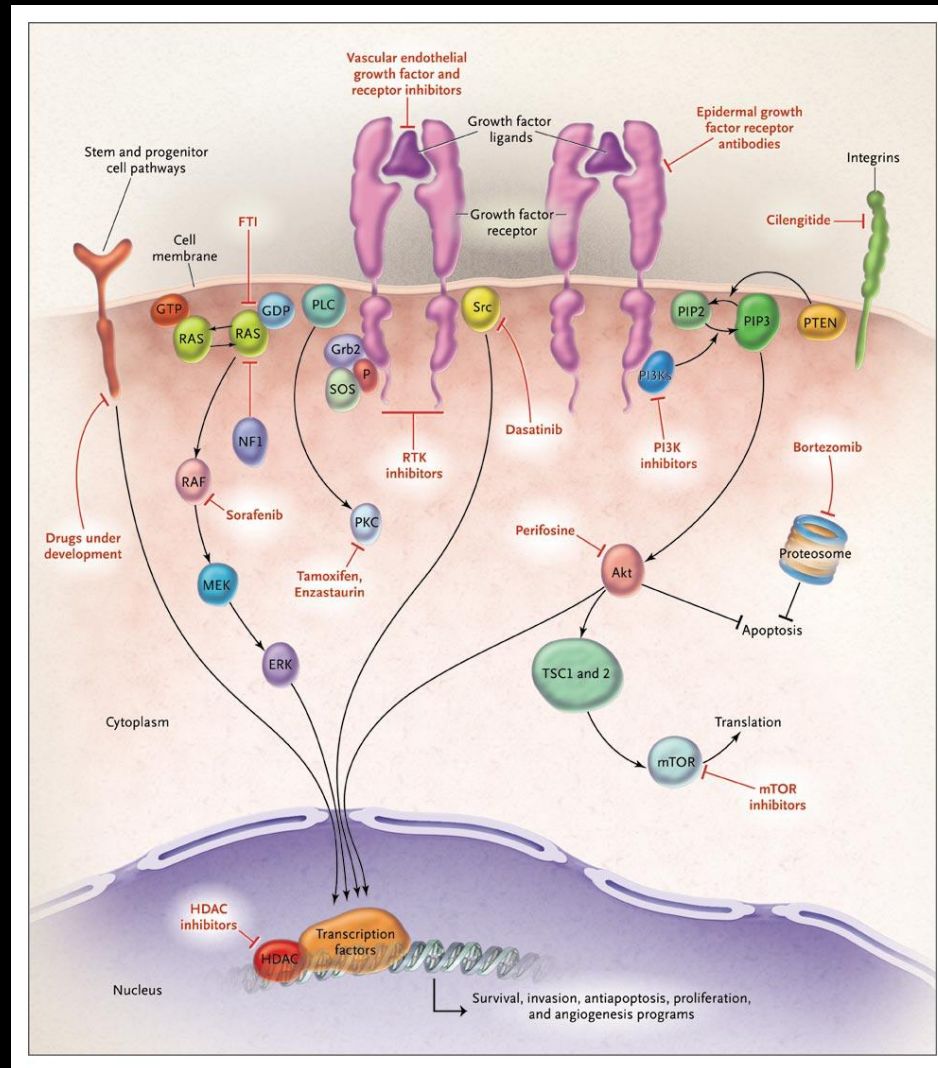
additional mutations

overexpression of
VEGF

= vascular endothelium
growth factor



Major Signaling Pathways in Malignant Gliomas and the Corresponding Targeted Agents in Development for Glioblastoma



Wen P and Kesari S. N Engl J Med 2008;359:492-507



The NEW ENGLAND
JOURNAL of MEDICINE

Grade	I	II	III	IV
			Anaplastic astrocytoma	Glioblastoma
	Diffuse astrocytoma		+ atypia + mitosis	+ vessels + necrosis
cellularity				
			additional mutations	
molecular		p53		



Astrocytes or precursor / stem cells



WHO grade II

Low-grade astrocytoma

TP53 mutation (59%)

5.1 years



WHO grade III

Anaplastic astrocytoma

TP53 mutation (53%)

1.9 years



WHO grade IV

Secondary glioblastoma

LOH 10q (63%)

EGFR Amplification (8%)

p16^{INK4a} Deletion (19%)

TP53 Mutation (65%)

PTEN Mutation (4%)

5% of cases

WHO grade II

WHO grade III

WHO grade IV

Astrocytes or precursor / stem cells

Low-grade astrocytoma

TP53 mutation (59%)

5.1 years

Anaplastic astrocytoma

TP53 mutation (53%)

1.9 years

Secondary glioblastoma

LOH 10q (63%)

EGFR Amplification (8%)

p16^{INK4a} Deletion (19%)

TP53 Mutation (65%)

PTEN Mutation (4%)

5% of cases

younger

> 50 years

Clinical history:

<3 months 68%

<6 months 84%

**Primary glioblastoma
*de novo***

LOH 10q (70%)

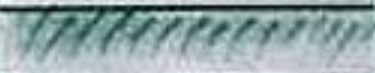
EGFR Amplification (36%)

p16^{INK4a} Deletion (31%)

TP53 Mutation (28%)

PTEN Mutation (25%)

95% of cases

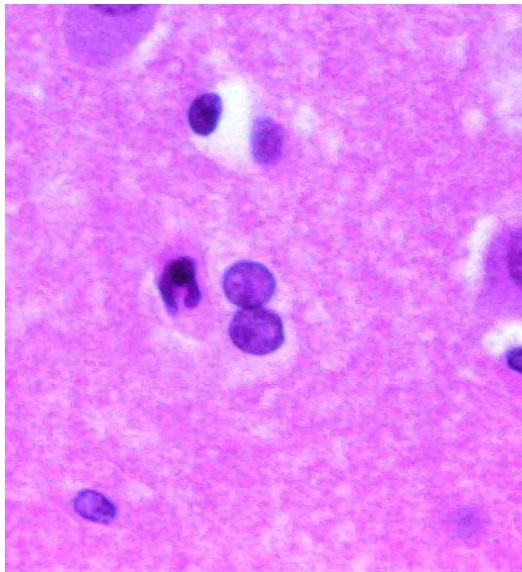
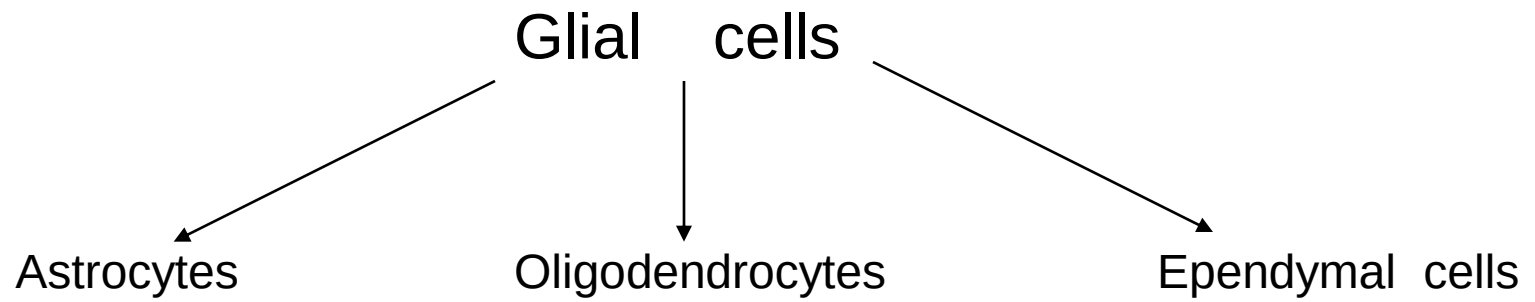
Grade	I	II	III	IV
			Anaplastic astrocytoma	Glioblastoma multiforme
	Diffuse astrocytoma		+ atypia + mitosis	+ vessels + necrosis
cellularity				

benign

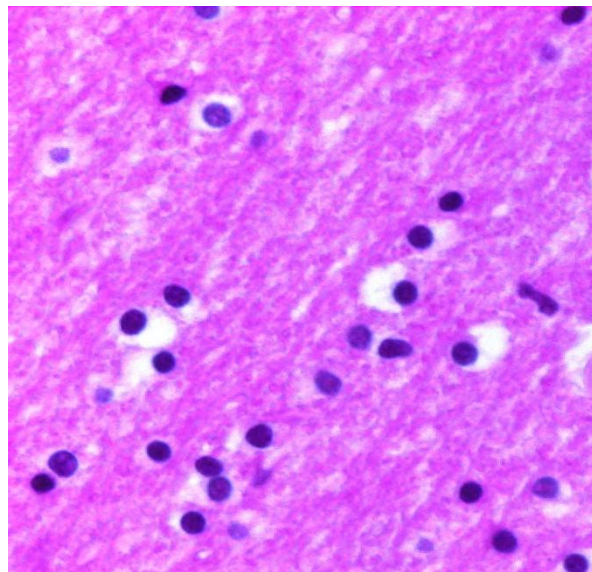
malignant

WHO grading (1979) = „malignancy scale“

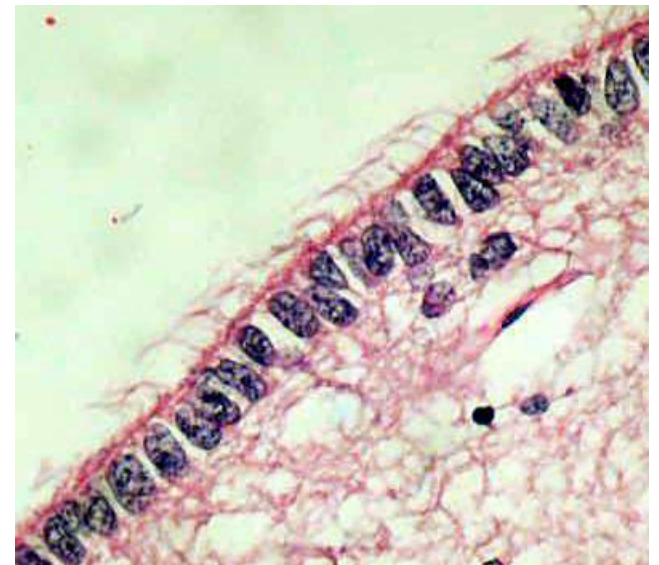
Grade	I	II	III	IV
Survival	cured	> 5 years	2-3 years	<1 year
	benign		malignant	



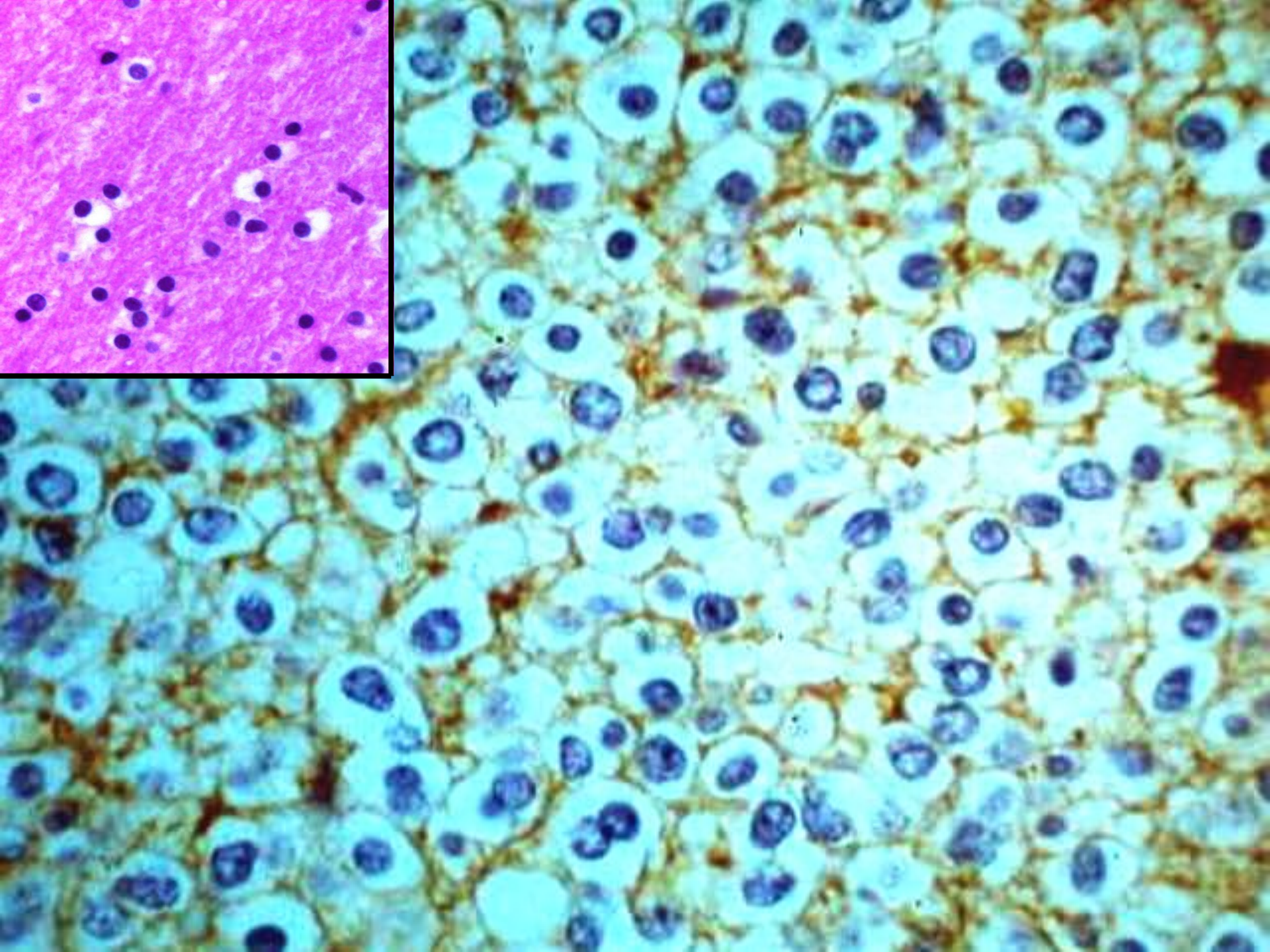
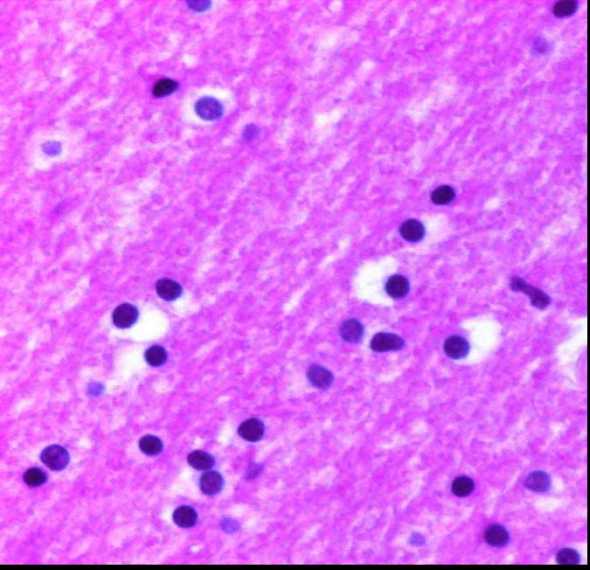
Astrocytoma

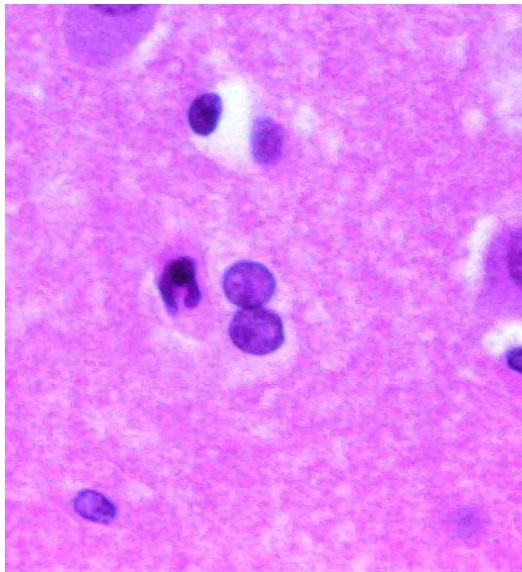
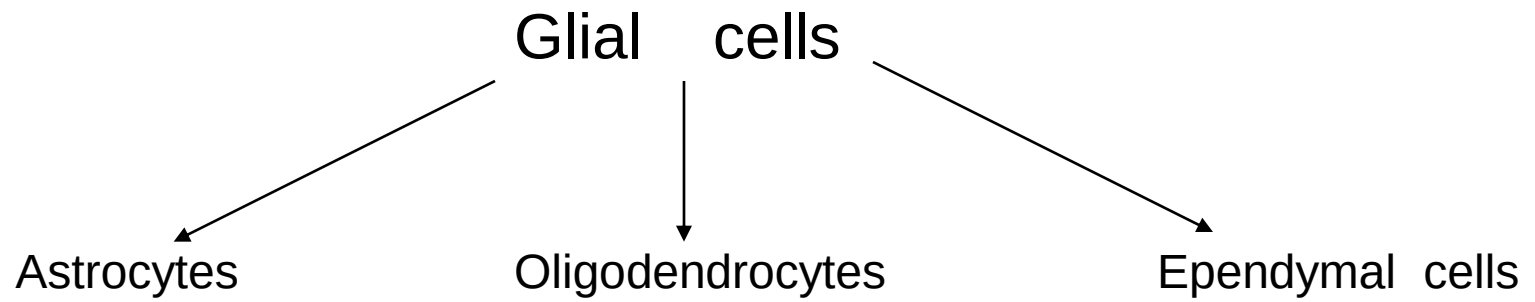


Oligodendrocytoma

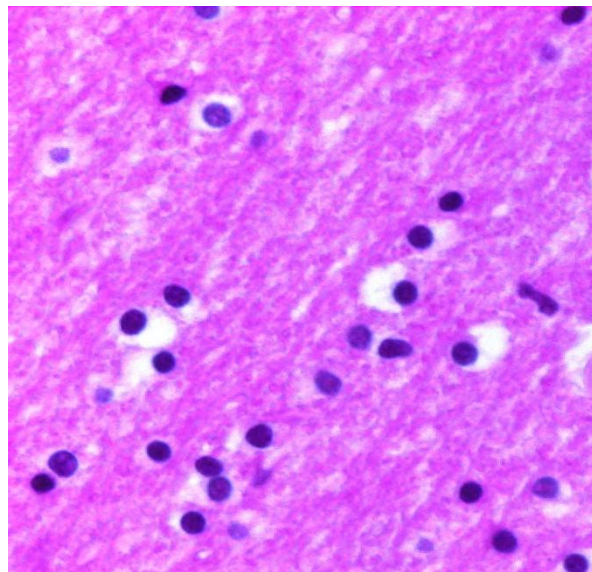


Ependymoma

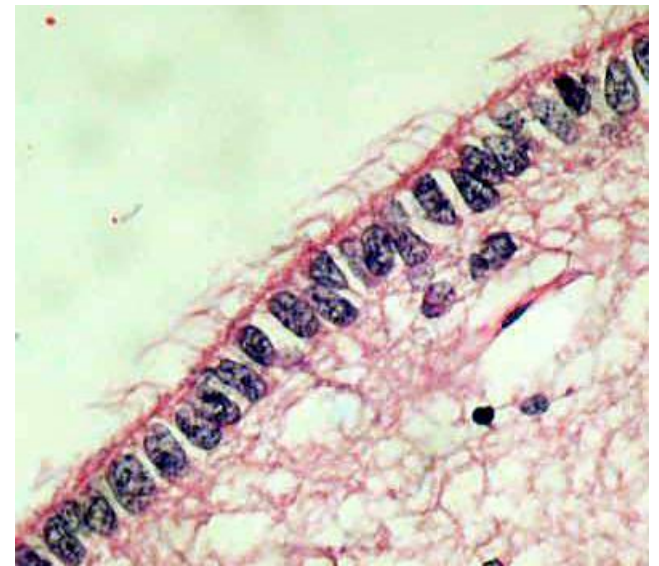




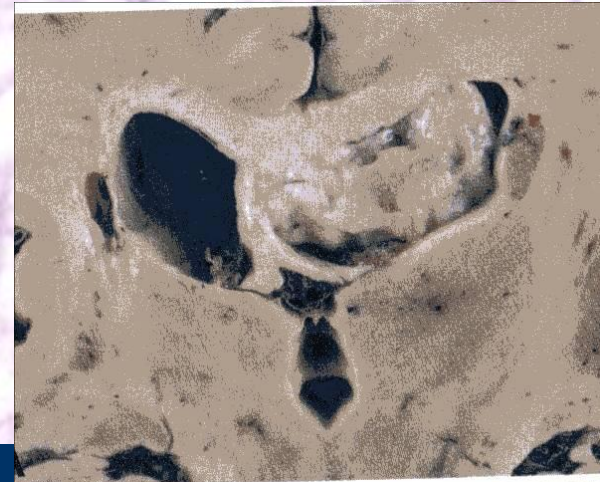
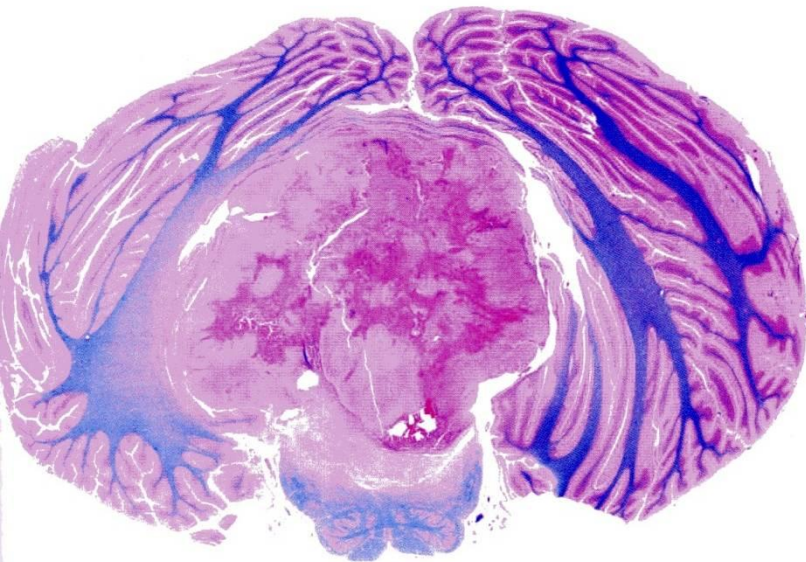
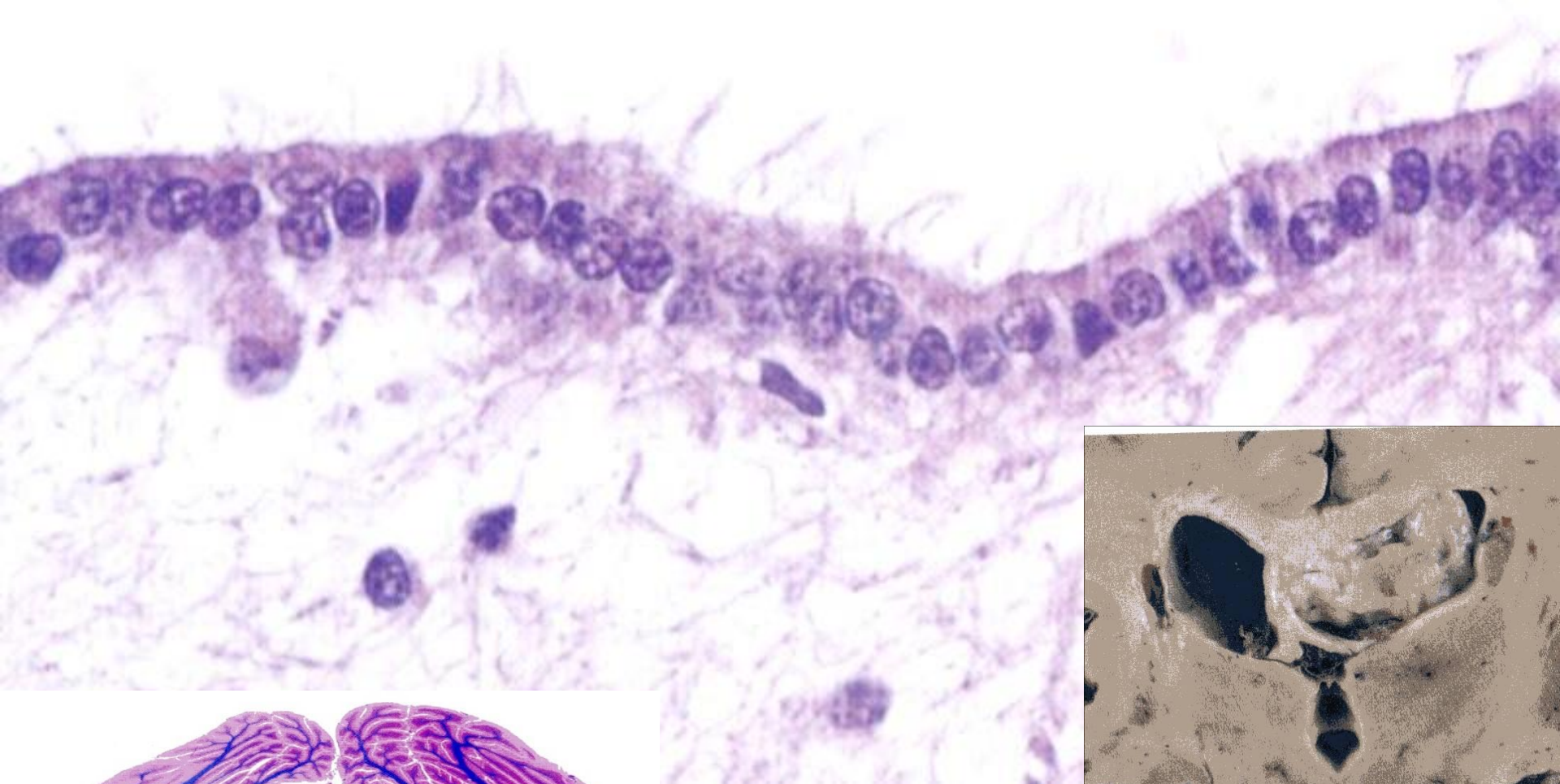
Astrocytoma



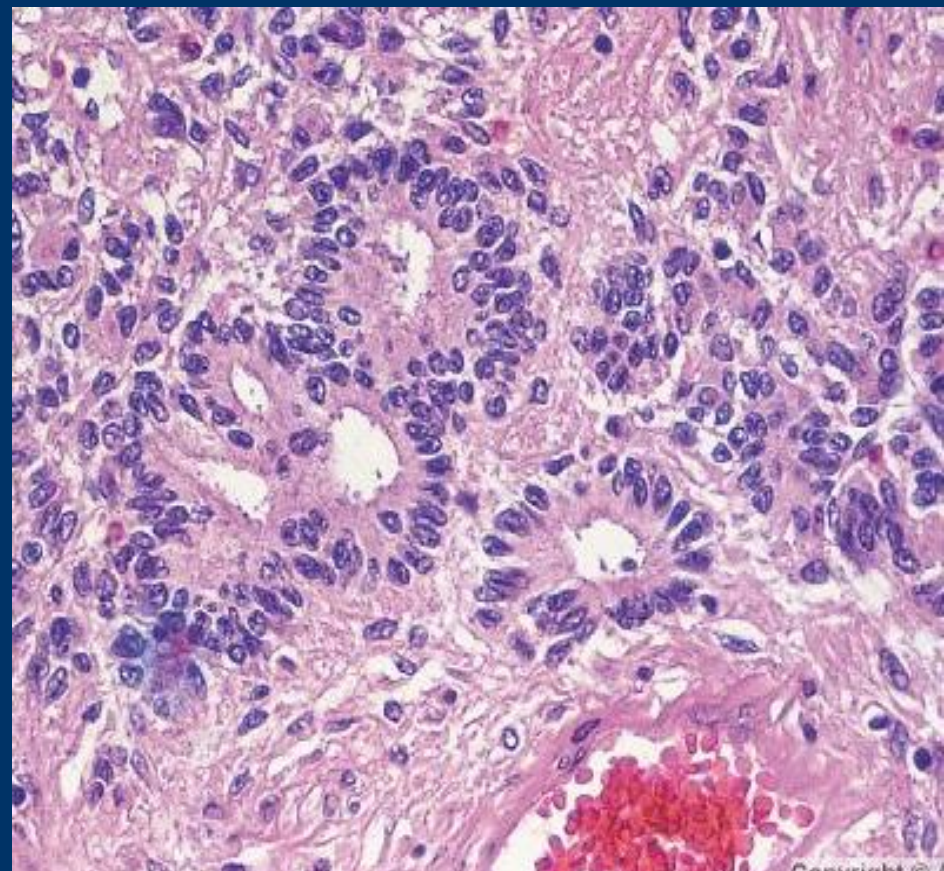
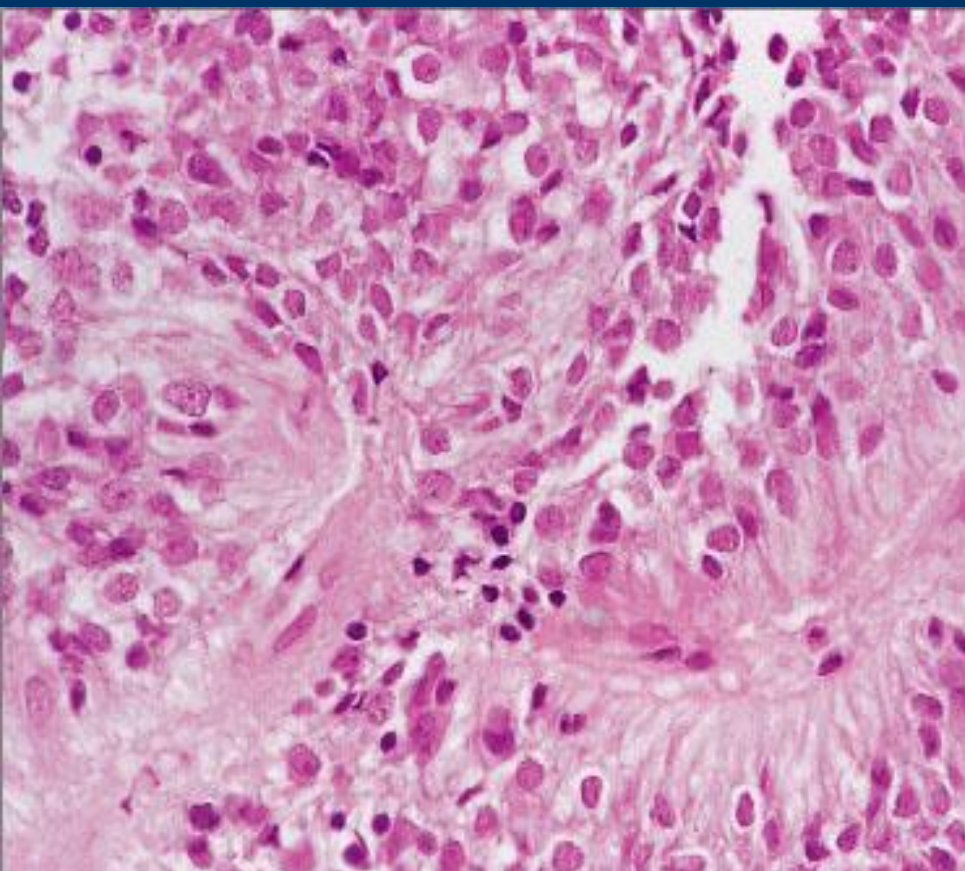
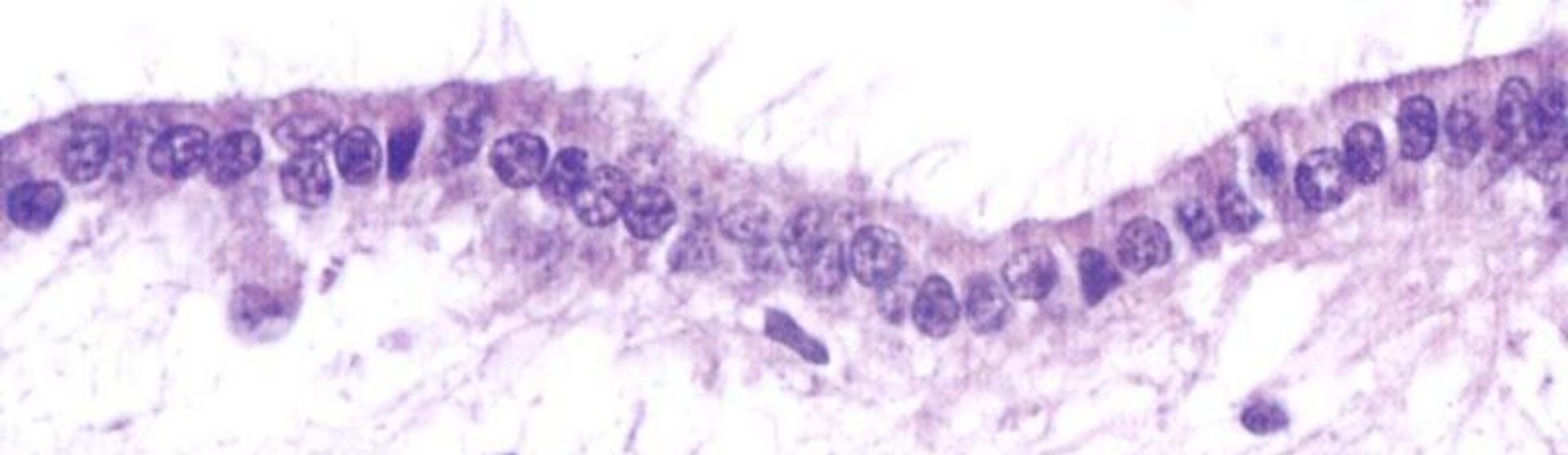
Oligodendroglioma

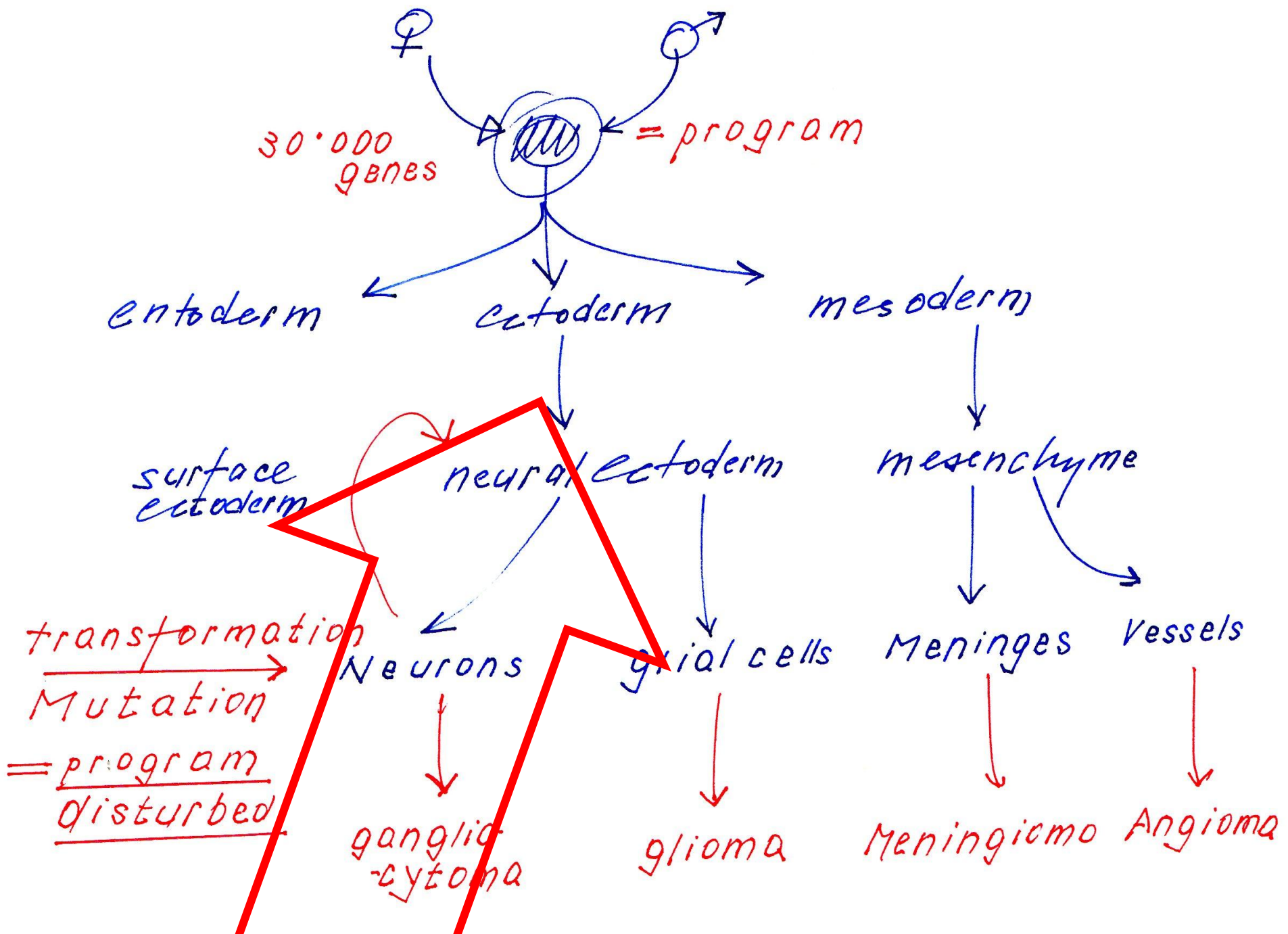


Ependymoma

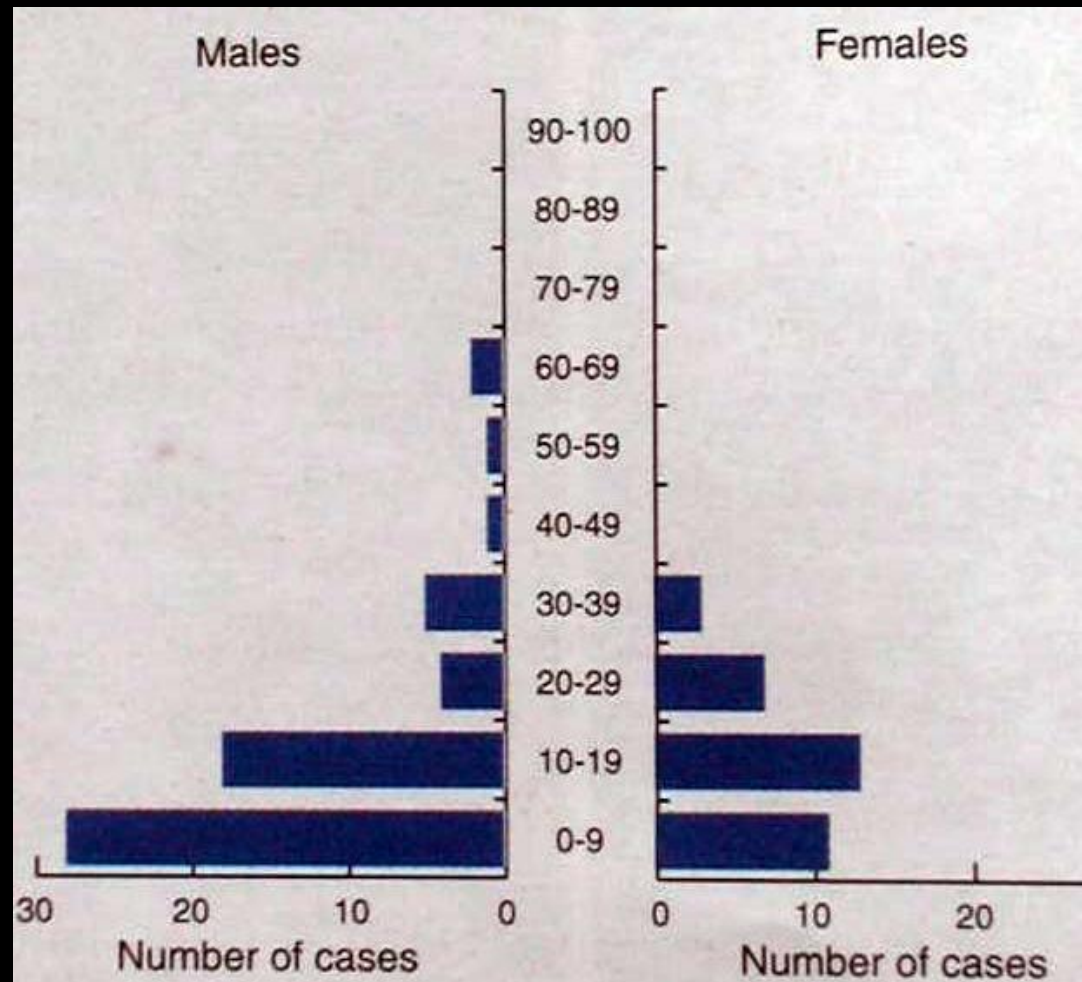
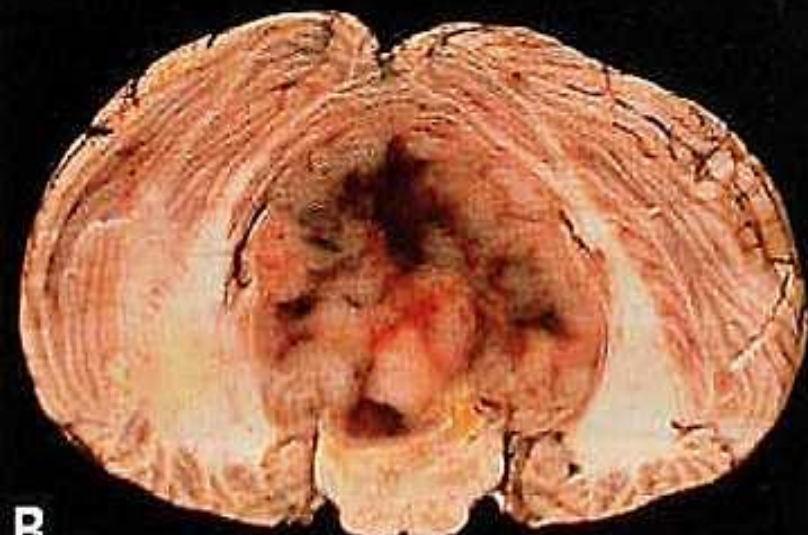
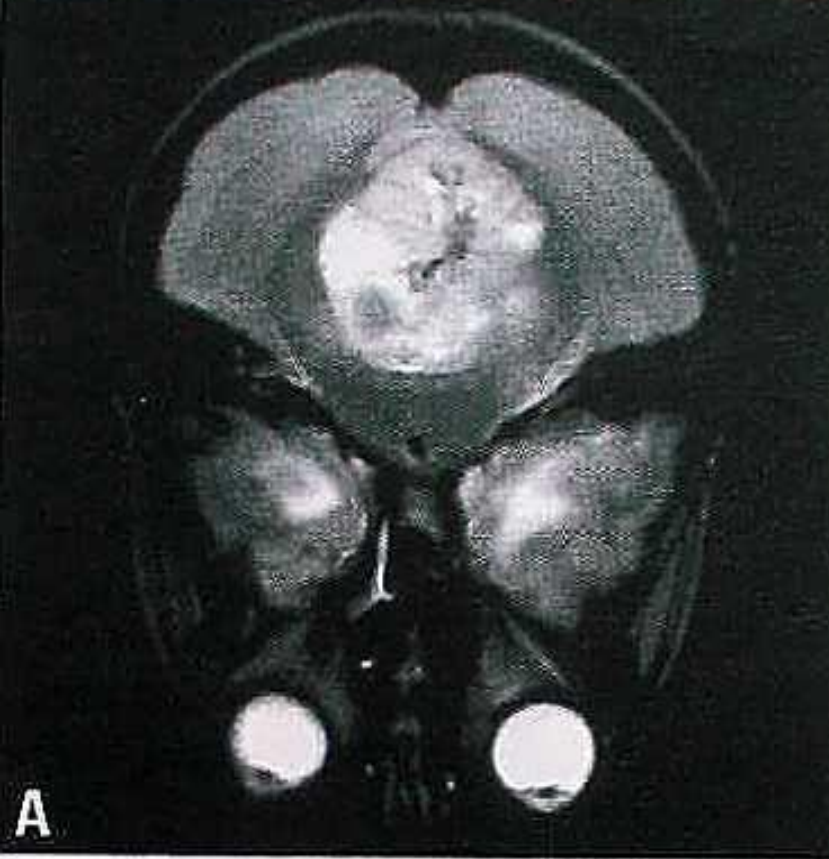


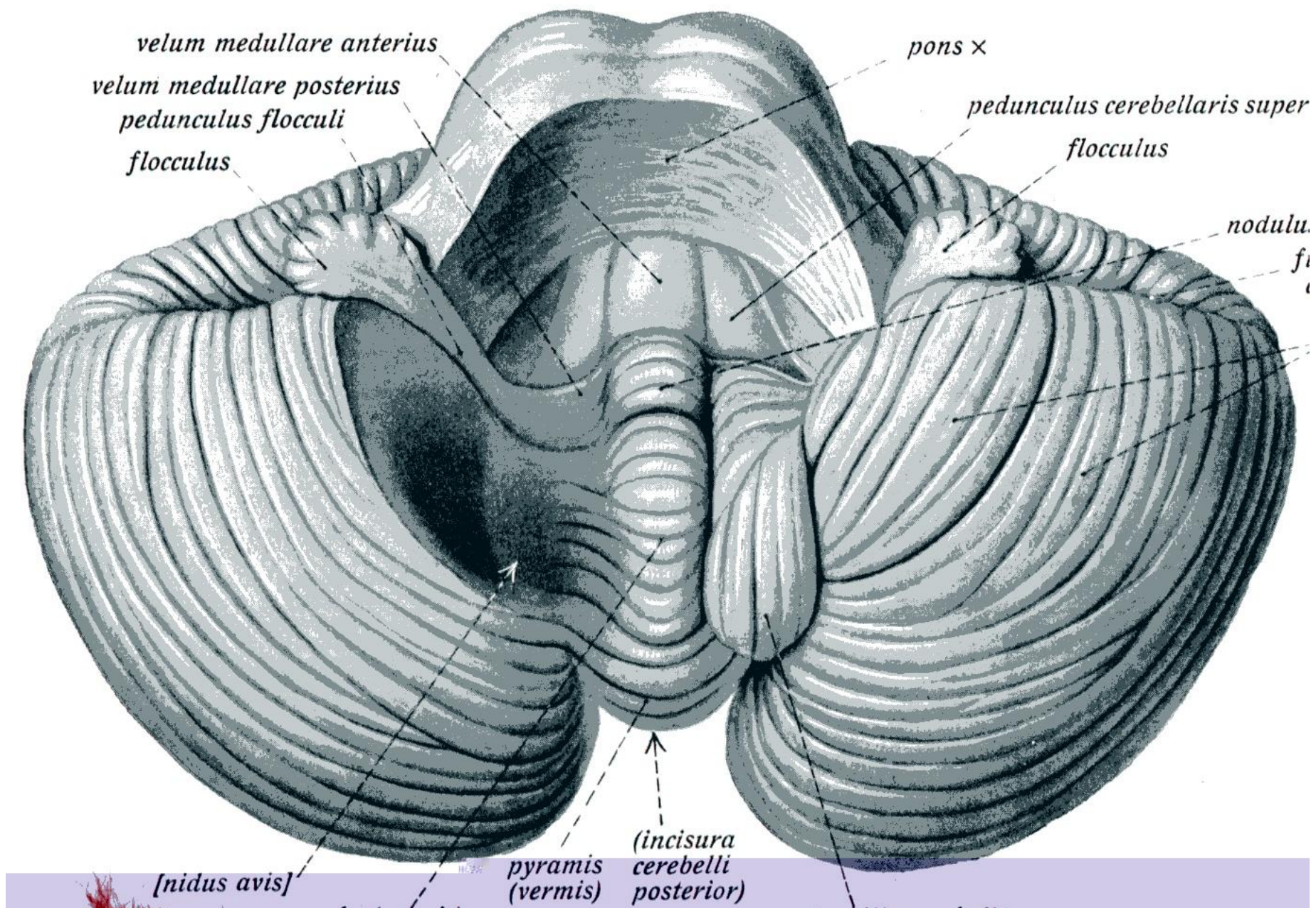
Ependymoma





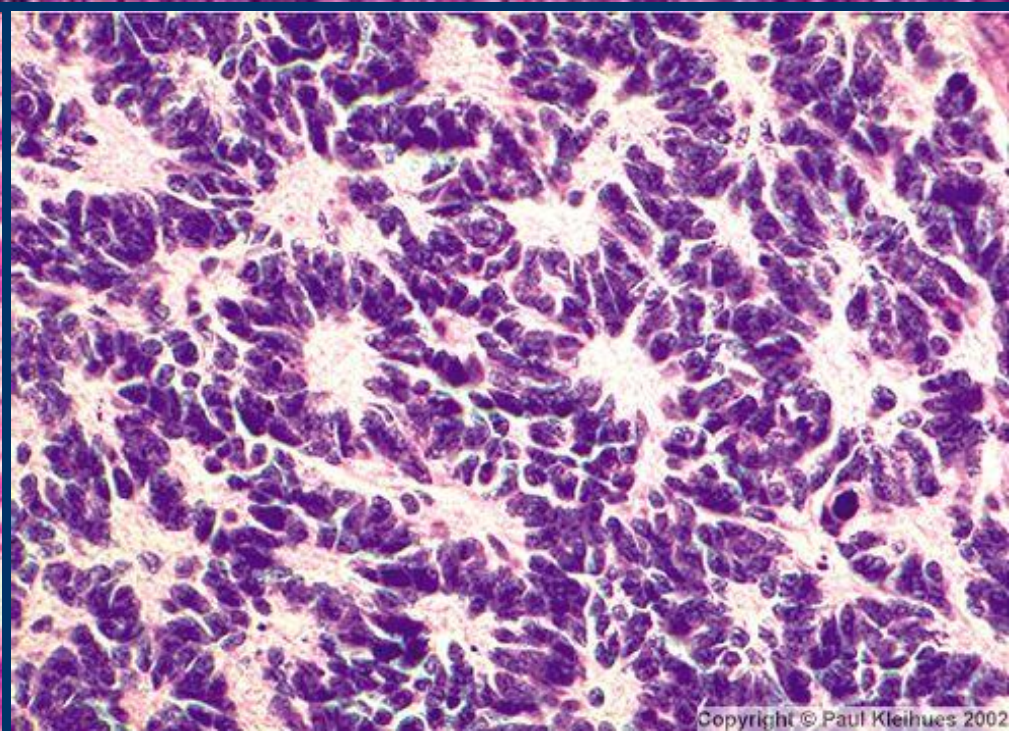
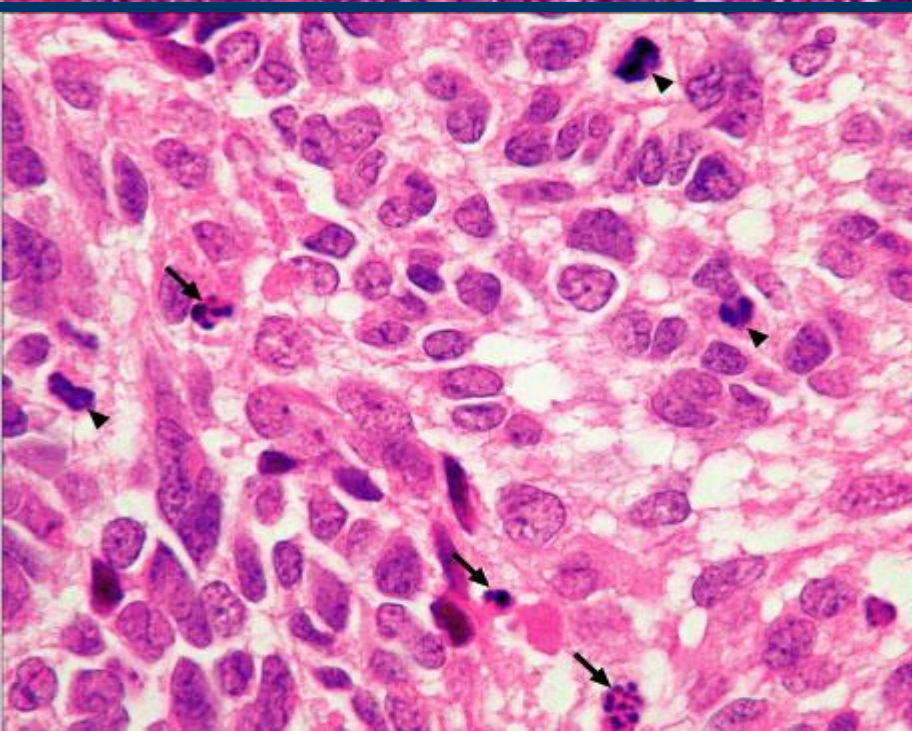
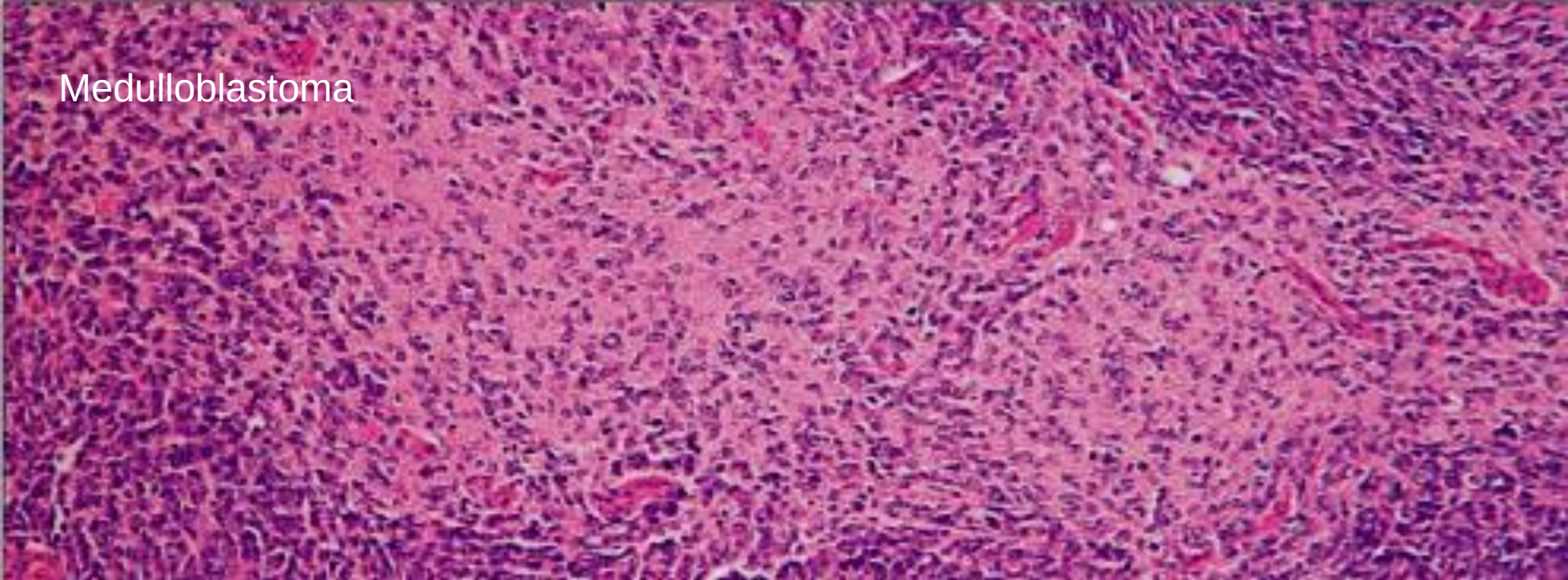
Medulloblastoma







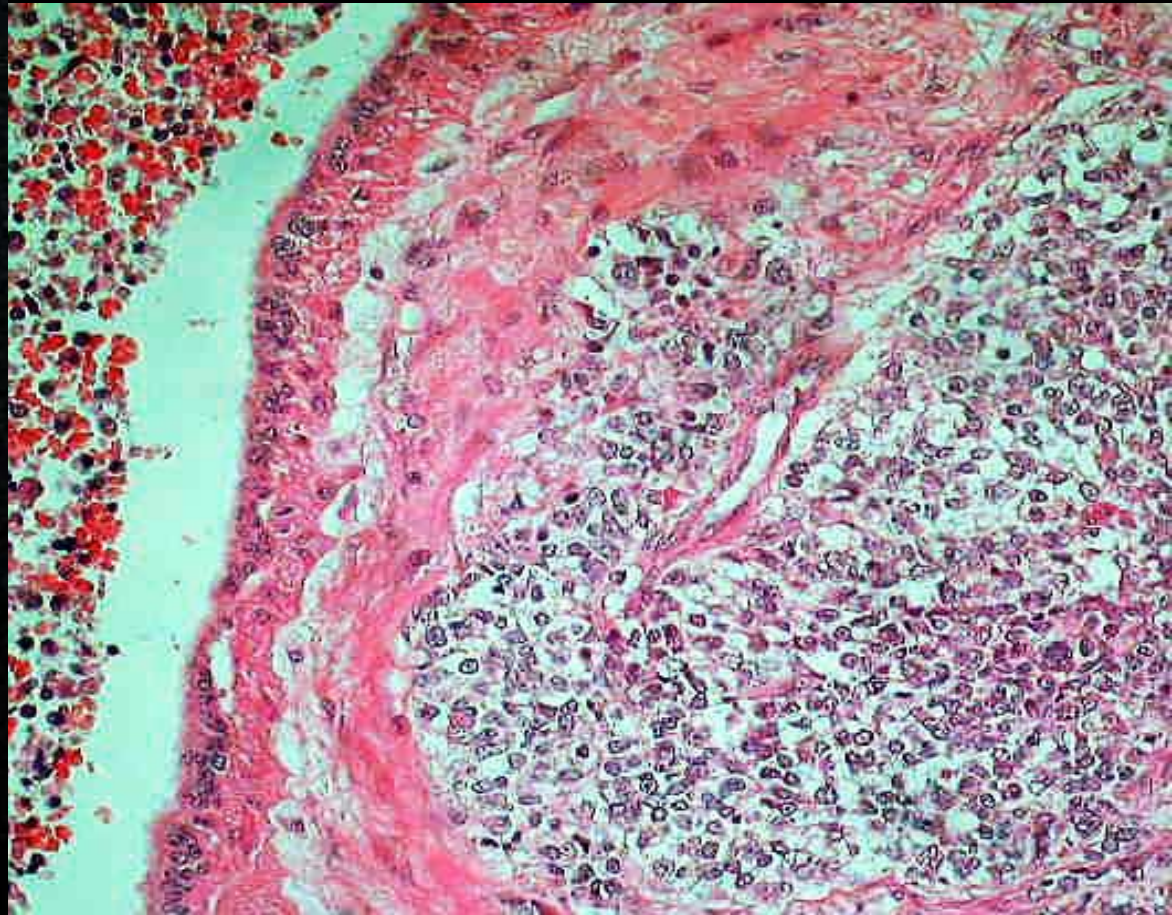
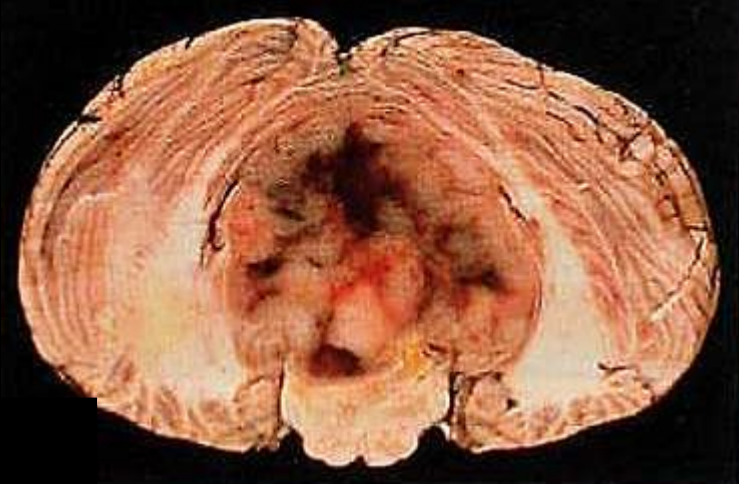
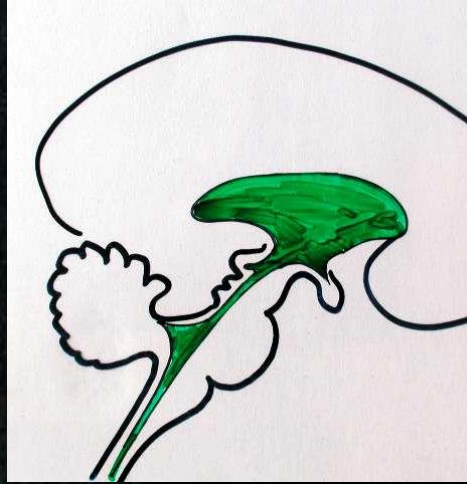
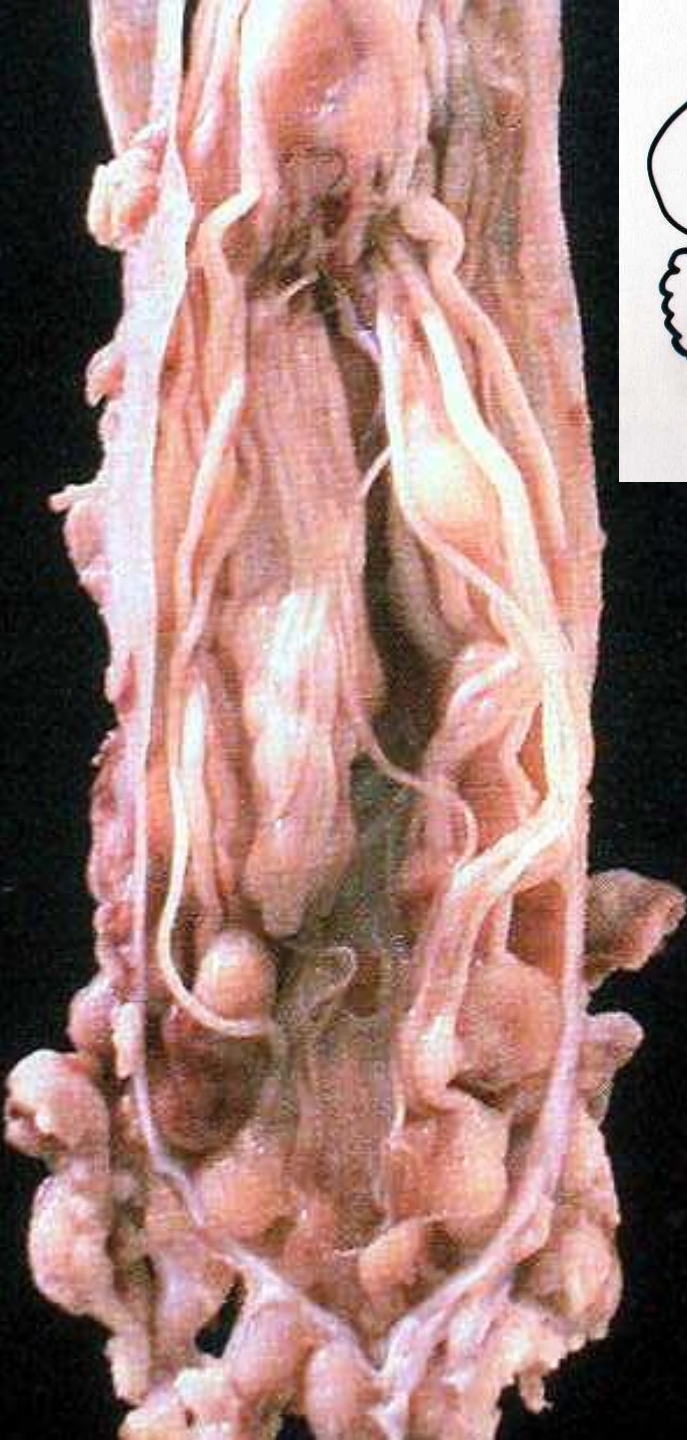
Medulloblastoma

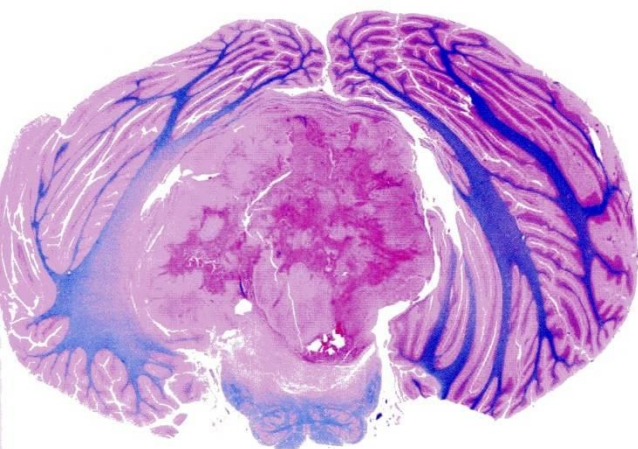


no
lymphatics

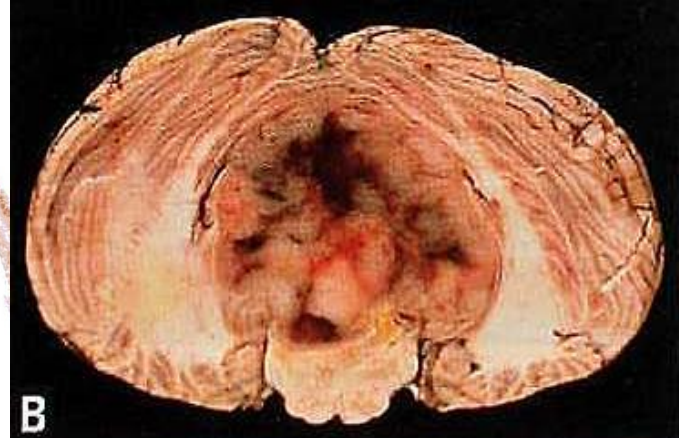
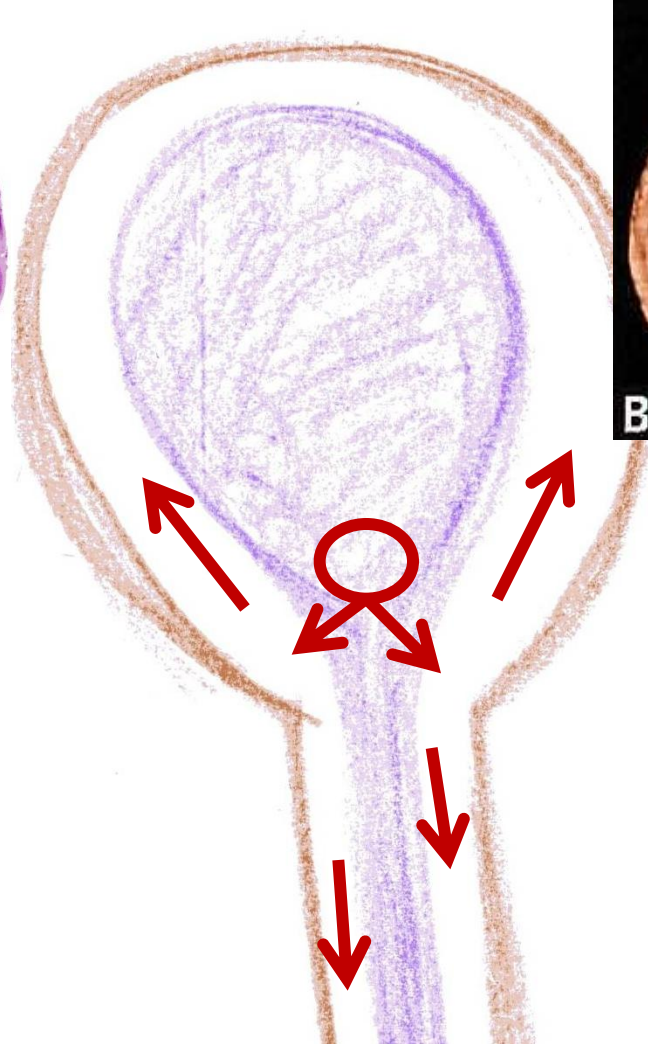


but CSF

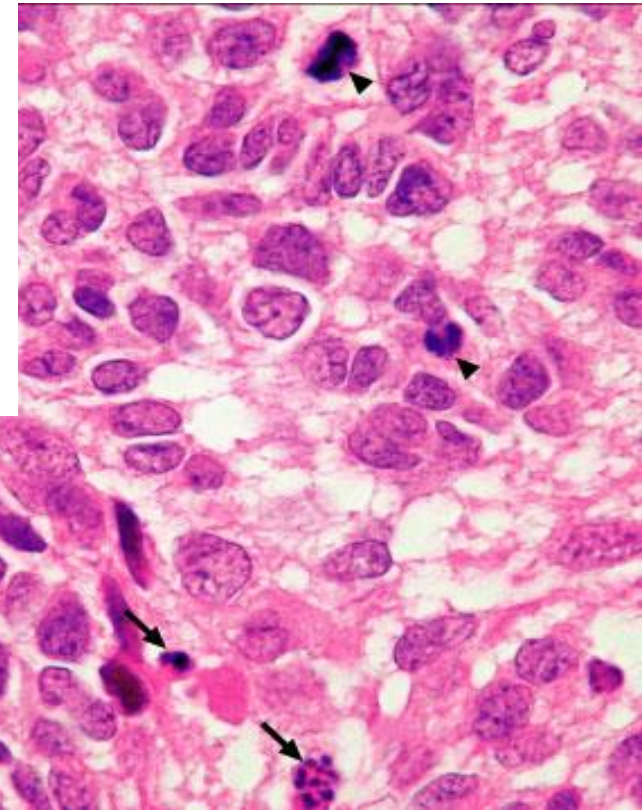
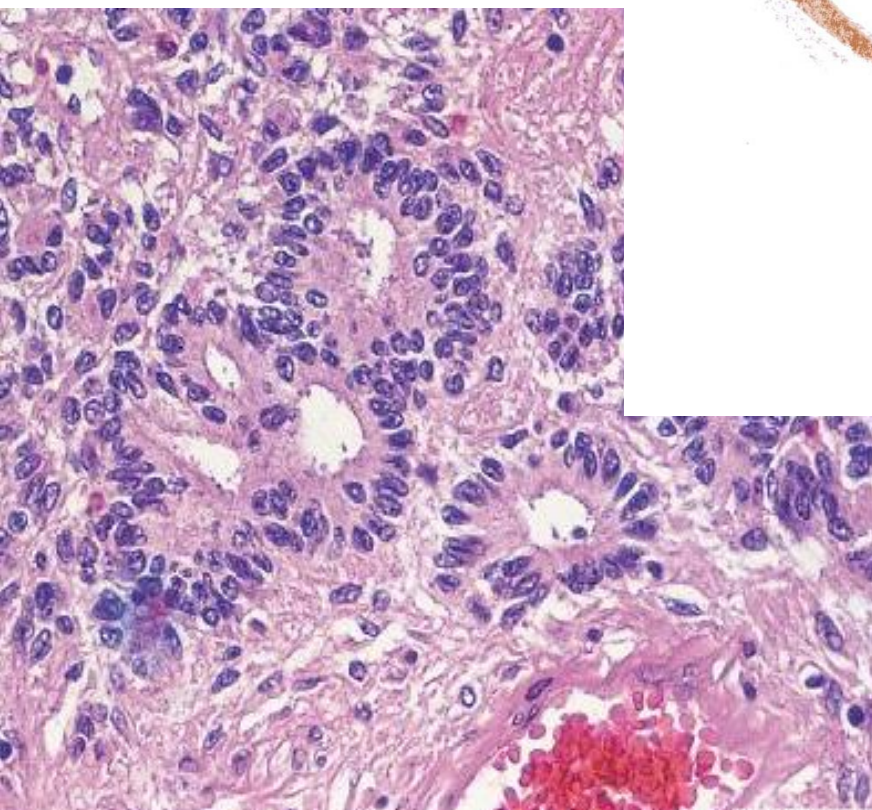


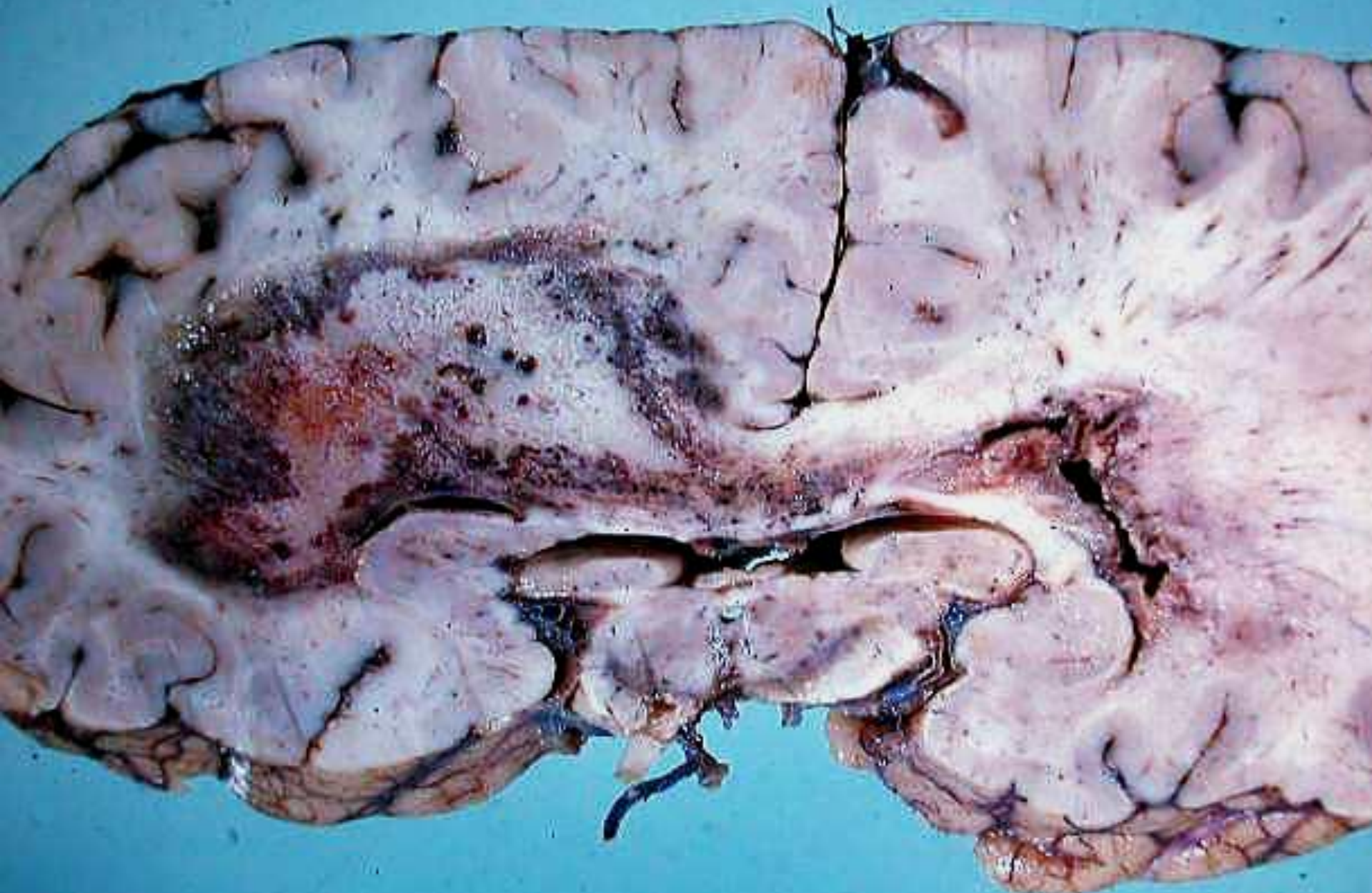


Ependymoma

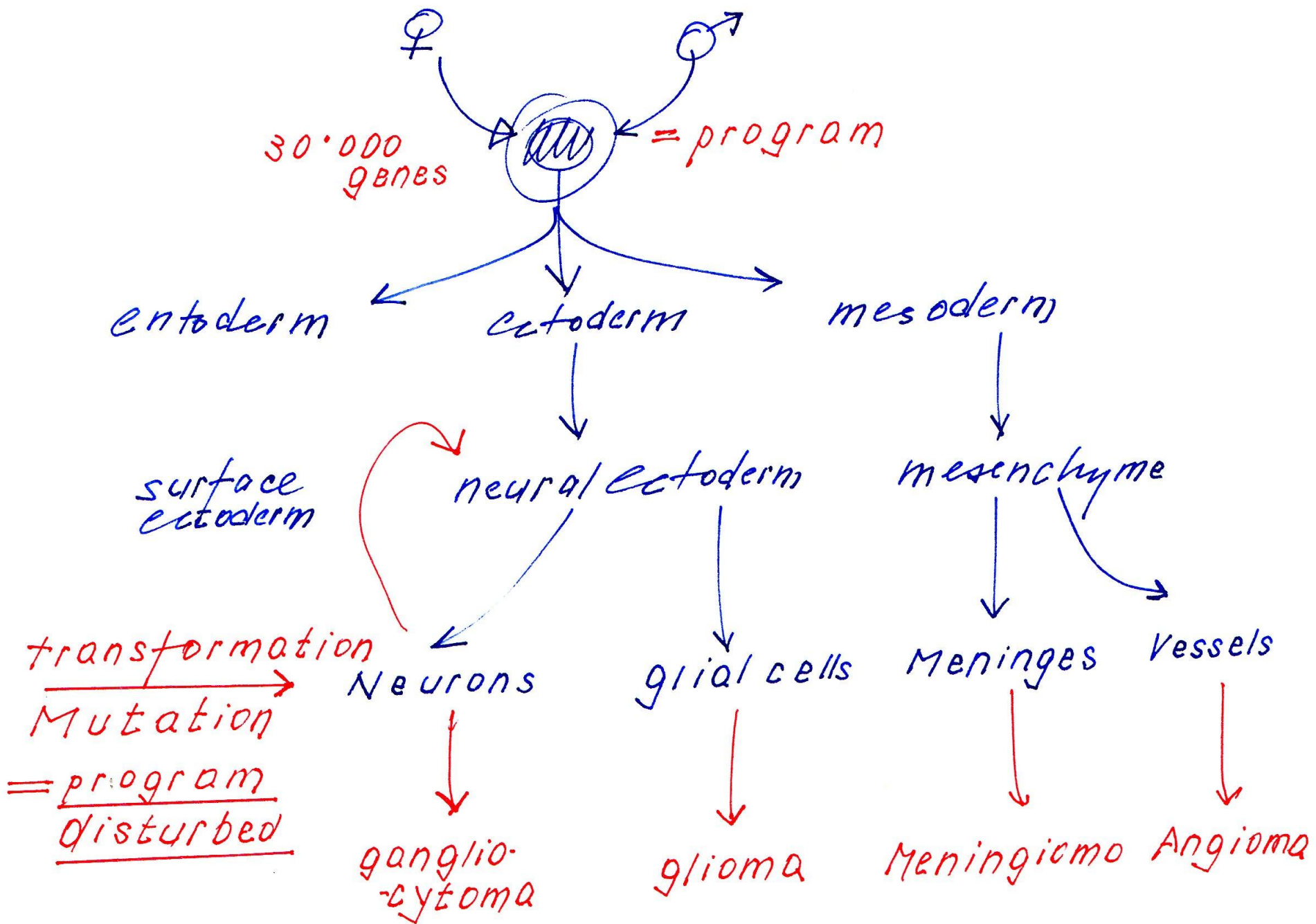


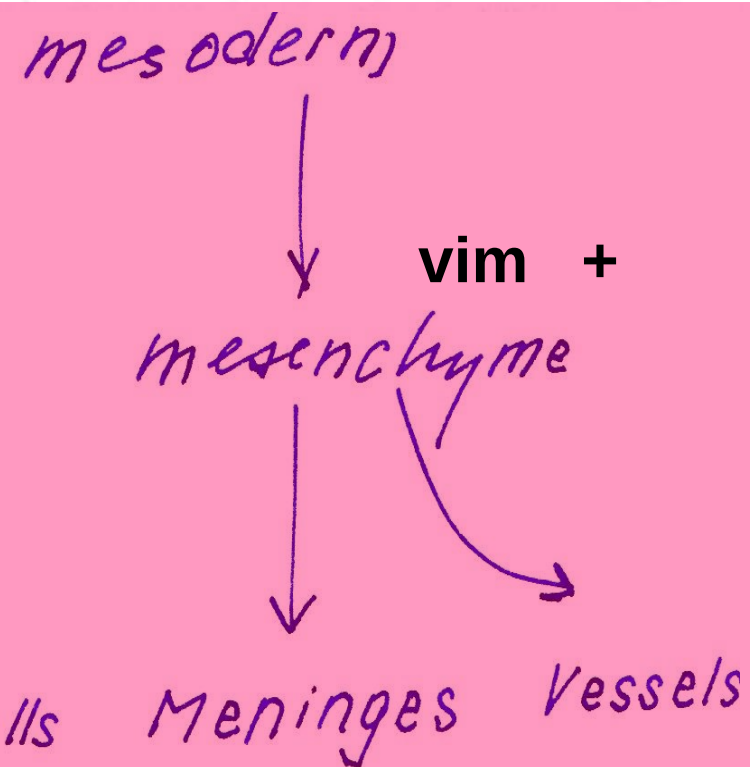
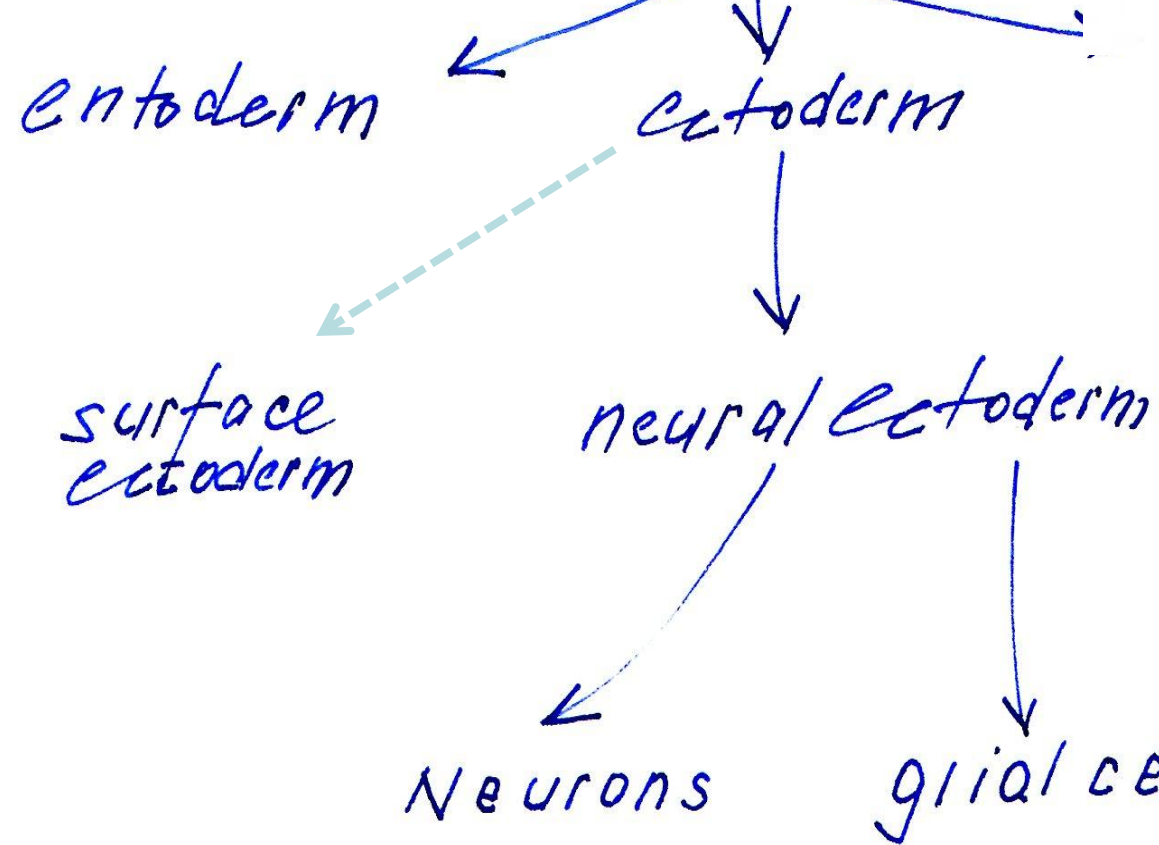
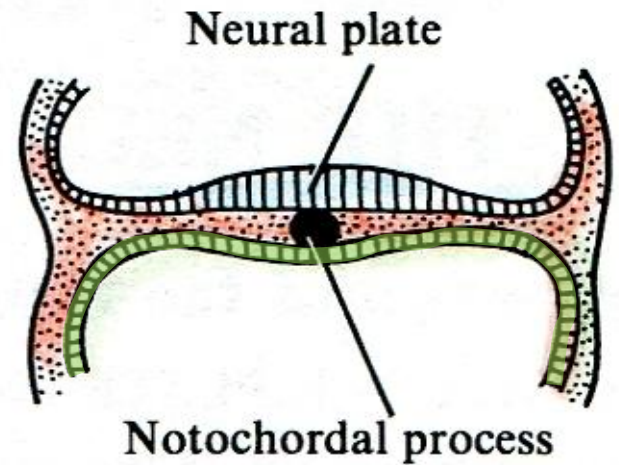
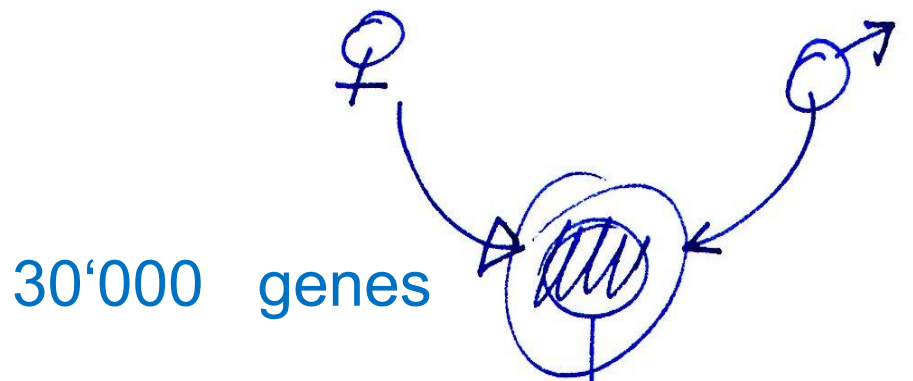
Medulloblastoma

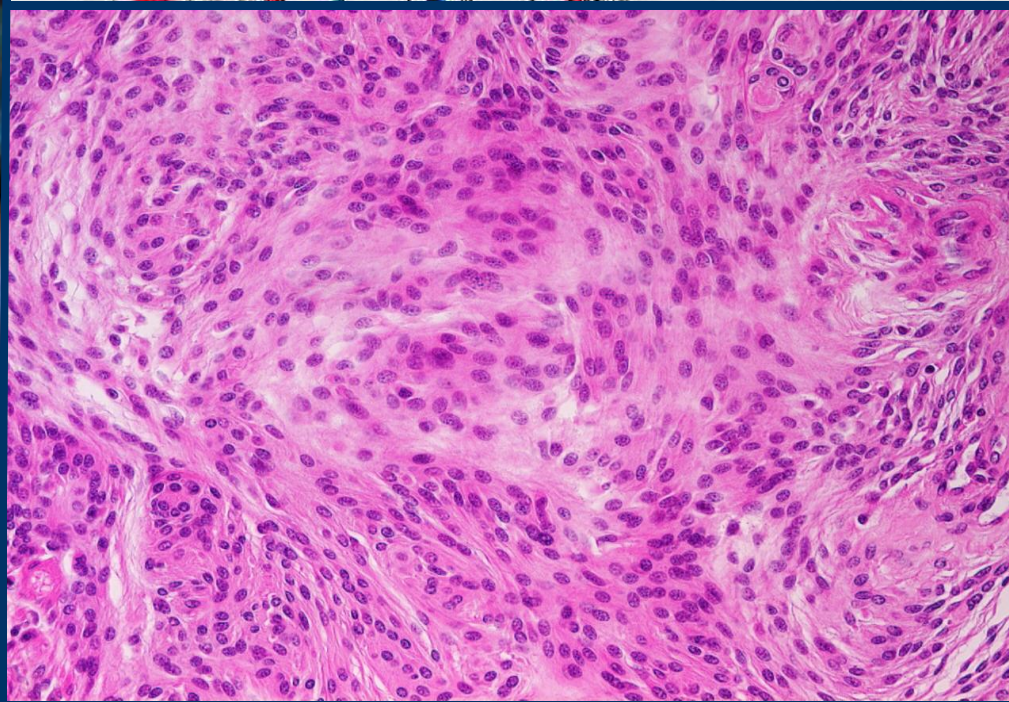
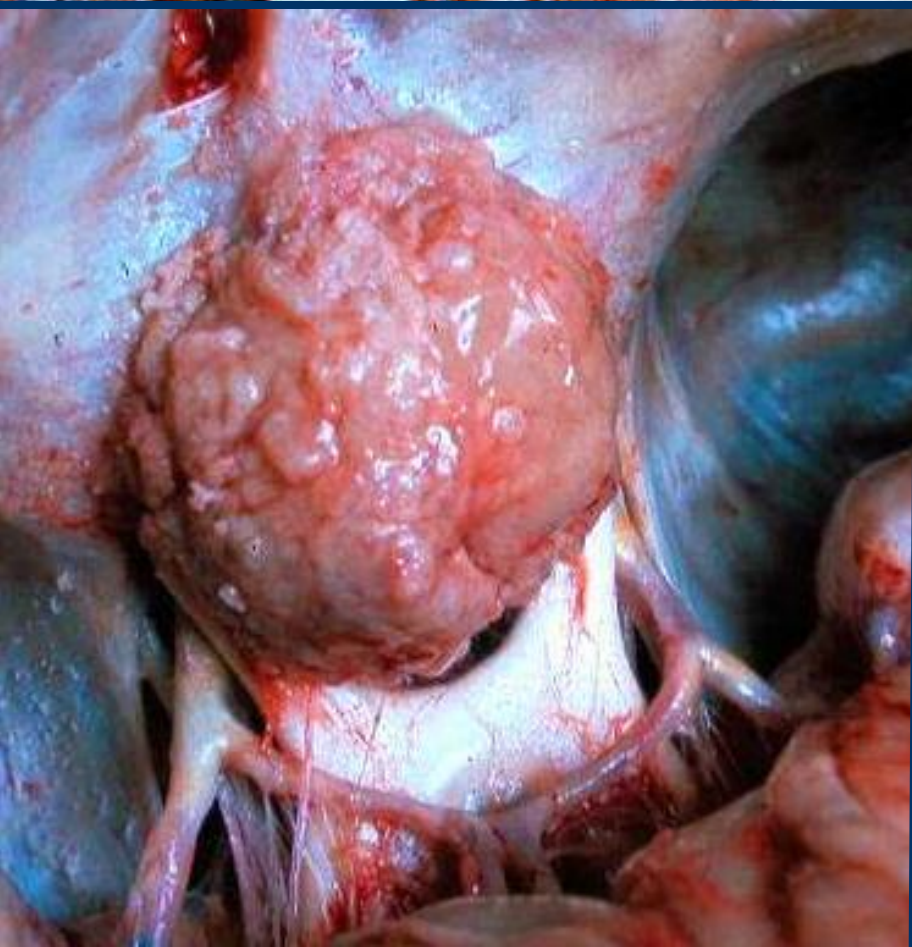
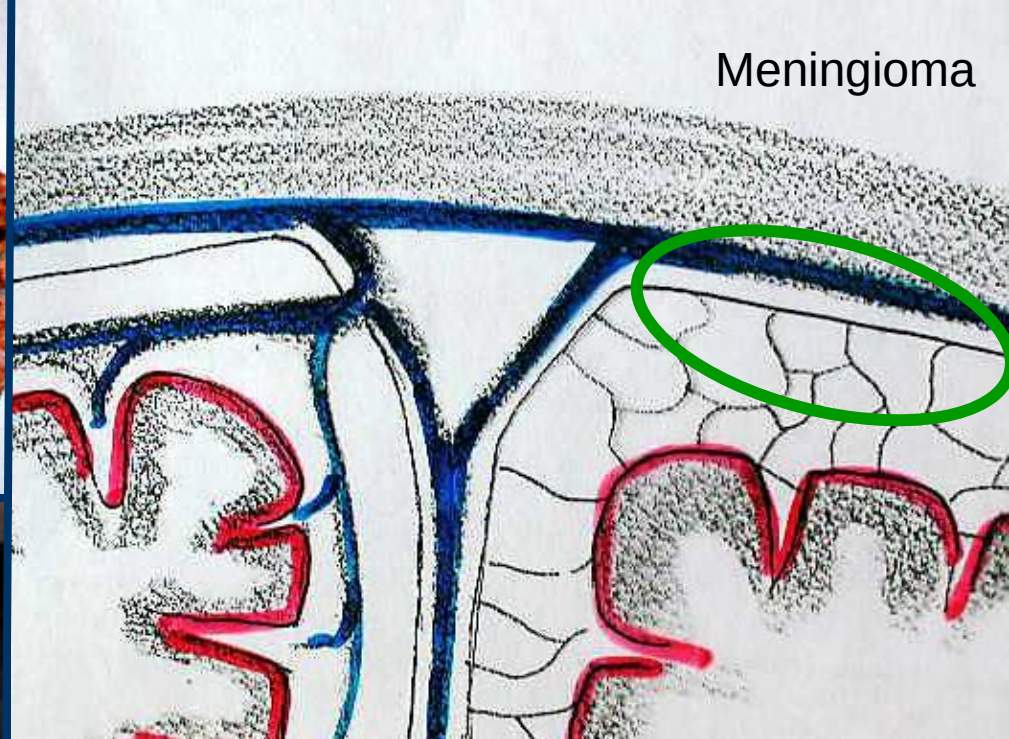
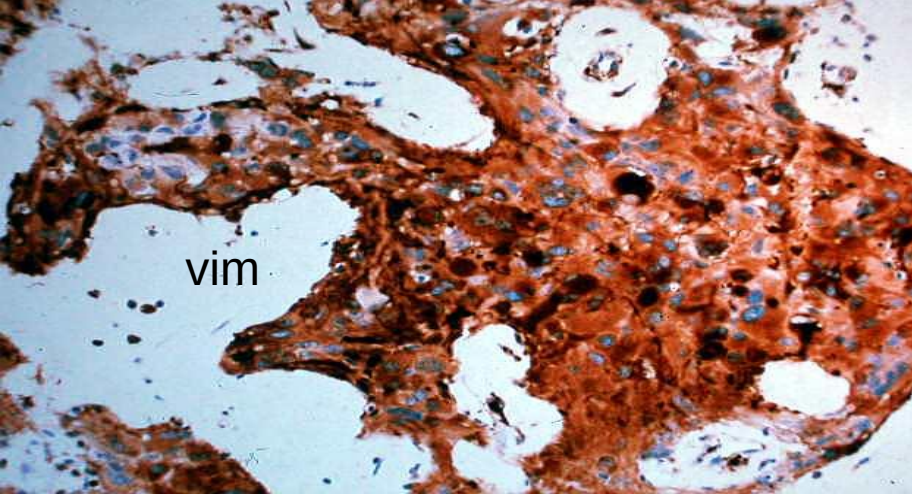


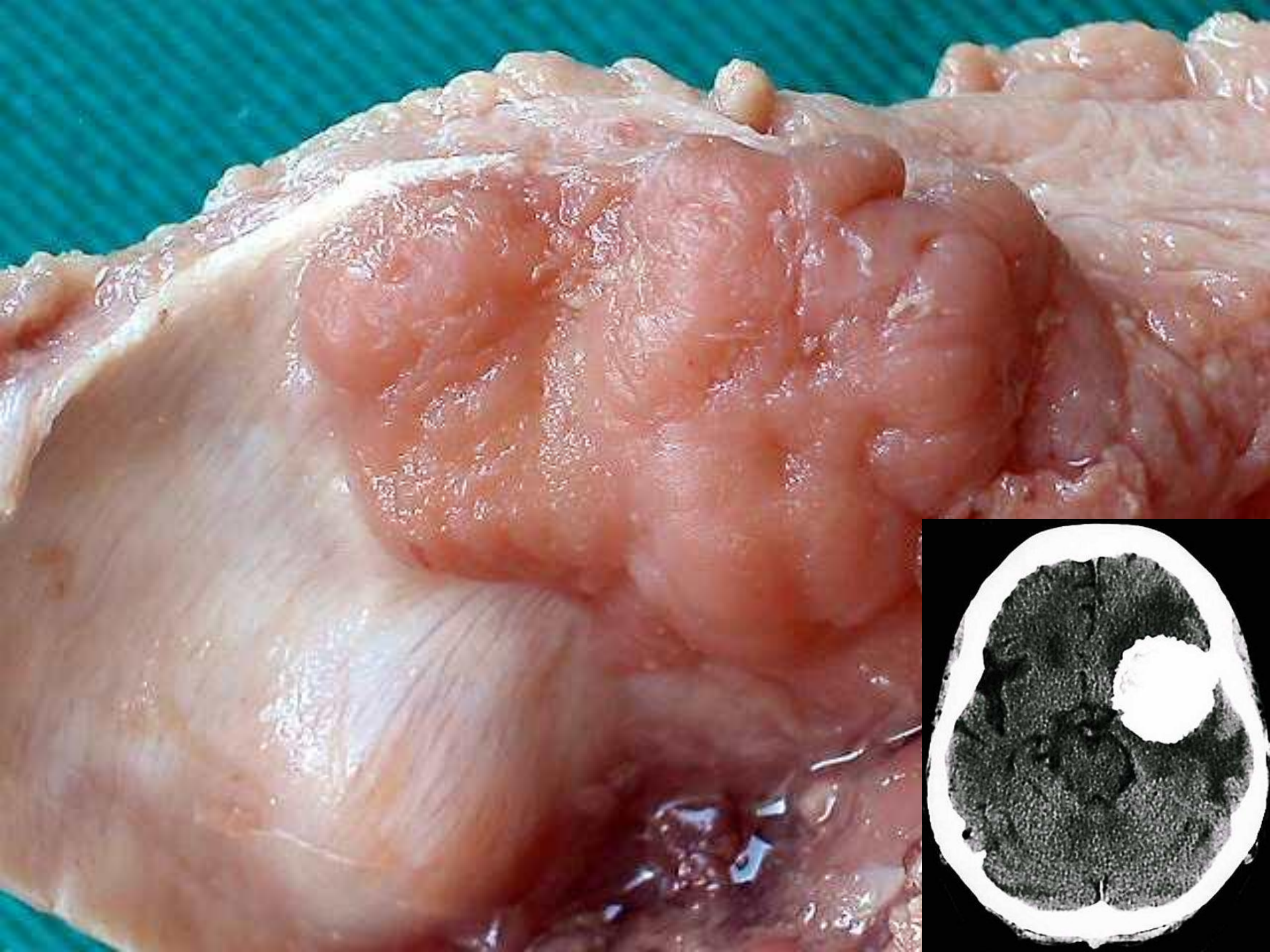


Glioblastoma → invades myelinated nerve tracts (callosum)

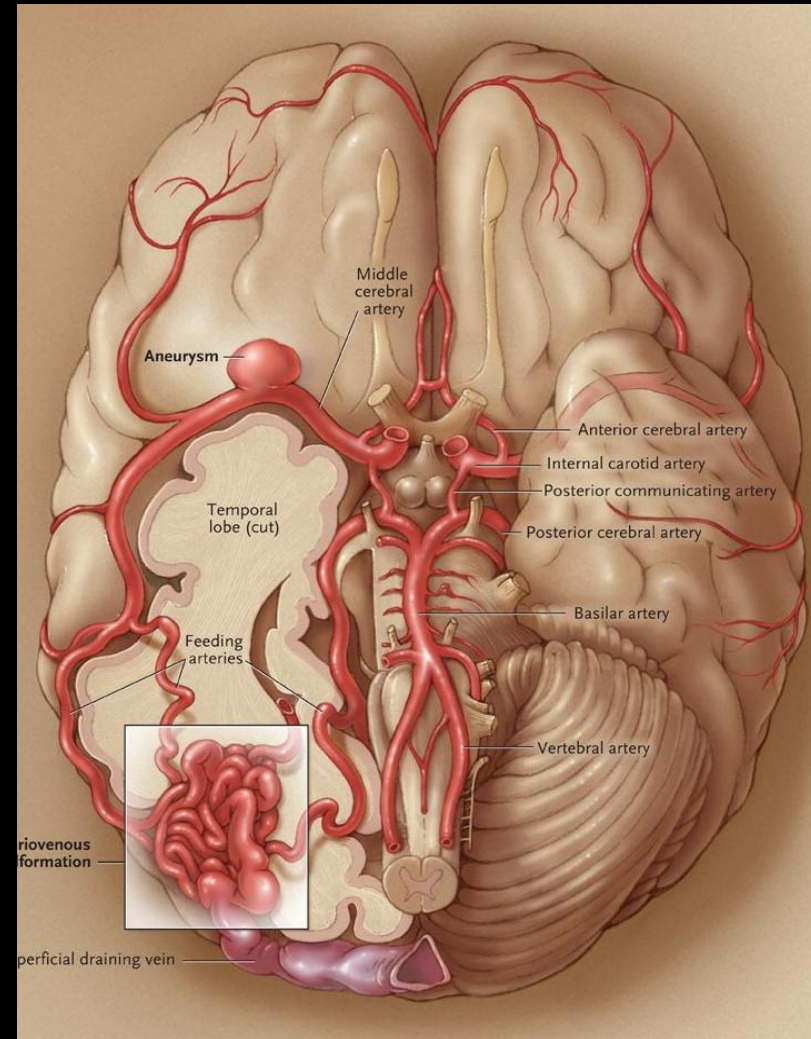








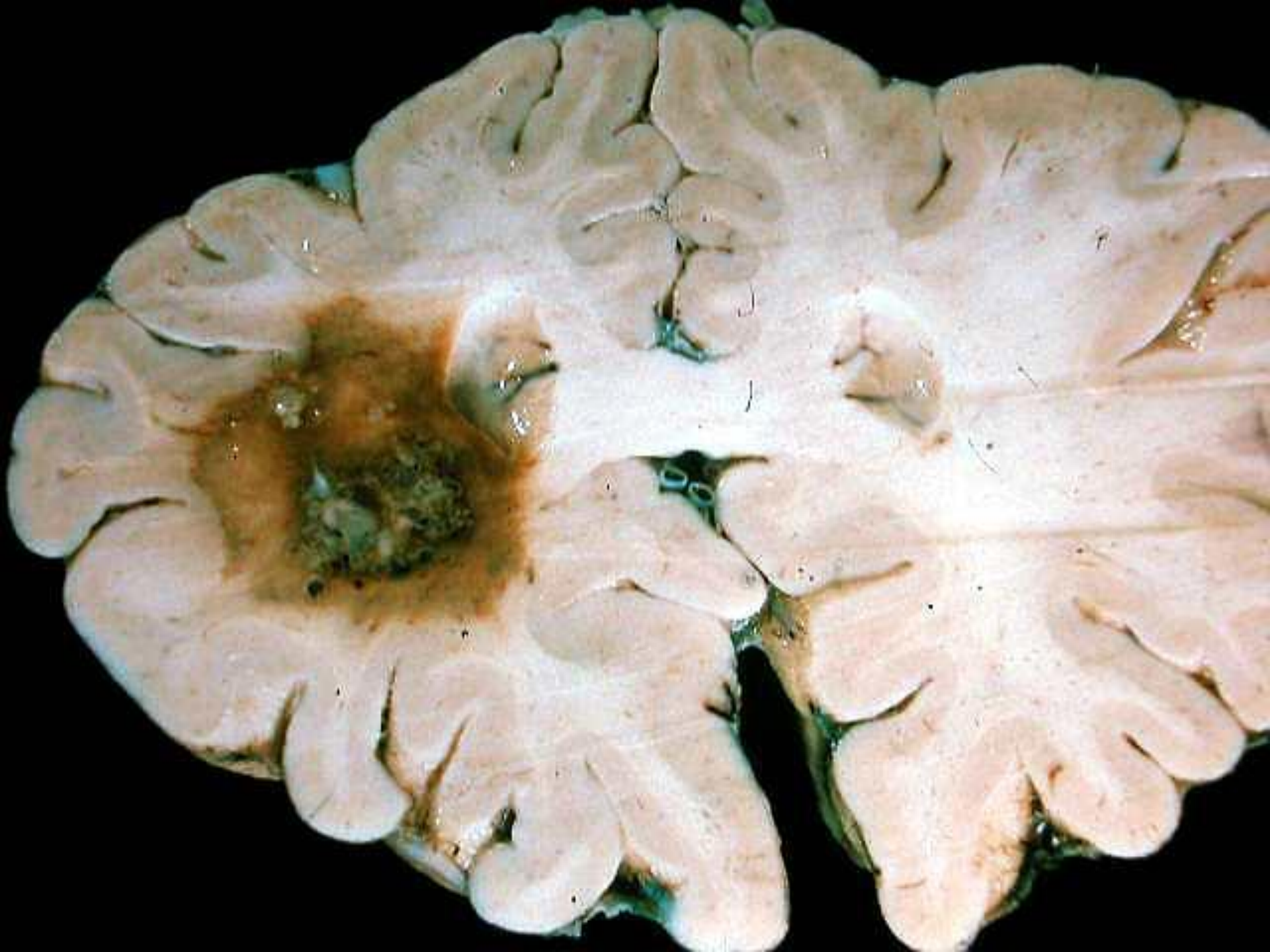
Imaging of Arteriovenous Malformations.

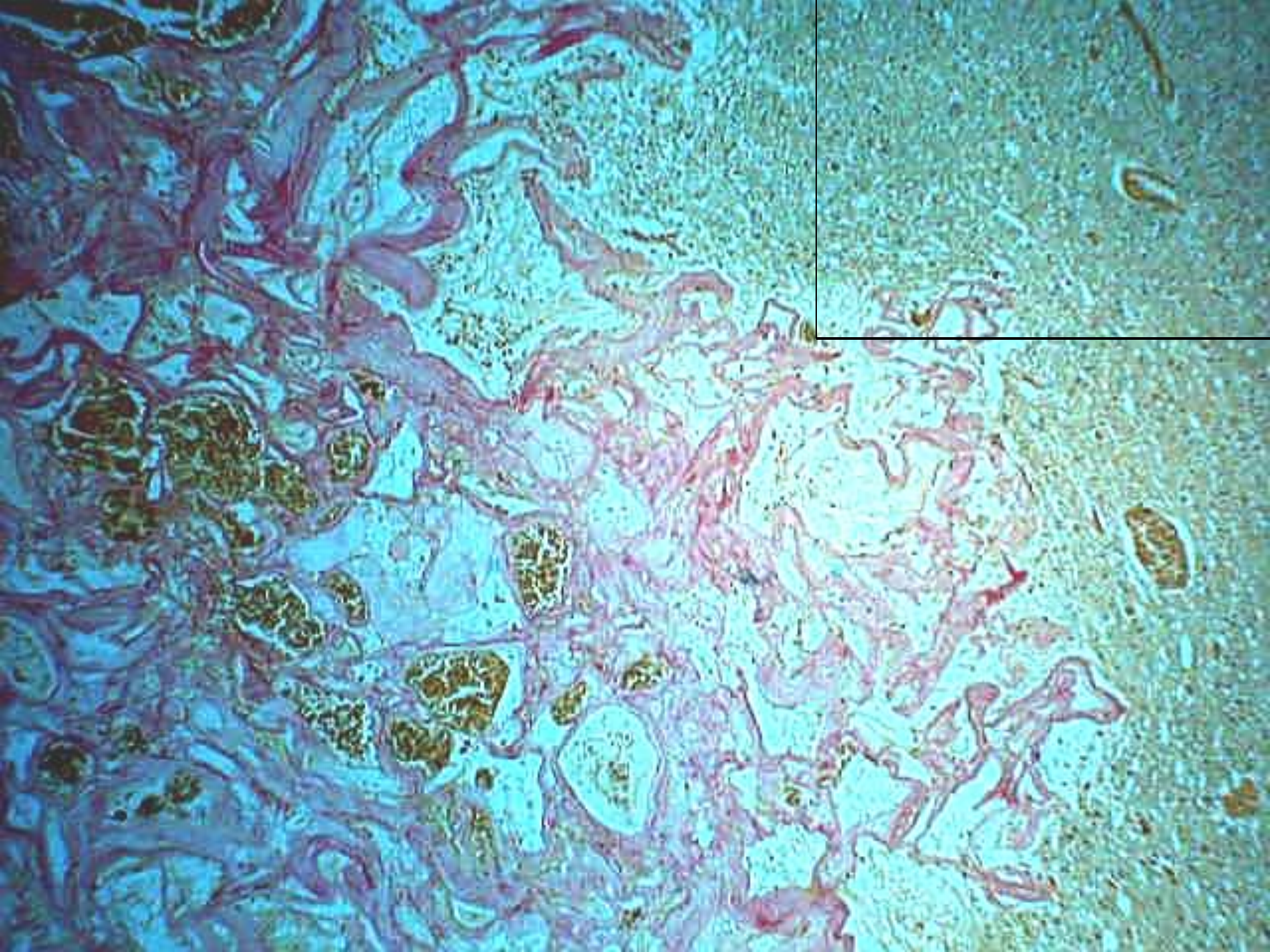


Friedlander RM. N Engl J Med 2007;356:2704-2712.



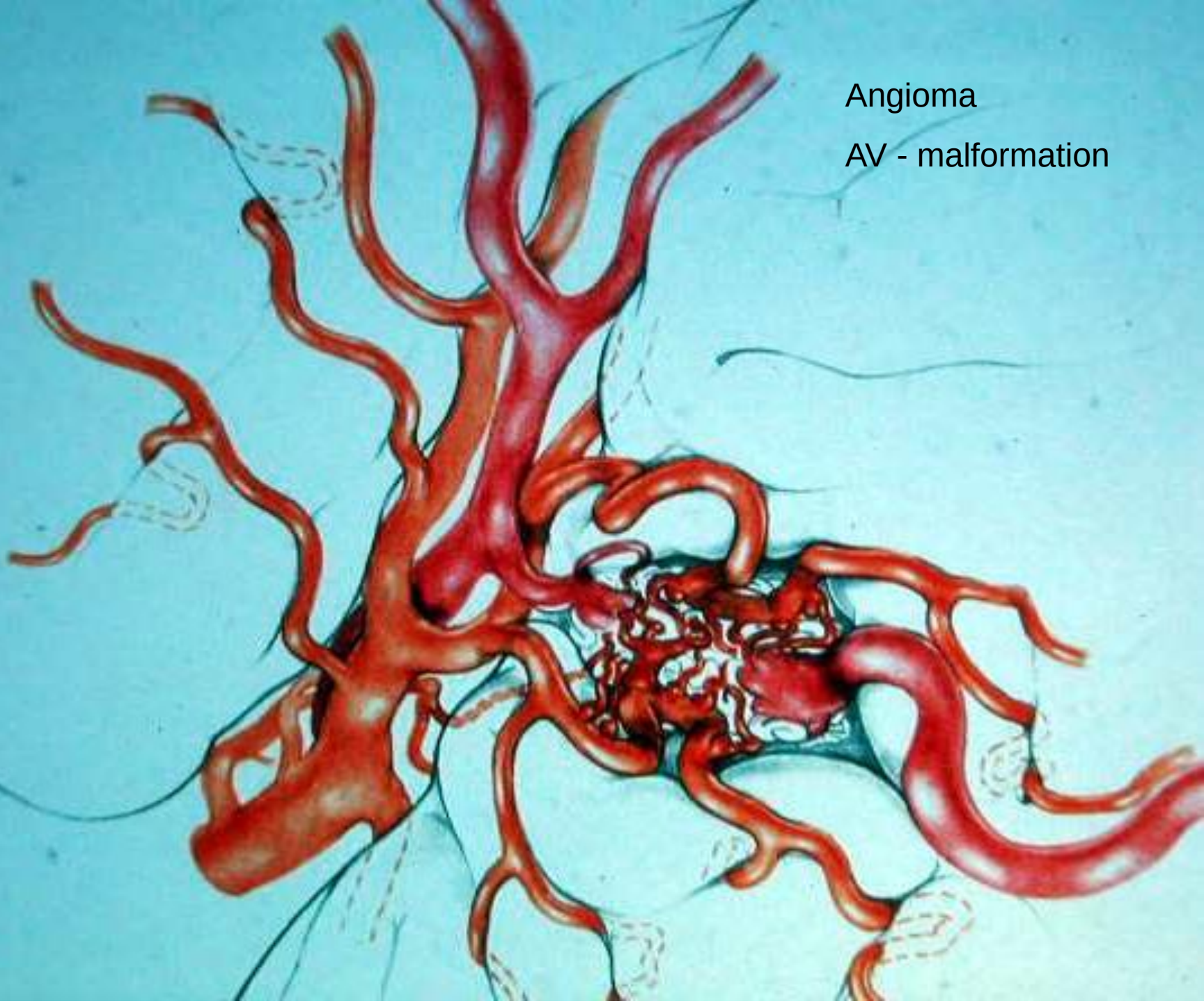
The NEW ENGLAND
JOURNAL of MEDICINE



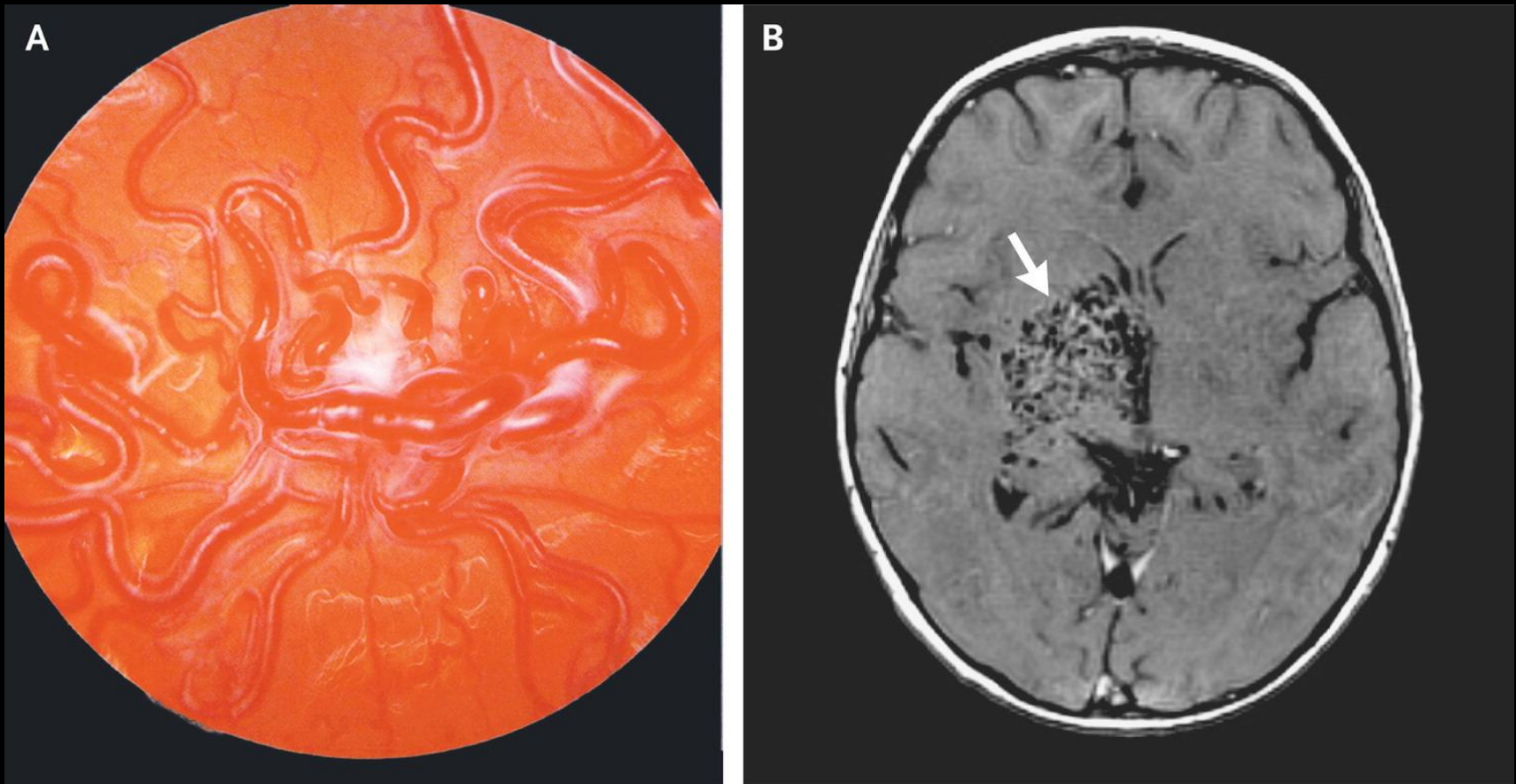


Angioma

AV - malformation



Orbital and Cerebral Arteriovenous Malformations



Choudhri O, Chang SD. N Engl J Med 2011;365:e28.



The NEW ENGLAND
JOURNAL of MEDICINE

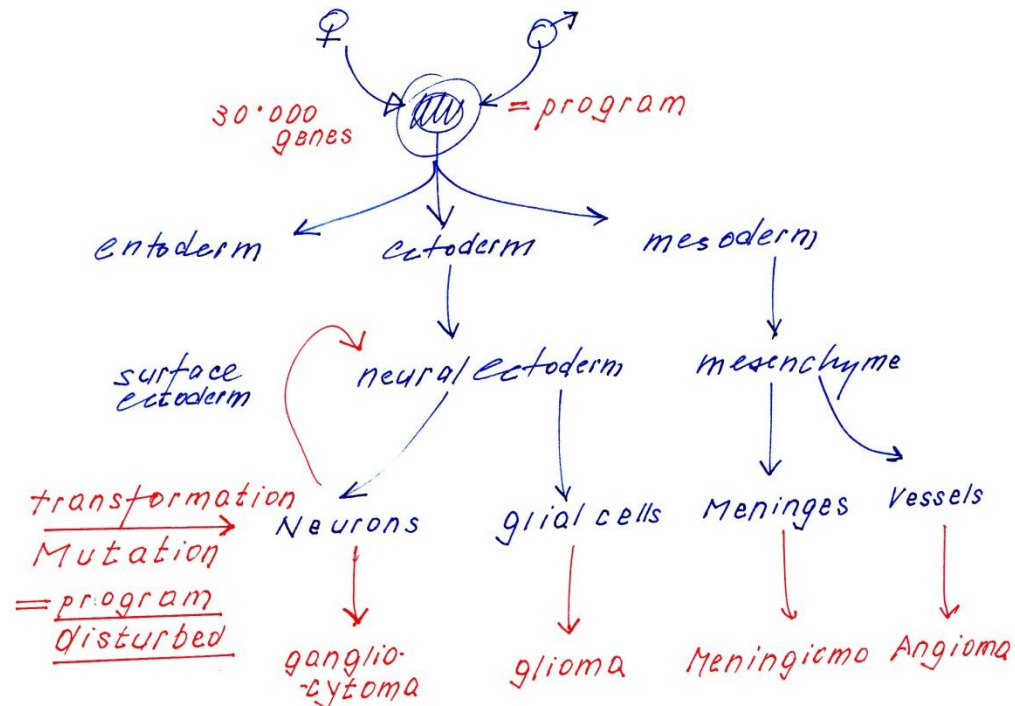
CNS and HIV

HIV

HIV-associated
[opportunistic]

CNS Lymphoma

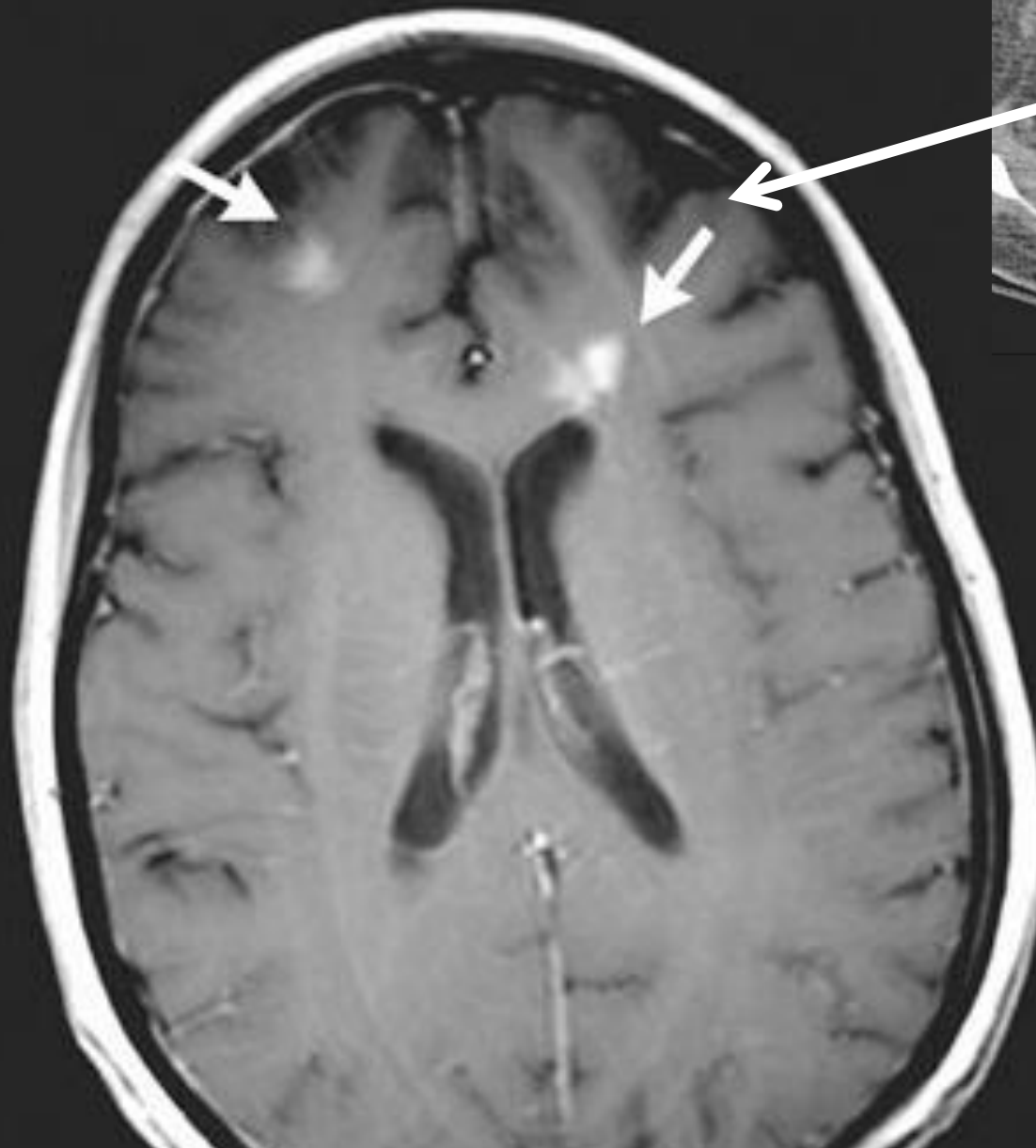
HIV meningoencephalitis
(subacute, giant cells)



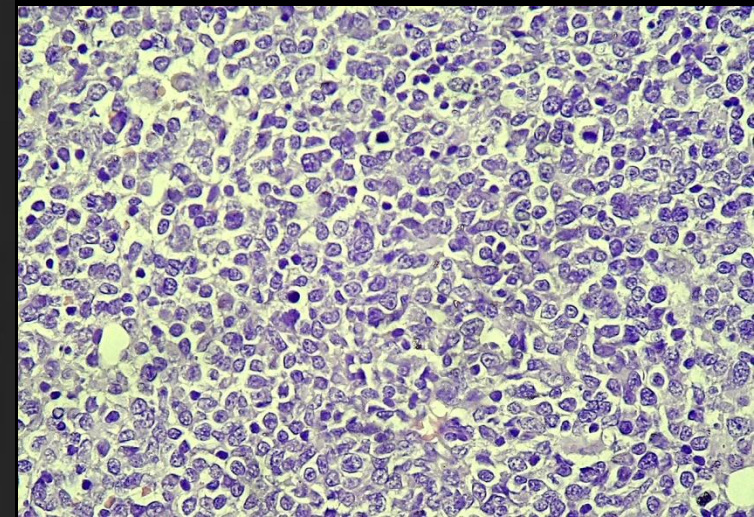
Primary lymphoma of the CNS

- in the absence of lymphoma outside the CNS = rare = 1% of all intracranial tumors
- 2% of all lymphomas
- secondary involvement of CNS in systemic lymphoma: 8 – 27%
- in childhood lymphomas 50%
- in high-grade NHL more common

Secondary non-Hodgkin lymphoma of the CNS



a 28-year-old pregnant women

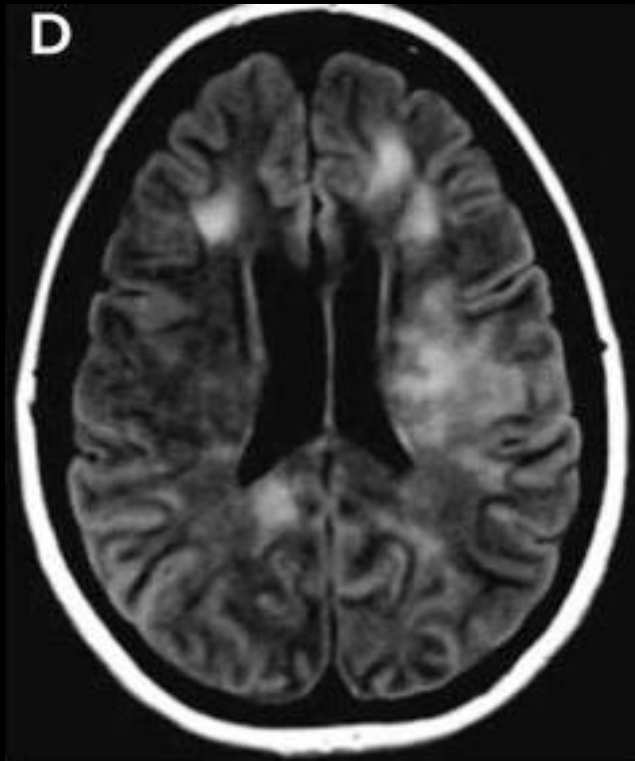


O'Gara P et al. N Engl J Med 2009;361:2462-2473



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JOURNAL of MEDICINE

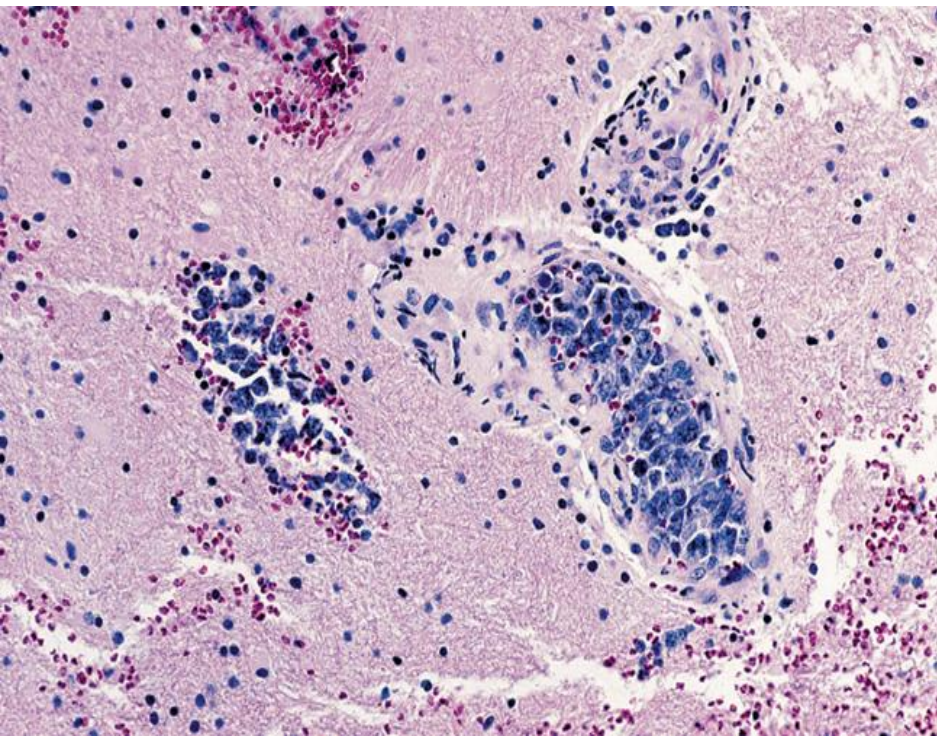
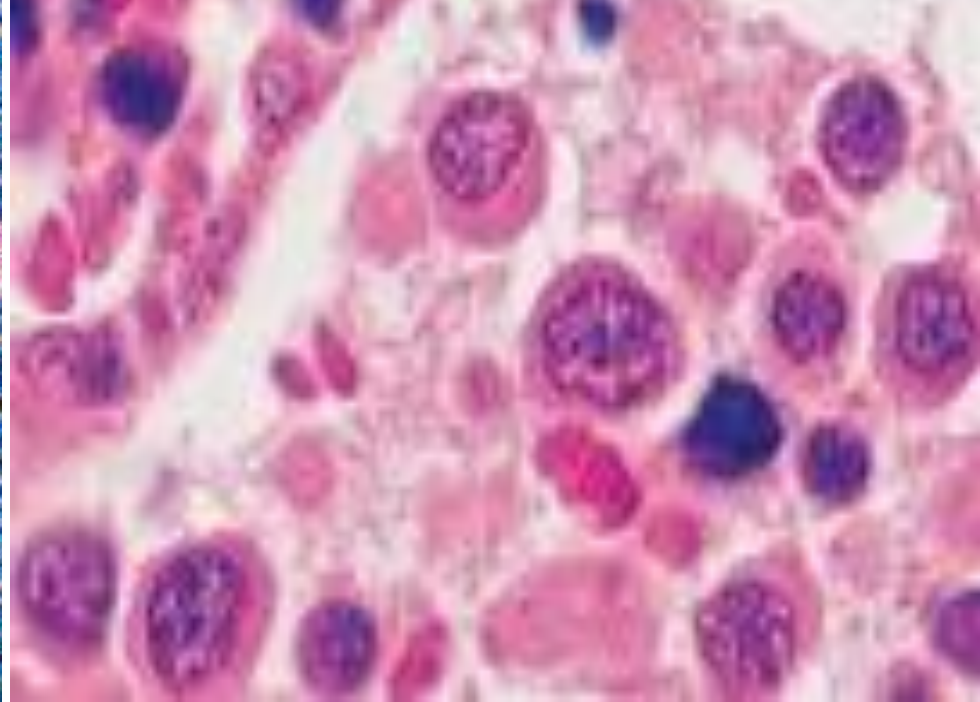
Primary lymphoma of the CNS



Pless ML et al. N Engl J Med 2010;362:1129-1138.



The NEW ENGLAND
JOURNAL of MEDICINE



Primary CNS lymphomas

- high-grade B-cell lymphomas
- angiocentric / intravascular

Primary lymphoma of the CNS

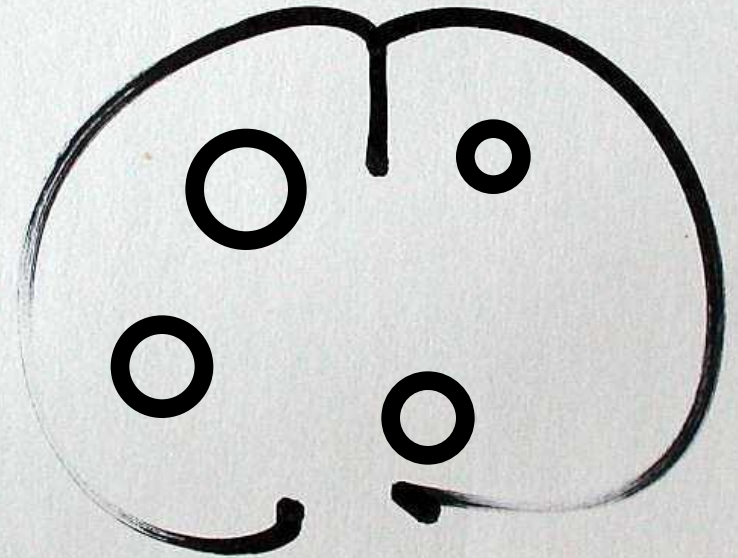
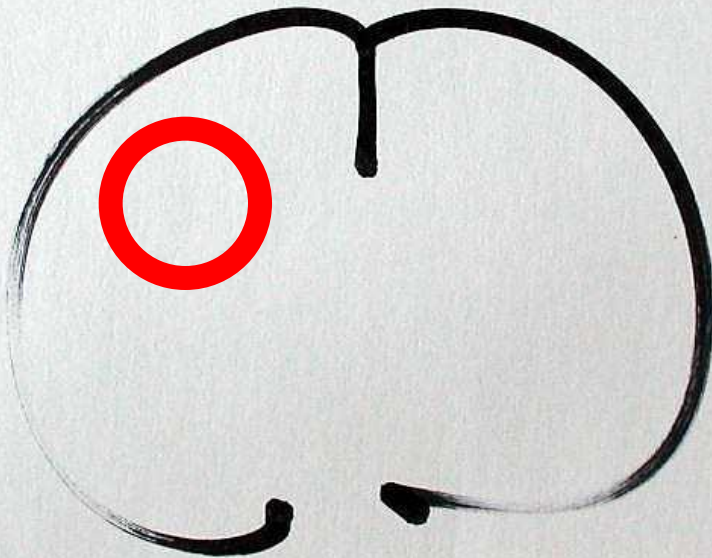
- in the absence of lymphoma outside the CNS
- **in AIDS: 4.7 / 1000 patients**

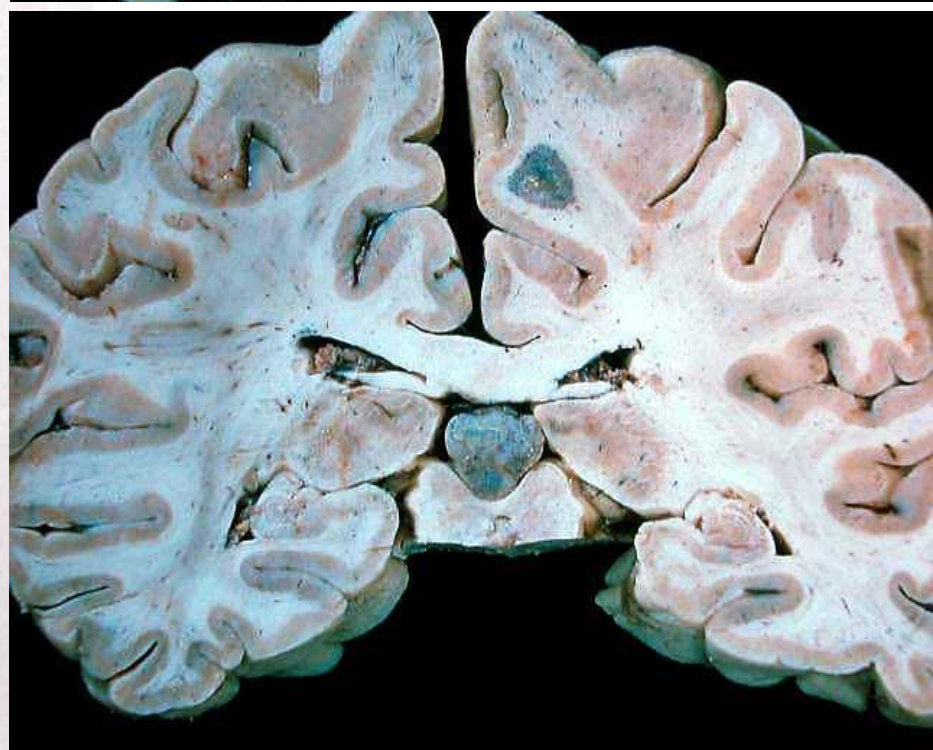
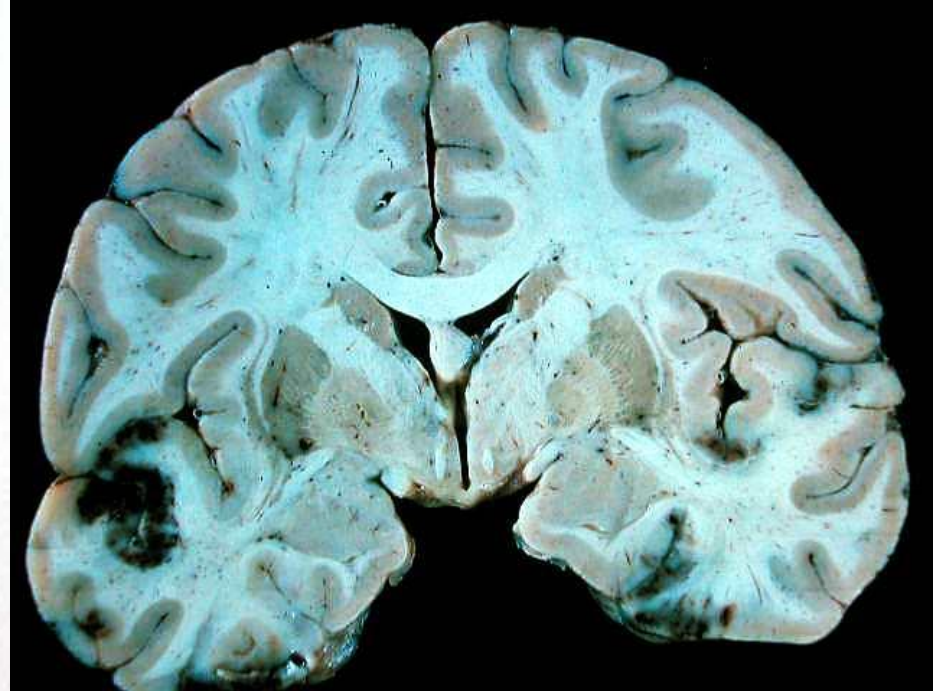
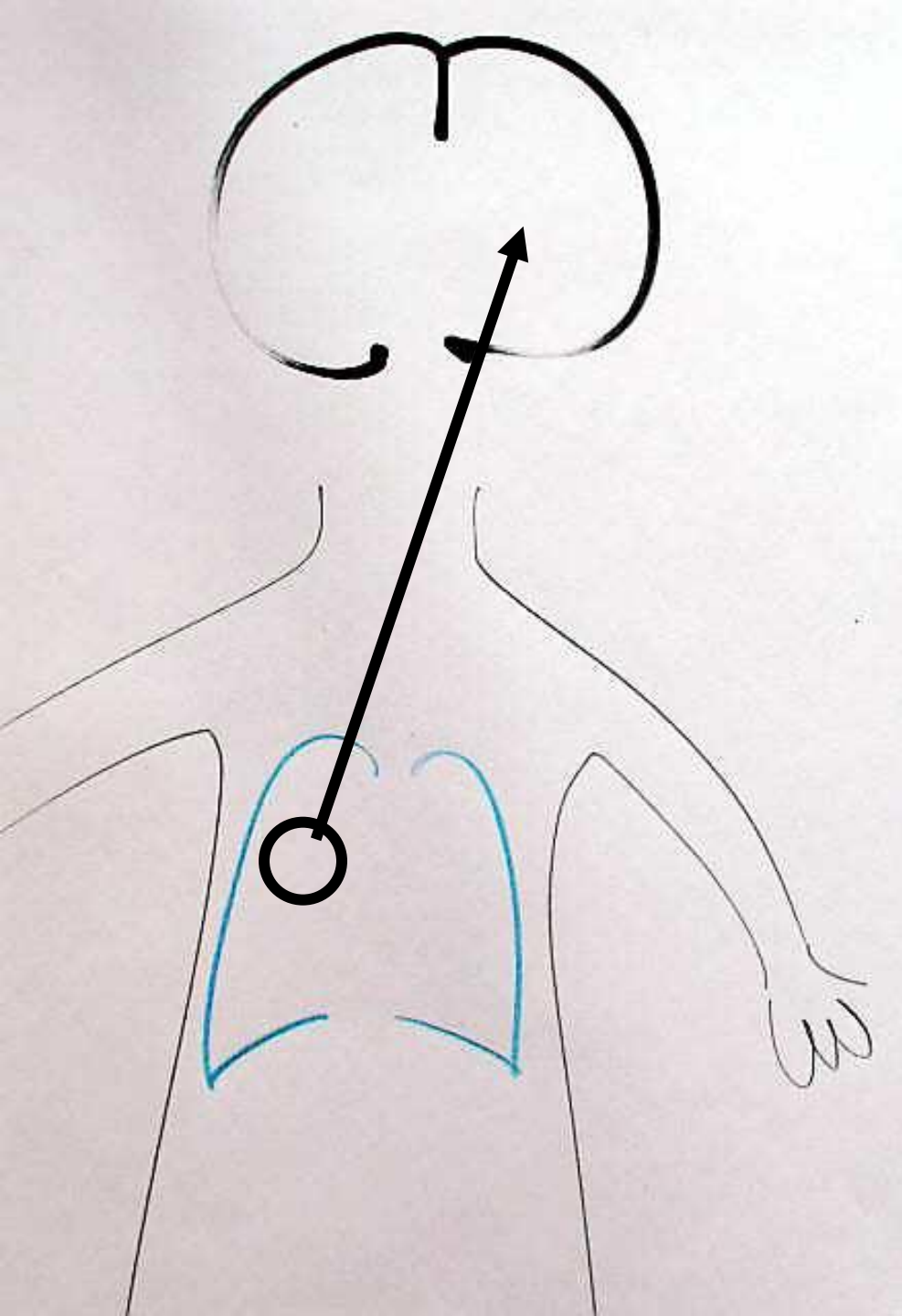
Primary lymphoma of the CNS

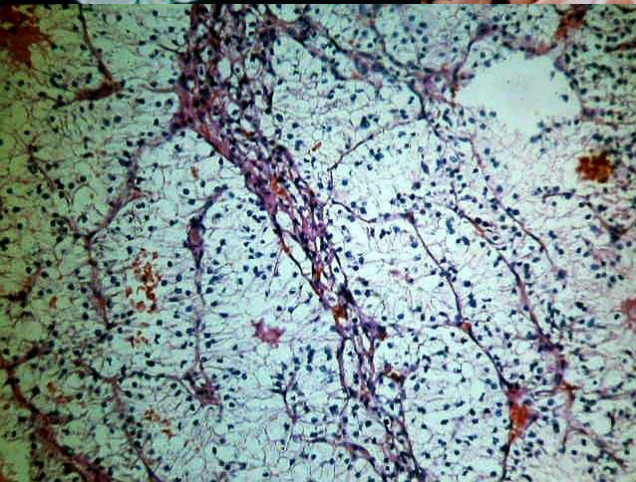
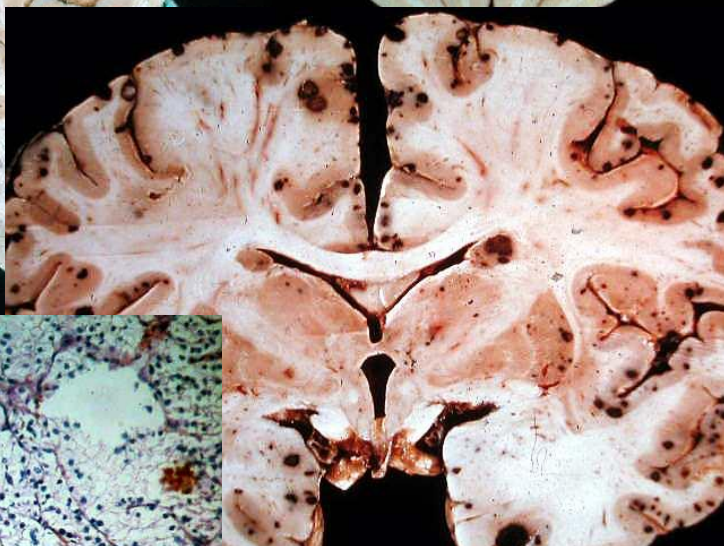
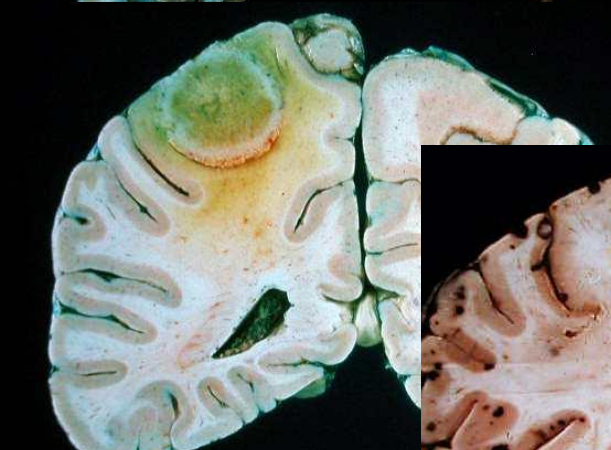
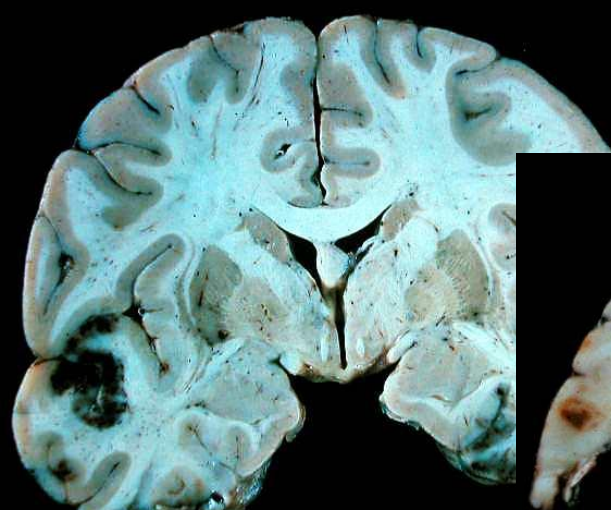
- in the absence of lymphoma outside the CNS
- in AIDS: 4.7 / 1000 patients
= 3600 x higher than in normal population
- Primary lymphomas
= the most common primary brain tumor in AIDS

Brain tumors

- About half of the brain tumors are primary, half metastatic







Lung

Breast

Colon

Melanoma

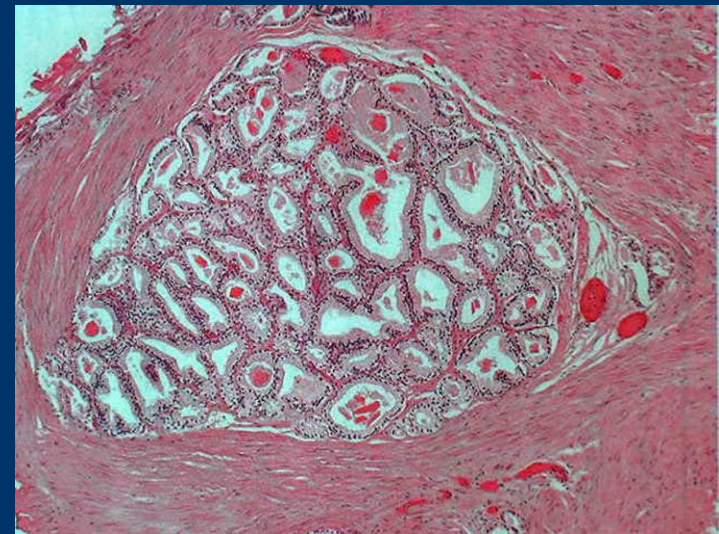
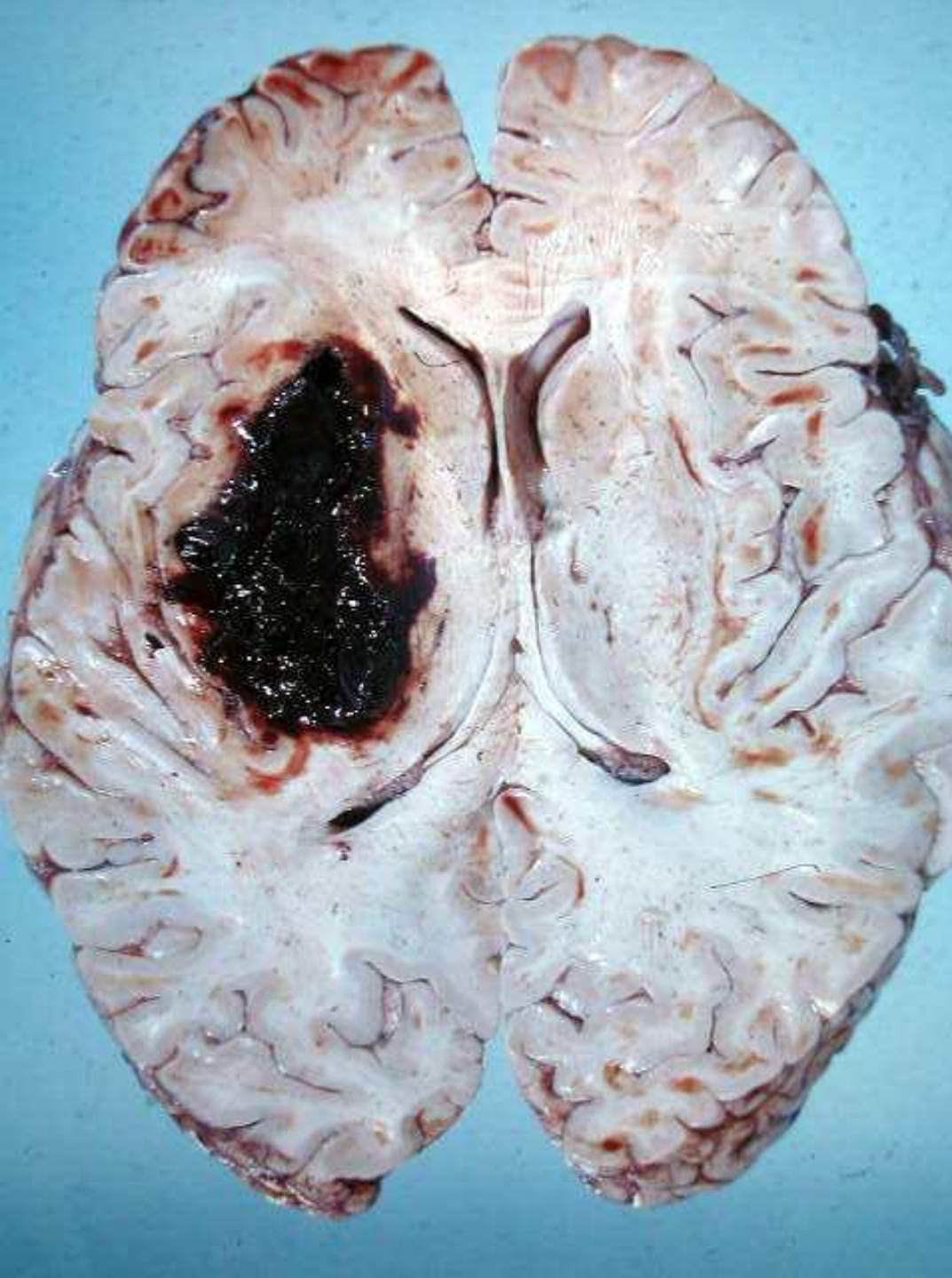
Renal cell ca

80%

Choriocarcinoma

common

Prostatic adenocarcinoma
rare



In children:

Leukemia

Lymphoma

Osteogenic sarcoma

Rhabdomyosarcoma

Ewing sarcoma

In adults:

Lung

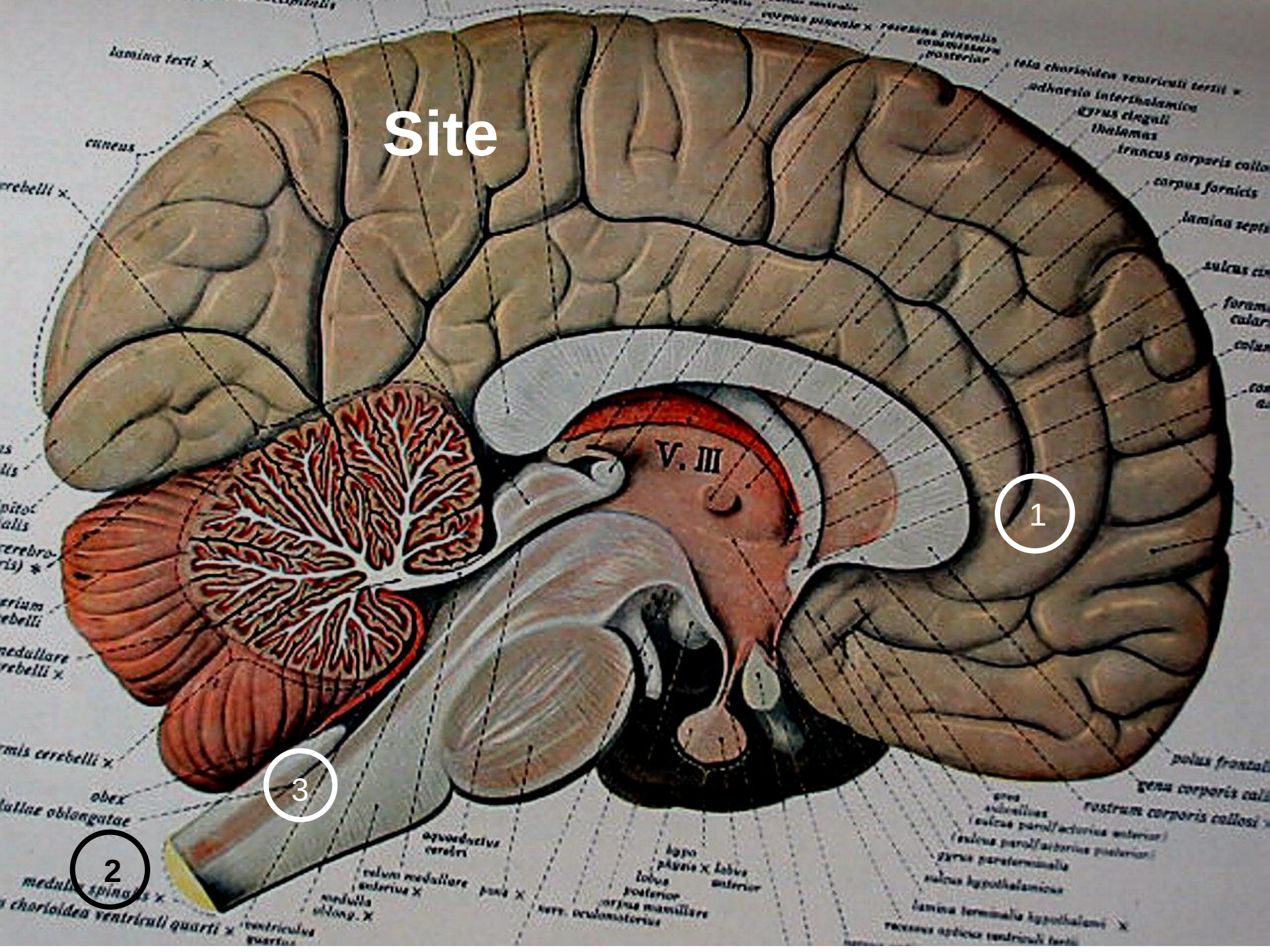
Breast

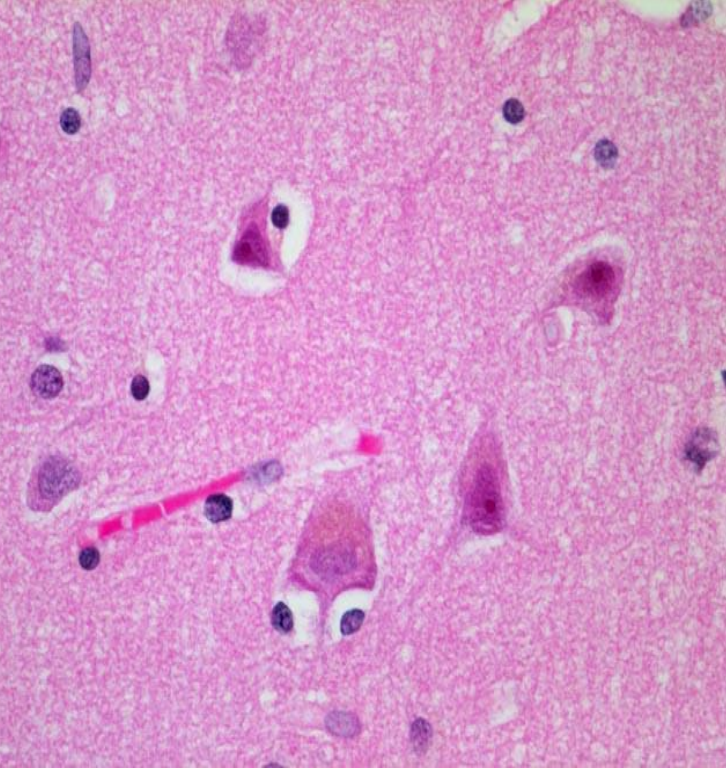
Colon

Melanoma

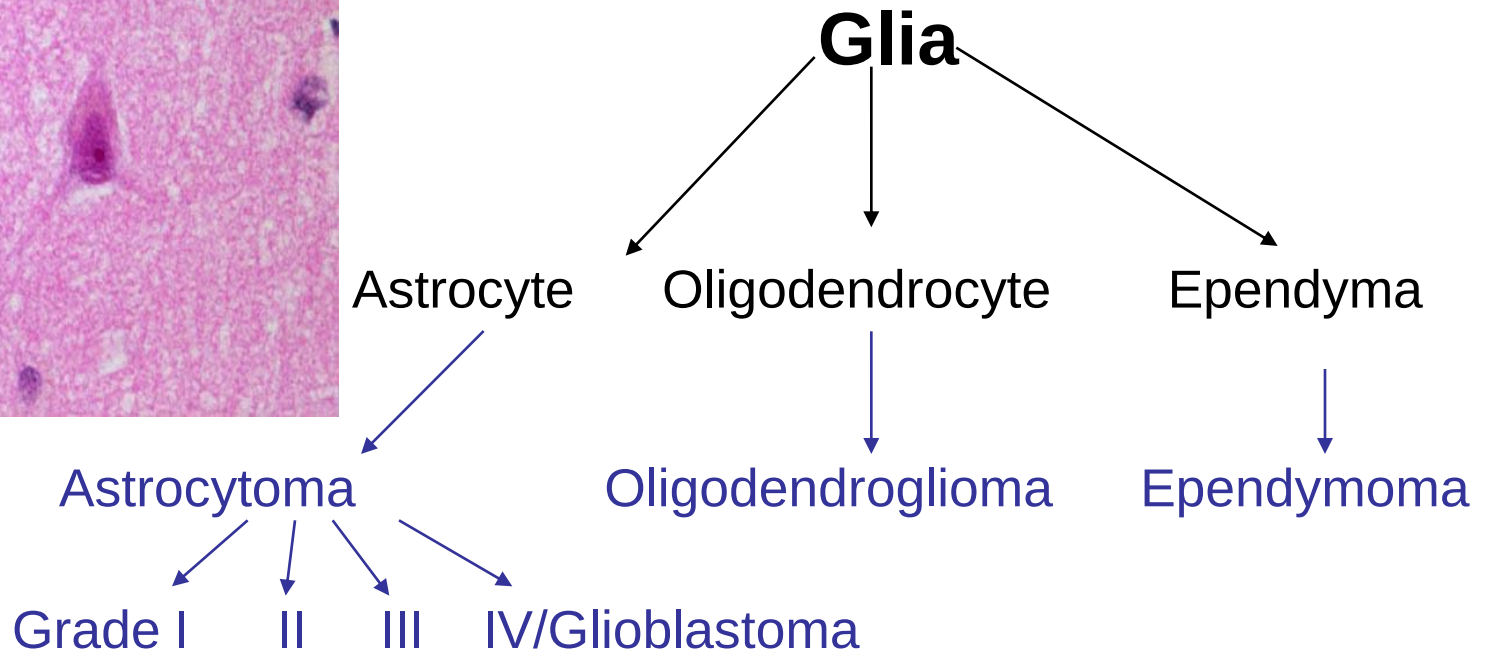
Renal cell ca

Site





Neurons → **Gangliocytoma**



Embryonal cells → **Medulloblastoma**

Leptomeninges (arachnoid) → **Meningioma**

Others: lymphoma, mesenchymal tumors (angioma)

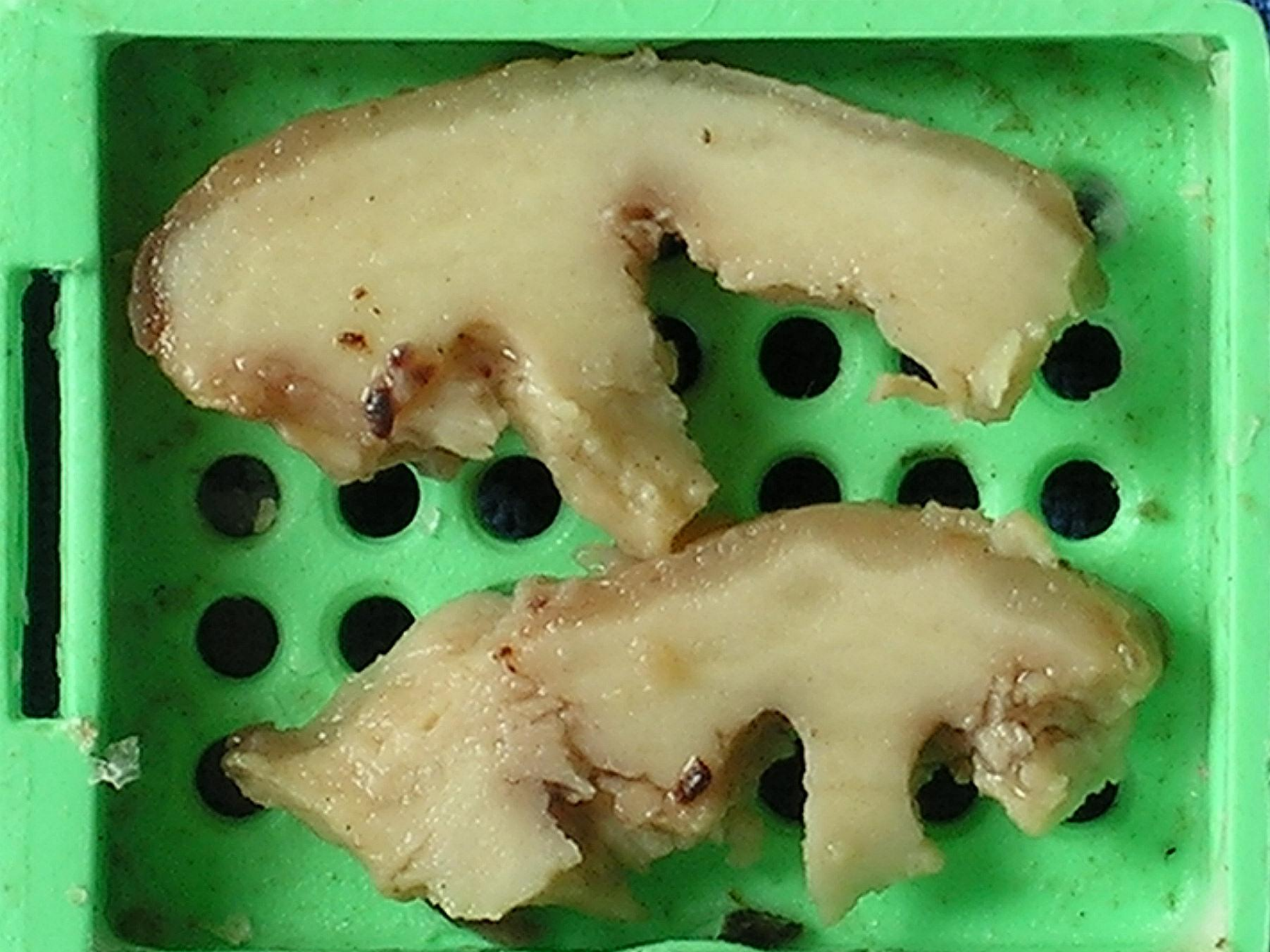
Brain tumors

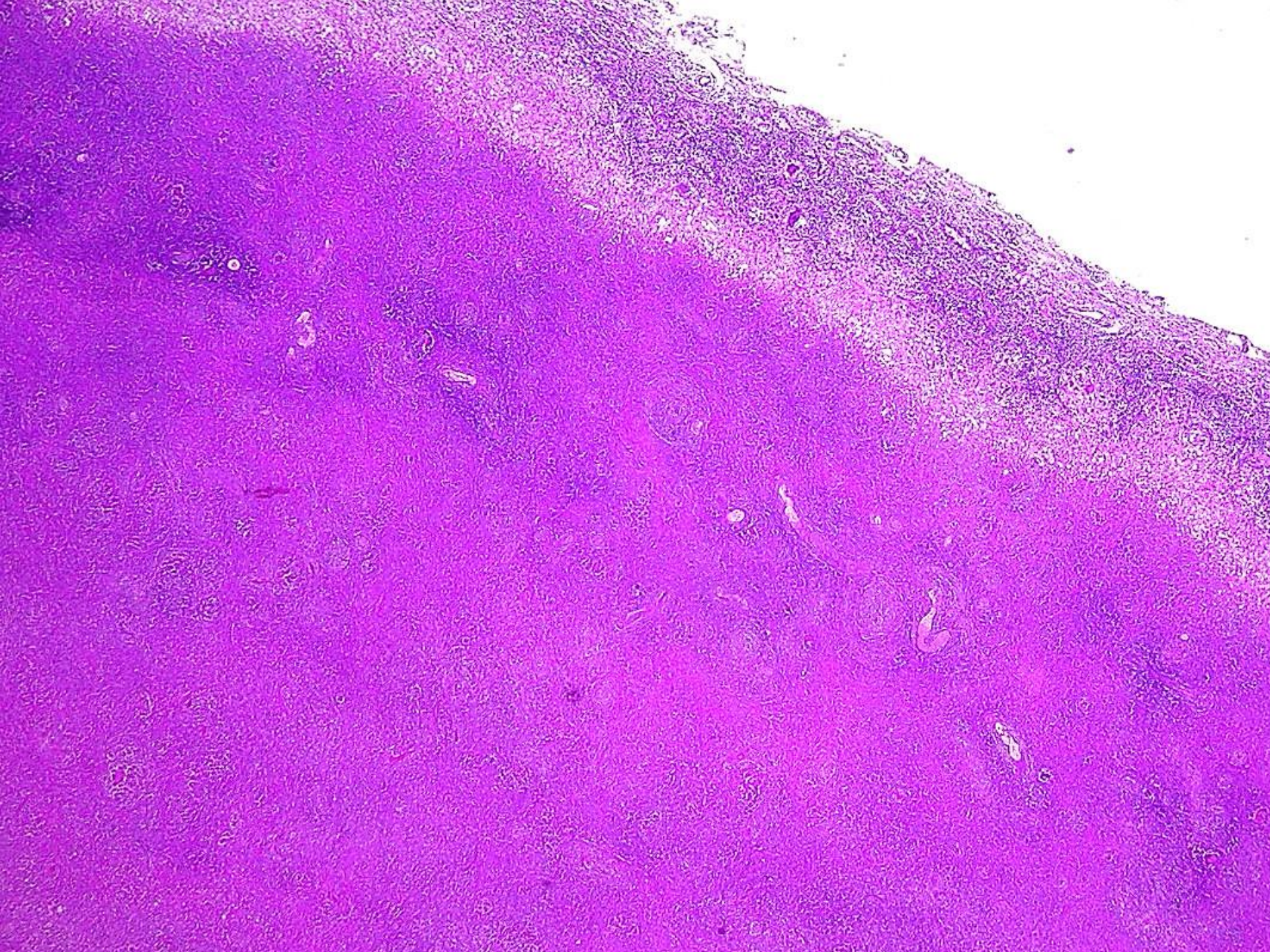
- About half of the brain tumors are primary, half metastatic
- Distinction between benign and malignant is less evident, gliding, [I – IV]
- and less relevant [site]
- Pattern of spread: no lymphatics
- but: CSF [medulloblastoma, ependymoma]
- Even grade IV tumors do not metastasize outside the brain

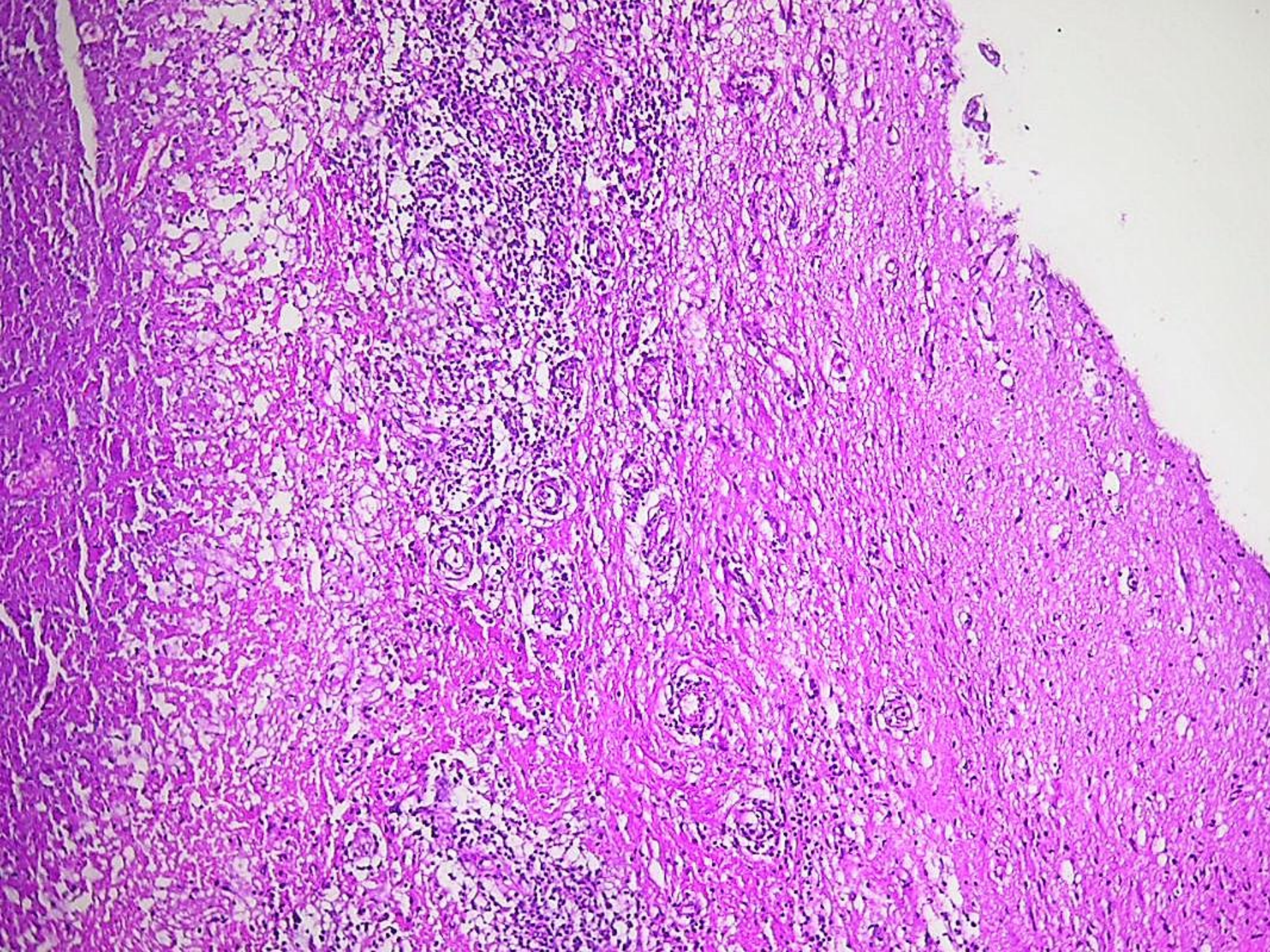
Practical Example

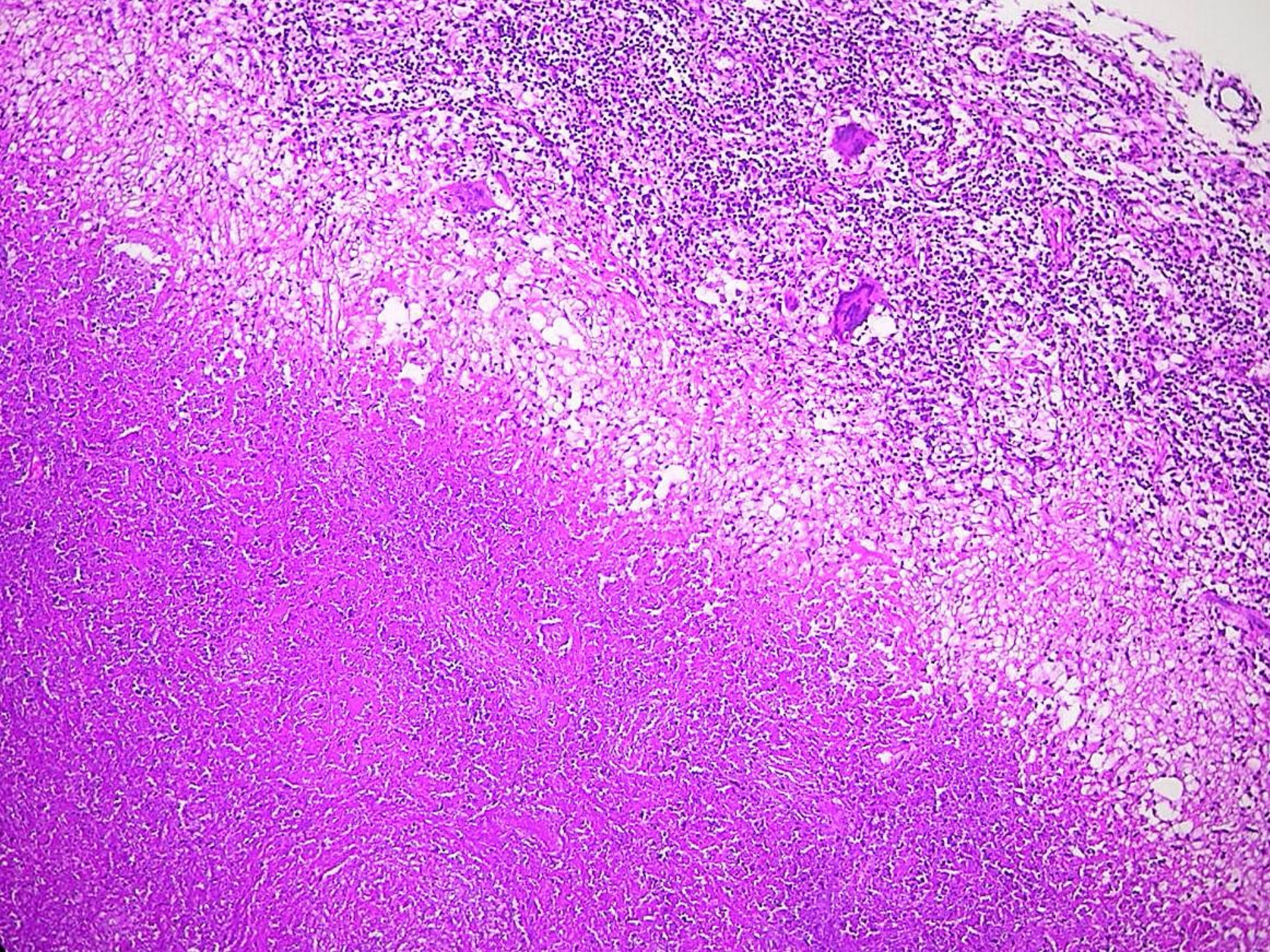
Number	11.2425
Sex / age	M 14
Clinic	Blindness 3 months. Right cerebellar mass with obstructive hydrocephalus.

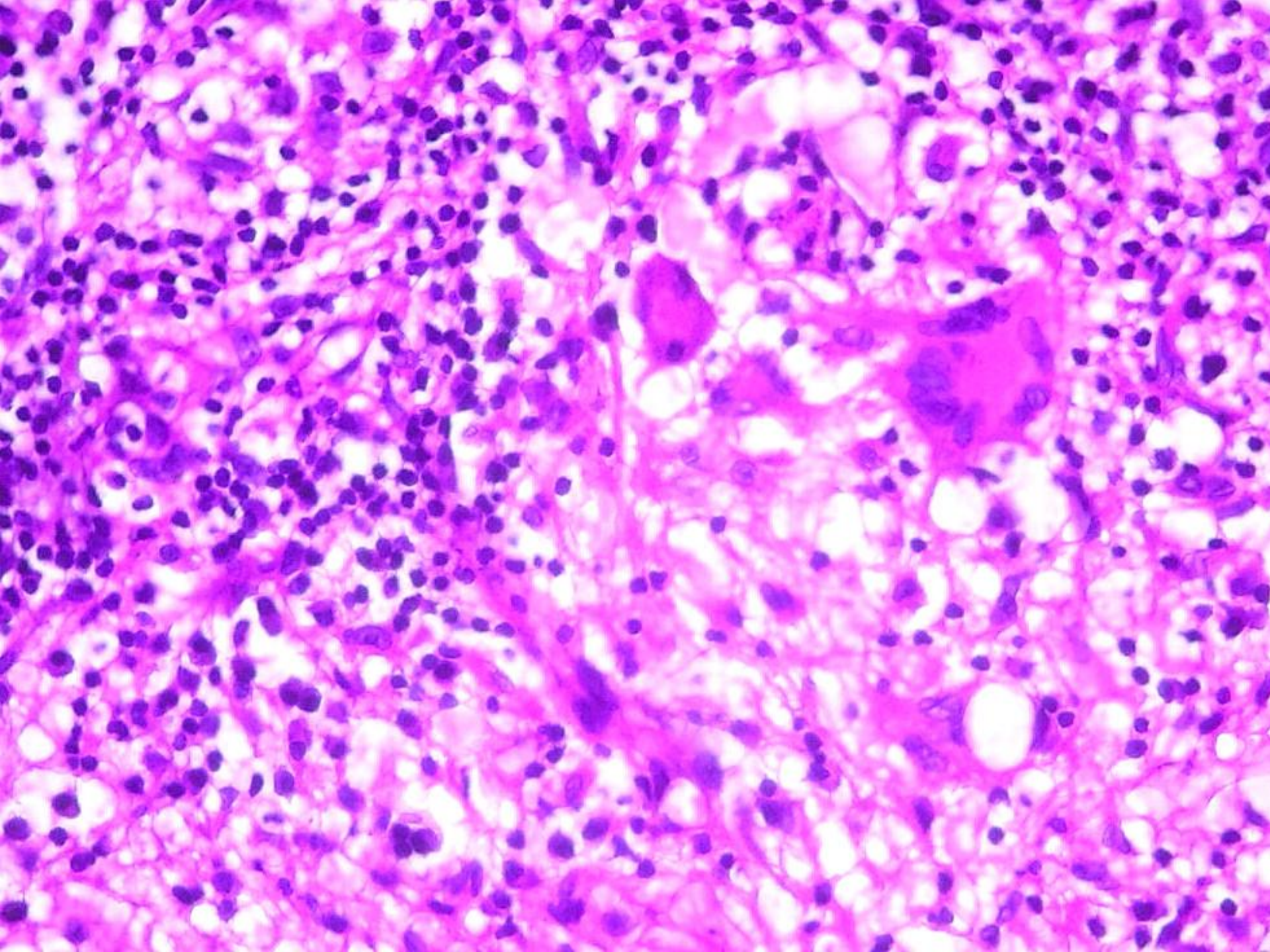












Number	11.2425
Sex / age	M 14
Clinic	Blindness 3 months. Right cerebellar mass with obstructive hydrocephalus.
Diagnosis	Tuberculoma
Comment	